MATH 3410: Complex Variables

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Preface

These are the first edition of these lecture notes for MATH 3410 (Complex Variables). Consequently, there may be several typographical errors, missing exposition on necessary background, and more advanced topics for which there will not be time in class to cover. Future iterations of these notes will hopefully be fairly self-contained provided one has the necessary background. If you come across any typos, errors, omissions, or unclear expositions, please feel free to contact me so that I may continually improve these notes.

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Chapter 1

Complex Numbers

The study of complex analysis begins with a deep understanding of complex numbers, which are the foundational objects upon which the entire theory is built. While real numbers suffice for most classical calculus, many mathematical phenomena and physical models demand a more expansive number system that can handle operations like taking square roots of negative numbers. This need leads naturally to the field of complex numbers.

Complex numbers extend the real number system by introducing a new unit, denoted by i, which satisfies the property $i^2 = -1$. The resulting system, denoted by $\mathbb{C}$, possesses rich algebraic and geometric structure. Unlike the real line, the complex numbers form a two-dimensional plane, allowing us to study functions, differentiation, and integration in multidimensional, yet highly structured setting.

This chapter introduces the algebraic and geometric properties of complex numbers, explores their representation in both rectangular and polar form, and discuss key operations such as complex conjugation, modulus, and argument. We also require the complex exponential function and Euler's identity, which provide powerful tools for expressing and manipulating complex quantities. These foundational tools will play a crucial role in the chapters to come.

1.1 Complex Numbers

The system of real numbers $\mathbb{R}$ is sufficient for solving many algebraic and analytical problems. However, even simple polynomial equations such as

$$x^2 + 1 = 0$$

have no solutions in $\mathbb{R}$. This limitation motivates the extension of the real numbers to a larger number system in which such equations admit solutions.

The complex numbers arise precisely from this need. By introducing a new number i satisfying

$$i^2 = -1$$

we obtain a system $\mathbb{C}$ that is algebraically complete: every nonconstant polynomial equation has a solution in $\mathbb{C}$.

Beyond algebra, complex numbers play a central role in analysis, geometry, differential equations, and many applied fields. In this section, we develop the algebraic structure of $\mathbb{C}$ and establish its fundamental properties.

Definition 1.1.1. A *complex number* is a number of the form

$$z = a + ib$$

where $a, b \in \mathbb{R}$ and i is the imaginary unit satisfying

$$i^2 = -1$$

The number a is called the *real part of* z, denoted by $\mathrm{Re}(z)$, and the number b is called the *imaginary part of* z, denoted by $\mathrm{Im}(z)$.

Remark 1.1.2. Given a complex number $z = a + ib$,

- If $b = 0$, then $z = a$ is a *real number*.
- If $a = 0$, then $z = ib$ is a *pure imaginary number*

The set of all complex numbers is denoted by

$$\mathbb{C} = \{a + ib : a, b \in \mathbb{R}\}$$

Definition 1.1.3. Let $z = a_1 + ib_1$ and $w = a_2 + ib_2$. Then $z = w$ if and only if $a_1 = a_2$ and $b_1 = b_2$. Equivalently,

$$z = w \iff \mathrm{Re}(z_1) = \mathrm{Re}(z_2), \quad \mathrm{Im}(z_1) = \mathrm{Im}(z_2)$$

Let $z = a_1 + ib_1$ and $w = a_2 + ib_2$. We introduce the operations of addition, multiplication, and division.

- Addition:

$$z + w = (a_1 + ib_1) + (a_2 + ib_2) = (a_1 + a_2) + i(b_1 + b_2)$$

- Multiplication:

$$zw = (a_1 + ib_1)(a_2 + ib_2) = (a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1)$$

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- Division (when $z_2 \neq 0$):

$$\frac{z}{w} = \frac{a_1 + ib_1}{a_2 + ib_2} = \frac{(a_1 + ib_1)(a_2 - ib_2)}{a_2^2 + b_2^2}$$

The process for dividing complex numbers involve multiplying the numerator and denominator by the complex conjugate, which we will talk about shortly.

The complex numbers satisfy the algebraic laws.

- Commutativity: For any $z_1, z_2 \in \mathbb{C}$,

$$z_1 + z_2 = z_2 + z_1$$
$$z_1 z_2 = z_2 z_1$$

- Associativity: For any $z_1, z_2, z_3 \in \mathbb{C}$,

$$z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$$
$$z_1(z_2 z_3) = (z_1 z_2) z_3$$

- Distributivity: For any $z_1, z_2, z_3 \in \mathbb{C}$,

$$z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$$

- Additive identity: There exists an additive identity $0 = 0 + 0i$ satisfying

$$z + 0 = z$$

  for any complex number z.
- Multiplicative identity: There exists a multiplicative identity $1 = 1 + 0i$ satisfying

$$z \cdot 1 = z$$

  for any complex number z.
- Additive inverse: For any complex number $z = a + ib$, there exists an additive inverse $-z = -a - ib$ such that

$$z + (-z) = 0$$

- Multiplicative inverse: For any nonzero complex number $z = a + ib$, there exists a multiplicative inverse

$$z^{-1} = \frac{a - ib}{a^2 + b^2}$$

  such that

$$z \cdot z^{-1} = 1$$

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Definition 1.1.4. Let $z = a + ib$. The *complex conjugate of* z is the complex number

$$\bar{z} = a - ib$$

Proposition 1.1.5. *Let $z, w \in \mathbb{C}$ be complex numbers. Then*

(1) $\overline{z + w} = \bar{z} + \bar{w}$
(2) $\overline{zw} = \bar{z}\bar{w}$
(3) $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$
(4) $\bar{\bar{z}} = z$
(5) $\mathrm{Re}(z) = \frac{z+\bar{z}}{2}$
(6) $\mathrm{Im}(z) = \frac{z-\bar{z}}{2}$
(7) $z\bar{z} = \mathrm{Re}^2(z) + \mathrm{Im}^2(z)$

Proof. The proof is straightforward and follows directly from the definitions. □

To divide complex numbers, multiply numerator and denominator by the conjugate

$$\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}$$

This eliminates the imaginary part in the denominator.

Remark 1.1.6. Although many algebraic properties carry over from $\mathbb{R}$ to $\mathbb{C}$, an important difference is that *there is no natural ordering on* $\mathbb{C}$. That is, expressions like $z < w$ are not defined in general. Complex numbers extend $\mathbb{R}$ algebraically, not order-theoretically. This distinction becomes crucial in analysis.

While complex numbers of the form $z = a + ib$ is useful in computation, it does not reveal the geometric structure underlying complex numbers.

A key insight is that each complex number can be identified with a point in the plane. This geometric viewpoint allows us to interpret algebraic operations visually and provides a foundation for much of complex analysis.

Every complex number

$$z = x + iy$$

can be identified with the ordered pair $(x, y) \in \mathbb{R}^2$. The *real part* corresponds to the x-coordinate, while the *imaginary part* corresponds to the y-coordinate. Thus, we establish a one-to-one correspondence:

$$\mathbb{C} \longleftrightarrow \mathbb{R}^2$$

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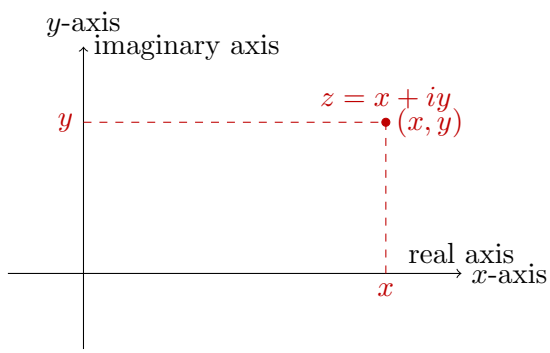
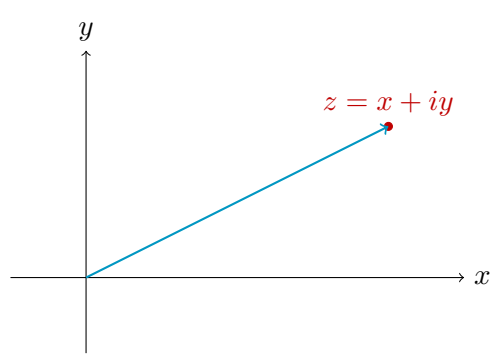
Definition 1.1.7. The *complex plane* (or *Argand plane*) is the plane in which the horizontal axis is the *real axis*, and the vertical axis is the *imaginary axis*. A complex number $z = x + iy$ is represented by the point (x, y).



A complex number $z = x + iy$ can also be viewed as a vector from the origin to the point (x, y). This interpretation leads to

- Vector addition $\iff$ complex addition
- Vector subtraction $\iff$ complex subtraction



If $z = (x_1, y_1)$ and $w = (x_2, y_2)$, then

$$z + w = (x_1 + x_2, y_1 + y_2)$$

Geometrically, this corresponds to the parallelogram law.

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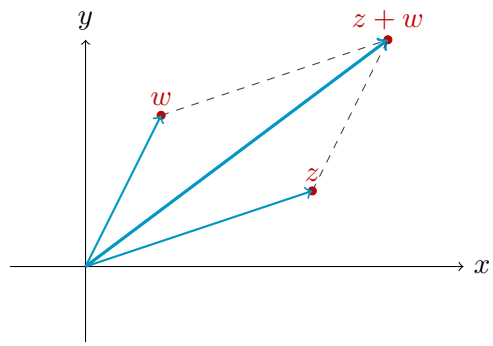


Definition 1.1.8. Let $z + x + iy$. The *modulus of* z is defined by

$$|z| = \sqrt{x^2 + y^2}$$

$|z|$ is the distance from the origin to the point (x, y). It is analogous to the length of a vector in $\mathbb{R}^2$.

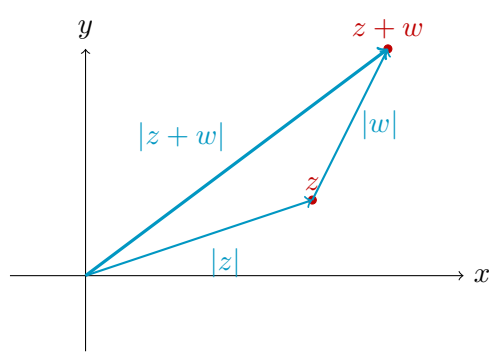
Proposition 1.1.9. *Let* $z, w \in \mathbb{C}$. *Then*

(1) $|z| \geq 0$
(2) $|z| = 0$ *if and only if* $z = 0$.
(3) $|zw| = |z||w|$
(4) $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$, *provided that* $w \neq 0$.

Proof. The proof is straightforward and follows directly from the definitions. □

Theorem 1.1.10 (Triangle Inequality). *For all* $z, w \in \mathbb{C}$,

$$|z + w| \leq |z| + |w|$$



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Observe that for any $z, w \in \mathbb{C}$, we can write

$$z = (z + w) + (-w)$$

Then applying the Triangle Inequality

$$|z| = |(z + w) + (-w)| \leq |z + w| + |-w|$$

But since $|-w| = |w|$, we have

$$|z| \leq |z + w| + |w|$$

Therefore

$$|z + w| \geq |w| - |z| = -(|z| - |w|)$$

Theorem 1.1.11 (**Lower Triangle Inequality**). *For any $z, w \in \mathbb{C}$,*

$$|z + w| \geq ||z| - |w||$$

That is, the length of one side of a triangle cannot exceed the sum of the other two. These inequalities are fundamental in analysis and will be used extensively later.

Example 1.1.12. Describe the set of points satisfying

$$|z| = |z - i|$$

Solution. Let $z = x + iy$ be arbitrary. Then note $z - i = x + iy - i = x + (y - 1)i$. Then using the definition of modulus,

$$|z| = \sqrt{x^2 + y^2}, \quad |z - i| = \sqrt{x^2 + (y - 1)^2}$$

and so squaring both quantities and equating gives

$$\begin{aligned} x^2 + y^2 &= x^2 + (y - 1)^2 \\ y^2 &= y^2 - 2y + 1 \\ y &= \frac{1}{2} \end{aligned}$$

Thus, the set is

$$z = x + \frac{1}{2}i$$

which represents a horizontal line.

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1.2 Polar Form, Powers, and Roots

In Section 1.1, complex numbers were expressed in *rectangular form*

$$z = x + iy$$

which is well-suited for addition and subtraction. However, this representation is less convenient when dealing with multiplication, division, and powers.

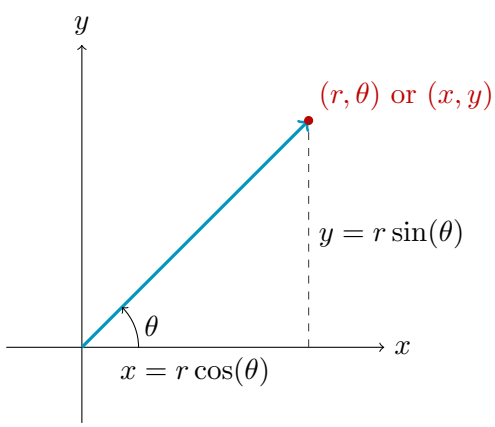
A more natural representation arises from geometry. Since every complex number corresponds to a point in the plane, it can also be described using *polar coordinates*. This leads to the *polar form* of a complex number, which encodes both its magnitude and direction.

This representation simplifies multiplication and division, and leads to powerful results such as *De Moivre's formula*, as well as explicit formulas for powers and roots of complex numbers. These ideas are fundamental in both theoretical and applied complex analysis.

Let $z = x + iy \neq 0$. From geometry,

$$x = r\cos(\theta), \quad y = r\sin(\theta)$$

where $r = |z| = \sqrt{x^2 + y^2}$ is the modulus, and θ is the *argument* (angle).



Definition 1.2.1. A nonzero complex number z can be written in the form

$$z = r(\cos(\theta) + i\sin(\theta))$$

where $r = |z| > 0$ and θ is an argument of z.

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An angle θ satisfying

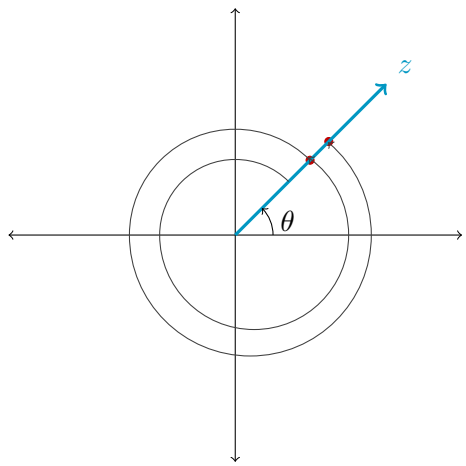
$$\cos(\theta) = \frac{x}{r}, \quad \sin(\theta) = \frac{y}{r}$$

is called an *argument of* z, denoted

$$\theta = \arg(z)$$

The argument is not unique, since for any $k \in \mathbb{Z}$,

$$\arg(z) = \theta + 2\pi k$$



Definition 1.2.2. The *principal argument of* z is the unique value $\mathrm{Arg}(z)$ such that

$$-\pi < \mathrm{Arg}(z) \leq \pi$$

Remark 1.2.3. Care must be taken when computing $\theta = \tan^{-1}\left(\frac{y}{x}\right)$.

- This only determines the angle up to quadrant ambiguity.
- The correct quadrant must be determined from (x, y).

Proposition 1.2.4. *Let* $z = r_1(\cos(\theta_1) + i\sin(\theta_1))$ *and* $w = r_2(\cos(\theta_2) + i\sin(\theta_2))$. *Then*

(1) $zw = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$
(2) $\frac{z}{w} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2))$

Proof. To see that (1) is true, multiplying the two complex numbers give

$$zw = r_1(\cos(\theta_1) + i\sin(\theta_1))r_2(\cos(\theta_2) + i\sin(\theta_2))$$

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$$
\begin{aligned}
&= r_1 r_2(\cos(\theta_1)\cos(\theta_2) + i\cos(\theta_1)\sin(\theta_2) + i\sin(\theta_1)\cos(\theta_2) + i^2\sin(\theta_1)\sin(\theta_2)) \\
&= r_1 r_2[(\cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2)) + i(\cos(\theta_1)\sin(\theta_2) + \sin(\theta_1)\cos(\theta_2))]
\end{aligned}
$$

Using the compound angle formulas

$$
\begin{aligned}
\cos(\theta_1 + \theta_2) &= \cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2) \\
\sin(\theta_1 + \theta_2) &= \sin(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2)
\end{aligned}
$$

the above gives

$$
zw = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))
$$

To see that (2) is true, dividing the two complex numbers give

$$
\begin{aligned}
\frac{z}{w} &= \frac{r_1(\cos(\theta_1) + i\sin(\theta_1))}{r_2(\cos(\theta_2) + i\sin(\theta_2))} \\
&= \frac{r_1}{r_2}\frac{\cos(\theta_1) + i\sin(\theta_1)}{\cos(\theta_2) + i\sin(\theta_2)} \cdot \frac{\cos(\theta_2) - i\sin(\theta_2)}{\cos(\theta_2) - i\sin(\theta_2)} \\
&= \frac{r_1}{r_2}\frac{\cos(\theta_1)\cos(\theta_2) - i\sin(\theta_2)\cos(\theta_1) + i\sin(\theta_1)\cos(\theta_2) - i^2\sin(\theta_1)\sin(\theta_2)}{\cos^2(\theta_2) + \sin^2(\theta_2)} \\
&= \frac{r_1}{r_2}[(\cos(\theta_1)\cos(\theta_2) + \sin(\theta_1)\sin(\theta_2)) + i(\sin(\theta_1)\cos(\theta_2) - \cos(\theta_1)\sin(\theta))]
\end{aligned}
$$

Using the compound angle formulas

$$
\begin{aligned}
\cos(\theta_1 - \theta_2) &= \cos(\theta_1)\cos(\theta_2) + \sin(\theta_1)\sin(\theta_2) \\
\sin(\theta_1 - \theta_2) &= \sin(\theta_1)\cos(\theta_2) - \cos(\theta_1)\sin(\theta_2)
\end{aligned}
$$

the above gives

$$
\frac{z}{w} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2))
$$

as desired. □

Corollary 1.2.5. *For any $z, w \in \mathbb{C}$,*

(1) $|zw| = |z||w|$
(2) $\arg(zw) = \arg(z) + \arg(w)$
(3) $\arg\left(\frac{z}{w}\right) = \arg(z) - \arg(w)$

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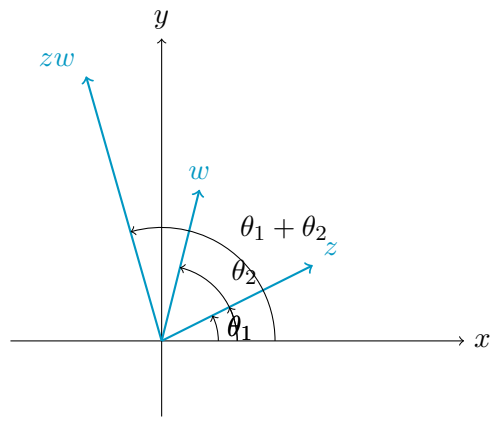

Theorem 1.2.6 (De Moivre's Formula). *Let* $z = r(\cos(\theta) + i\sin(\theta))$. *Then for any* $n \in \mathbb{Z}$,

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

That is, the magnitude is raised to the nth power, and the argument is multiplied by n.

Example 1.2.7. Let $z = \cos(\theta) + i\sin(\theta)$. Then

$$(\cos(\theta) + i\sin(\theta))^n = \cos(n\theta) + i\sin(n\theta)$$

For negative integers, we have

$$z^{-n} = \frac{1}{z^n} = r^{-n}(\cos(-n\theta) + i\sin(-n\theta))$$

Using the facts that $\cos(-\theta) = \cos(\theta)$ and $\sin(-\theta) = -\sin(\theta)$, we have

$$z^{-n} = r^{-n}(\cos(n\theta) - i\sin(n\theta))$$

We now solve complex equations of the form

$$w^n = z$$

Let

$$z = r(\cos(\theta) + i\sin(\theta))$$

where $r > 0$. Write w in polar form $w = \rho(\cos(\phi) + i\sin(\phi))$ and applying De Moivre's formula,

$$w^n = \rho^n(\cos(n\phi) + i\sin(n\phi))$$

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Setting $w^n = z$, we have

$$\rho^n = r, \quad n\phi = \theta + 2\pi k$$

Therefore, the solutions of w^n are

$$\rho = \sqrt[n]{r}, \quad \phi = \frac{\theta + 2\pi k}{n}$$

for $k \in \mathbb{Z}$, or

$$w_k = \sqrt[n]{r}\left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i\sin\left(\frac{\theta + 2\pi k}{n}\right)\right], \quad k = 0, 1, 2, ..., n-1$$

There are exactly n distinct roots, in which all roots have the same modulus $|w_k| = \sqrt[n]{r}$, and all arguments are equally spaced $\frac{2\pi}{n}$. Geometrically, the nth roots lie on a circle of radius $\sqrt[n]{r}$ and form a regular n-gon.

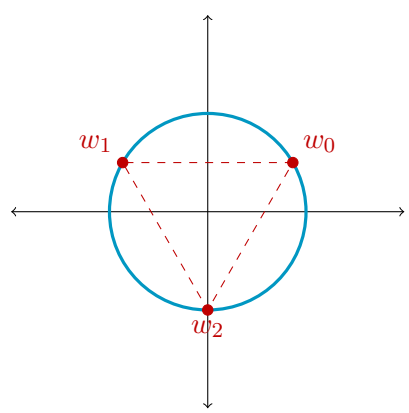
Example 1.2.8. Find the three cube roots of $z = i$.

Solution. We solve the equation $w^3 = i$. In this case, we have $r = 1$ and $\theta = \frac{\pi}{2}$. With $n = 3$, we have $\rho = 1$ and $\phi = \frac{\pi}{6} + \frac{2\pi k}{3}$. Thus, the solutions of $w^3 = i$ are given by

$$w_k = \cos\left(\frac{\pi}{6} + \frac{2\pi k}{3}\right) + i\sin\left(\frac{\pi}{6} + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2$$

Explicitly,

$$\begin{aligned}
w_0 &= \cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} + \frac{1}{2}i, \\
w_1 &= \cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2} + \frac{1}{2}i, \\
w_2 &= \cos\left(\frac{3\pi}{2}\right) + i\sin\left(\frac{3\pi}{2}\right) = -i
\end{aligned}$$



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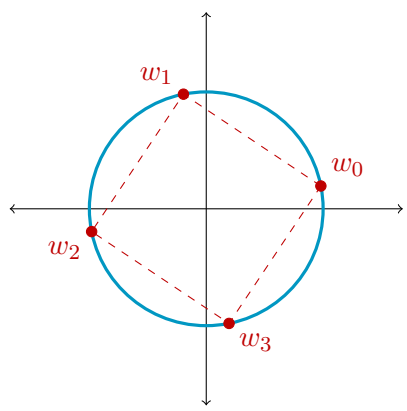
Example 1.2.9. Find the four fourth roots of $z = 1 + i$.

Solution. We solve the equation $w^4 = 1 + i$. In this case, we have $r = \sqrt{2}$ and $\theta = \frac{\pi}{4}$. With $n = 4$, we have $\rho = \sqrt[4]{2}$ and $\phi = \frac{\pi}{16} + \frac{\pi k}{2}$. Thus, the solutions of $w^4 = 1 + i$ are given by

$$w_k = \sqrt[4]{2}\left(\cos\left(\frac{\pi}{16} + \frac{\pi k}{2}\right) + i\sin\left(\frac{\pi}{16} + \frac{\pi k}{2}\right)\right), \quad k = 0, 1, 2, 3.$$

Explicitly,

$$\begin{aligned}
w_0 &= \sqrt[4]{2}\left(\cos\left(\frac{\pi}{16}\right) + i\sin\left(\frac{\pi}{16}\right)\right) \approx 1.1664 + 0.2320i,\\
w_1 &= \sqrt[4]{2}\left(\cos\left(\frac{9\pi}{16}\right) + i\sin\left(\frac{9\pi}{16}\right)\right) \approx -0.2320 + 1.1664i,\\
w_2 &= \sqrt[4]{2}\left(\cos\left(\frac{17\pi}{16}\right) + i\sin\left(\frac{17\pi}{16}\right)\right) \approx -1.1664 - 0.2320i,\\
w_3 &= \sqrt[4]{3}\left(\cos\left(\frac{25\pi}{16}\right) + i\sin\left(\frac{25\pi}{16}\right)\right) \approx 0.2320 - 1.1664i.
\end{aligned}$$



Definition 1.2.10. The *principal nth root* corresponds to $k = 0$

$$w_0 = \sqrt[n]{r}\left[\cos\left(\frac{\mathrm{Arg}(z)}{n}\right) + i\sin\left(\frac{\mathrm{Arg}(z)}{n}\right)\right]$$

Example 1.2.11. Solve $z^n = 1$.

Solution. Since $1 = \cos(0) + i\sin(0)$,

$$z_k = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right), \quad k = 0, 1, 2, ..., n-1$$

These are the nth roots of unity. All roots lie on the unit circle, are equally spaced, and they form a regular polygon.

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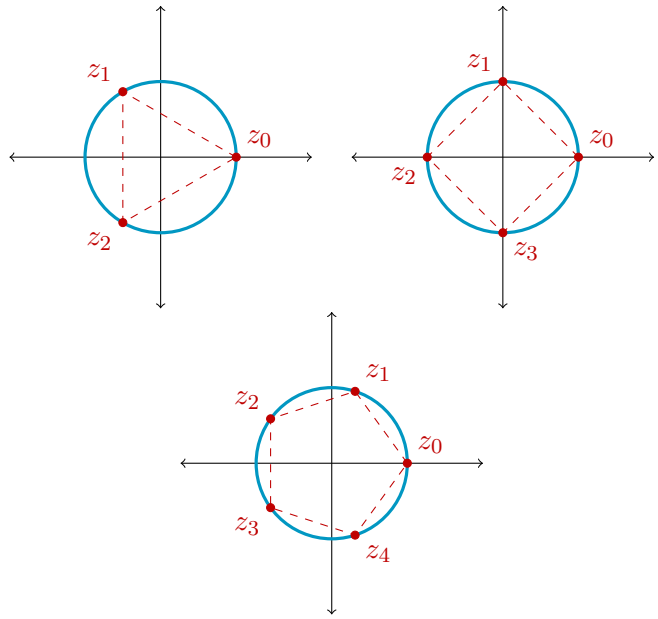

Remark 1.2.12. Care must be taken

$$\arg(z) \neq \operatorname{Arg}(z)$$

This is because $\operatorname{Arg}(z)$ is single-valued, while $\arg(z)$ is multi-valued.

Remark 1.2.13. In general,

$$\operatorname{Arg}(zw) \neq \operatorname{Arg}(z) + \operatorname{Arg}(w)$$

This equality holds modulo 2π, but not necessarily for principal values.

Remark 1.2.14. Unlike $\mathbb{R}$, the complex numbers are closed under taking roots:

$$z \in \mathbb{C} \implies \text{all roots of } z \text{ lie in } \mathbb{C}$$

1.3 Euler's Formula and the Exponential Form of Complex Numbers

In Section 1.2, we expressed complex numbers in the form

$$z = r(\cos(\theta) + i\sin(\theta))$$

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which greatly simplified multiplication, division, and powers. However, this representation can be made even more compact and powerful through the introduction of the complex exponential function.

A remarkable identity, known as Euler's formula, establishes a deep connection between exponential and trigonometric functions:

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

This allows us to rewrite polar form as

$$z = re^{i\theta}$$

which is called the exponential form of a complex number.

Euler's formula is one of the most important results in mathematics. It provides a unified framework for working with complex numbers and plays a central role in complex analysis, differential equations, Fourier analysis, and engineering applications.

We extend the exponential function to complex numbers.

Definition 1.3.1. For $z = x + iy \in \mathbb{C}$, define

$$e^z = e^{x+iy} = e^x(\cos(y) + i\sin(y))$$

If $z = i\theta$, then

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

This is *Euler's Formula.*

Theorem 1.3.2 (Euler's Formula). *For all $\theta \in \mathbb{R}$,*

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

Proof. For any complex number z, we have the power series formula for e^z given as

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

Hence, replacing z with $i\theta$ gives

$$\begin{aligned} e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\ &= 1 + i\theta + \frac{(i\theta)^2}{2} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \cdots \end{aligned}$$

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$$= 1 + i\theta - \frac{\theta^2}{2} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} + \cdots$$
$$= \left(1 - \frac{\theta^2}{2} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)$$

Using the power series formula for $\cos(\theta)$ and $\sin(\theta)$ given by

$$\cos(\theta) = \sum_{n=0}^{\infty} \frac{(-1)^n \theta^{2n}}{(2n)!}, \quad \sin(\theta) = \sum_{n=0}^{\infty} \frac{(-1)^n \theta^{2n+1}}{(2n+1)!}$$

we have

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

as desired. □

Corollary 1.3.3. *For any* $\theta \in \mathbb{R}$,
(1) $\cos(\theta) = \frac{e^{i\theta}+e^{-i\theta}}{2}$
(2) $\sin(\theta) = \frac{e^{i\theta}-e^{-i\theta}}{2}$

Proof. By Euler's formula, we have

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

Also, write

$$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta) = \cos(\theta) - i\sin(\theta)$$

Then adding the two equations, we get

$$2\cos(\theta) = e^{i\theta} + e^{-i\theta}$$
$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

Similarly, subtracting the two equations, we get

$$2i\sin(\theta) = e^{i\theta} - e^{-i\theta}$$
$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

as desired. □

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By setting $\theta = \pi$, we have the famous identity

$$e^{i\pi} + 1 = 0$$

This identity connects the constants e, i, π, 1, and 0.

Using Euler's formula, the polar form becomes

$$z = r(\cos(\theta) + i\sin(\theta)) = re^{i\theta}$$

Again, $r = |z|$ represents the magnitude, and $\theta = \arg(z)$ represents the angle.

Since $\arg(z)$ is multi-valued, the complex number z is not unique, and thus,

$$z = re^{i(\theta+2\pi k)}$$

for $k \in \mathbb{Z}$. However, since $\operatorname{Arg}(z)$ is single-valued,

$$z = re^{i\operatorname{Arg}(z)}$$

is a unique.

Proposition 1.3.4. *For any $z, w \in \mathbb{C}$,*

$$e^{z+w} = e^z e^w$$

Proof. Let $z = x_1 + iy_1$ and $w = x_2 + iy_2$. Then $z + w = (x_1 + x_2) + i(y_1 + y_2)$ and so by definition of the complex exponential

$$\begin{aligned}
e^{z+w} &= e^{x_1+x_2}(\cos(y_1 + y_2) + i\sin(y_1 + y_2)) \\
&= e^{x_1}e^{x_2}(\cos(y_1)\cos(y_2) - \sin(y_1)\sin(y_2) + i\sin(y_1)\cos(y_2) + i\cos(y_1)\sin(y_2)) \\
&= e^{x_1}e^{x_2}[\cos(y_1)(\cos(y_2) + i\sin(y_2)) + i\sin(y_1)(\cos(y_2) + i\sin(y_2))] \\
&= e^{x_1}e^{x_2}(\cos(y_1) + i\sin(y_1))(\cos(y_2) + i\sin(y_2)) \\
&= e^{x_1}(\cos(y_1) + i\sin(y_1))e^{x_2}(\cos(y_2) + i\sin(y_2)) \\
&= e^z e^w
\end{aligned}$$

□

Corollary 1.3.5. *For any $\theta_1, \theta_2 \in \mathbb{R}$,*

(1) $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$

(2) $\frac{e^{i\theta_1}}{e^{i\theta_2}} = e^{i(\theta_1-\theta_2)}$

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Multiplication by $e^{i\theta}$ corresponds to a rotation by angle θ.

Let $z = re^{i\theta}$. By raising both sides to n, we obtain

$$z^n = r^n e^{in\theta}$$

In connection to De Moivre's Formula, this is equivalent to

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

Therefore,

$$z^n = r^n e^{in\theta} = r^n(\cos(n\theta) + i\sin(n\theta))$$

To solve the complex equation

$$w^n = z = re^{i\theta}$$

we write

$$w = \rho e^{i\phi}$$

Then the solution to the complex equation are

$$w_k = \sqrt[n]{r}e^{i\frac{\theta+2\pi k}{n}}, \quad k = 0, 1, 2, ..., n-1$$

1.4 Basic Topology on $\mathbb{C}$

In previous sections, we developed both the algebraic and geometric structure of complex numbers. We saw that $\mathbb{C}$ can be identified with the plane $\mathbb{R}^2$, and that notions such as distance and magnitude arise naturally through the modulus.

To study complex functions rigorously, we must now introduce the language of *topology*, which formalizes ideas such as "closeness", "continuity", and "boundary". In particular, we will define important classes of sets in the complex plane, including open sets, neighbourhoods, domains, and regions.

These concepts are essential for understanding limits, continuity, and "holomorphicity" in later chapters. Although they may appear abstract at first, they are deeply geometric and can be visualized in terms of familiar objects such as disks and circles.

Definition 1.4.1. Let $z_0 = x_0 + iy_0 \in \mathbb{C}$ and $\rho > 0$. The set of points $z \in \mathbb{C}$ satisfying

$$|z - z_0| = \rho$$

is called a *circle of radius* ρ centered at z_0.

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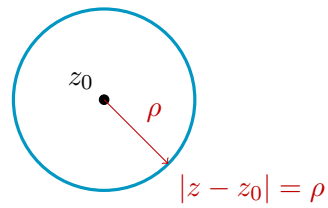
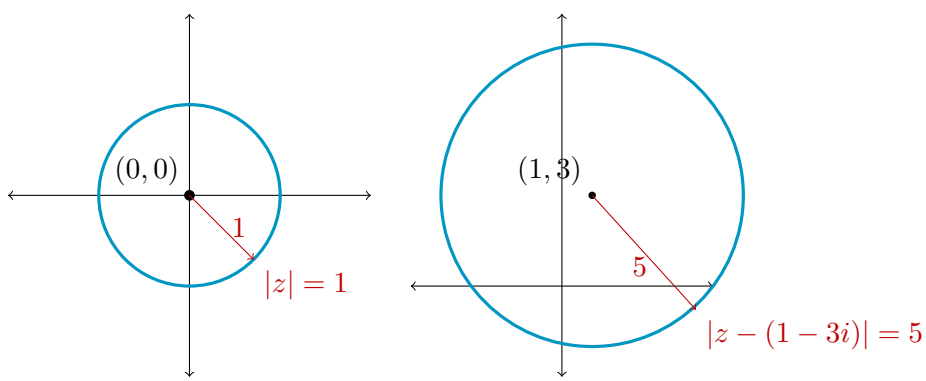
Since
$$|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2}$$
this is precisely the usual equation of a circle in $\mathbb{R}^2$.



Example 1.4.2. (1) $|z| = 1$ is the unit circle centered at the origin.
(2) $|z - (1 - 3i)| = 5$ is the circle centered at $1 - 3i$ with radius 5.



Definition 1.4.3. For $z_0 \in \mathbb{C}$ and $\rho > 0$, the *open disk of radius ρ centered at z_0* is the set
$$D(z_0, \rho) = \{z \in \mathbb{C} : |z - z_0| < \rho\}$$
and the *closed disk of radius ρ centered at z_0* is the inequality
$$D[z_0, \rho] = \{z \in \mathbb{C} : |z - z_0| \leq \rho\}$$

Definition 1.4.4. A *neighbourhood of z_0* is any open disk centered at z_0. A *deleted neighbourhood of z_0* is
$$0 < |z - z_0| < \rho$$
which excludes the center point.

Example 1.4.5. (1) $|z| < 1$: neighbourhood of the origin.
(2) $0 < |z - (2 + 2i)| < 0.5$: deleted neighbourhood of $z = 2 + 2i$.

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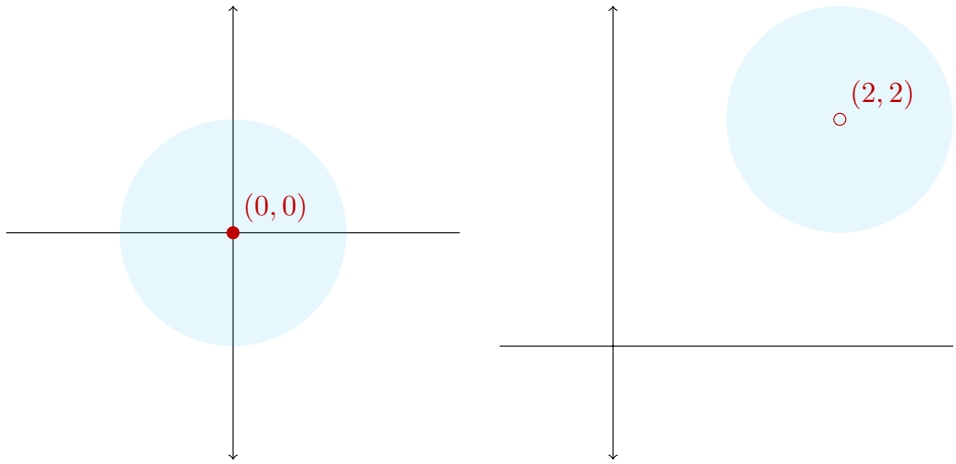
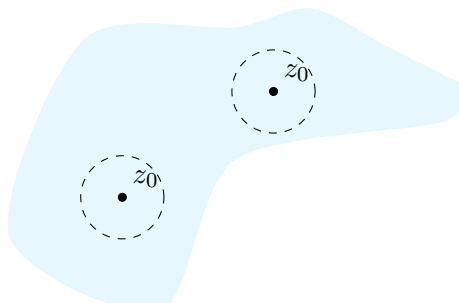
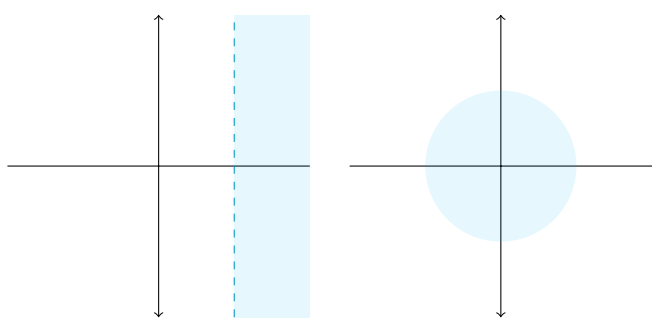


Definition 1.4.6. A set $S \subseteq \mathbb{C}$ is *open* if for every point $z_0 \in S$, there exists a neighbourhood of z_0 entirely contained in S.

Every point of an open set is an interior point. There are no "edge-only" points.



Example 1.4.7. (1) The half-plane $\{z \in \mathbb{C} : \mathrm{Re}(z) > 1\}$ is an open set.
(2) The disk $\{z \in \mathbb{C} : |z| < 2\}$ is an open set.
(3) The strip $\{z \in \mathbb{C} : -1 < \mathrm{Re}(z) < 1\}$ is an open set.



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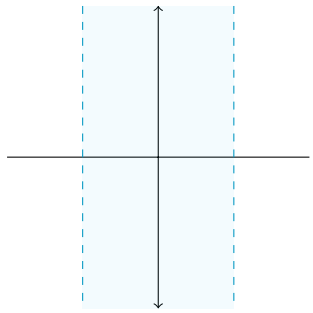
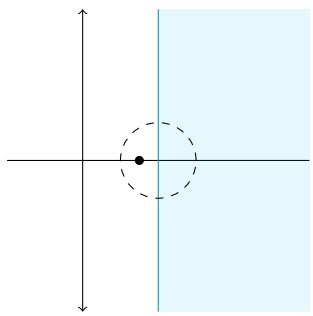
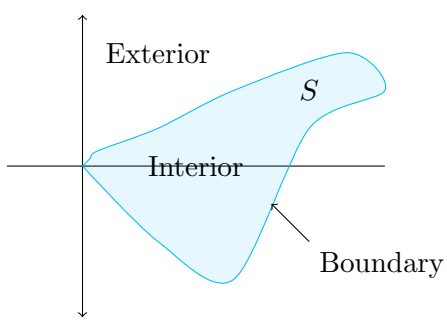


Example 1.4.8. The half-plane $\{z \in \mathbb{C} : \mathrm{Re}(z) \geq 1\}$ is not an open set since boundary points have no neighbourhood entirely inside the set.



Definition 1.4.9. Let $S \subseteq \mathbb{C}$.

(1) A point $z_0 \in S$ is an *interior point* if some neighbourhood of z_0 lies entirely in S.
(2) A point $z_0 \in S$ is a *boundary point* of S if every neighbourhood of z_0 contains points in S and points not in S. The set of all boundary points is called the boundary of S, denoted ∂S.
(3) A point $z_0 \in S$ is an *exterior point* if there exists a neighbourhood of z_0 containing no points of S.



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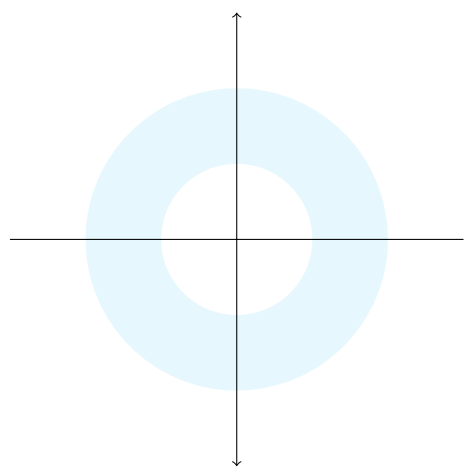
Definition 1.4.10. Let $0 < \rho_1 < \rho_2$. The set

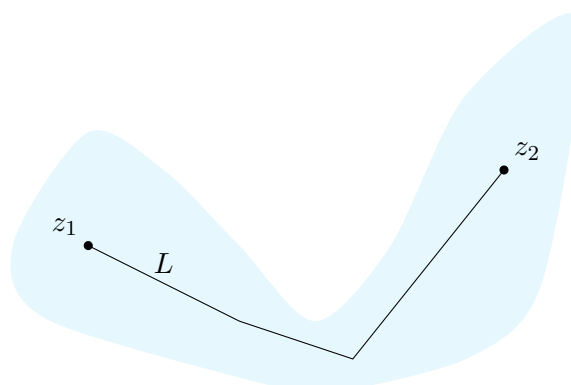
$$A(z_0, \rho_1, \rho_2) = \{z \in \mathbb{C} : \rho_1 < |z - z_0| < \rho_2\}$$

is called an annulus centered at z_0.

An annulus is simply a region between two circles, or an open circular "ring". In the case when $\rho_1 = 0$, we simply get a deleted neighbourhood of z_0.



Definition 1.4.11. A set S is *connected* if any two points in S can be joined by a path lying entirely in S. That is, any polygonal line L connecting any two points is contained inside S.



Definition 1.4.12. A *domain* is a set that is open and connected.

Example 1.4.13. (1) The open disk is a domain.

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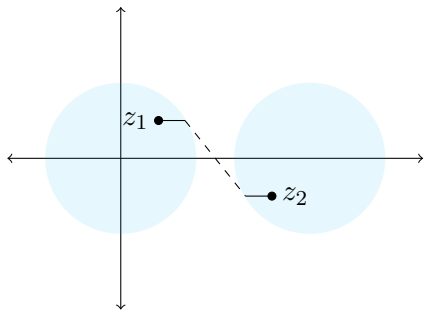
(2) The annulus is a domain.
(3) Two disjoint disks is not a domain since they are not connected.



Definition 1.4.14. A *region* is a set consisting of a domain together with some or all of its boundary points.

Example 1.4.15. (1) The open disk is a region.
(2) The closed disk is a region.
(3) The annulus (open or closed) is a region.

Definition 1.4.16. A set $S \subseteq \mathbb{C}$ is *bounded* if there exists $R > 0$ such that $|z| < R$ for all $z \in S$.

Example 1.4.17. (1) Disks are bounded sets
(2) Annuli are bounded sets
(3) $\mathbb{C}$ is not a bounded set.

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Chapter 2

Complex Functions and Holomorphic Functions

In Chapter 1, we introduced the complex numbers and developed the geometric language needed to work in the complex plane. We learned how to represent complex numbers algebraically and geometrically, how to interpret modulus and argument, and how to describe regions and sets in $\mathbb{C}$. With that foundation in place, we now turn to one of the central themes of complex analysis: functions of a complex variable.

At first glance, complex functions may seem similar to the real-valued functions studied in calculus. After all, a complex function is simply a rule that assigns to each complex number z another complex number $f(z)$. However, the behaviour of such functions is far richer and more rigid than the real-variable setting. In real calculus, differentiability is a local linear approximation along the real line. In complex analysis, differentiability must hold in every direction in the plane simultaneously. This stronger requirement leads to deep and beautiful consequences, and it is the reason why complex analysis is one of the most elegant and powerful branches of mathematics.

The main goal of this chapter is to introduce the notion of a holomorphic function, which is the complex analogue of a differentiable function in real calculus. Holomorphic functions form the core class of functions in complex analysis. They are not only differentiable, but in fact possess remarkable properties that far exceed those of ordinary differentiable real functions. Later chapters will show that holomorphic functions are automatically infinitely differentiable, locally represented by power series, and strongly constrained by the geometry of the complex plane.

To reach this important concept, we proceed gradually. We begin by

defining what it means to view a function as a mapping from subsets of $\mathbb{C}$ into $\mathbb{C}$ and by exploring how such functions can be expressed in terms of their real and imaginary parts. We then develop the notions of limit and continuity for complex functions, extending familiar ideas from real analysis to the two-dimensional setting of the complex plane. Once these foundational ideas are in place, we introduce complex differentiability, and from there, the concept of holomorphicity.

A crucial result of the chapter is the appearance of the Cauchy–Riemann equations, which provide a bridge between the algebraic structure of complex numbers and the calculus of functions of two real variables. These equations give necessary conditions, and under suitable assumptions sufficient conditions, for a complex function to be holomorphic. They reveal that holomorphicity is not merely a formal extension of real differentiability, but rather a highly structured condition linking the real and imaginary parts of a function in a precise way.

The chapter concludes with an introduction to harmonic functions, which arise naturally from holomorphic functions. Harmonic functions satisfy Laplace's equation and play an important role not only in complex analysis, but also in physics, engineering, and potential theory. The connection between holomorphic and harmonic functions is one of the first signs of the deep unity underlying the subject.

2.1 Complex Functions

One of the central ideas in all of mathematics is that of a function. In its most basic form, a function is a rule that assigns to each input exactly one output. In elementary calculus, we are usually concerned with functions whose inputs and outputs are real numbers, such as

$$f(x) = x^2, \quad f(x) = \sin(x), \quad f(x) = e^x$$

In complex analysis, we extend this idea by allowing both the inputs and outputs to be complex numbers.

This seemingly simple generalization leads to a much richer theory. A complex function does not merely assign one number to another; it acts on points in the complex plane and can therefore be interpreted geometrically as a mapping from one region of the plane to another. This geometric viewpoint is one of the distinguishing and powerful features of complex analysis.

In this section, we introduce the basic notion of a complex function and examine how such functions are described algebraically. In particular, we

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will see that every complex function

$$w = f(z)$$

can be written in terms of its real and imaginary parts, thereby connecting complex analysis with the theory of functions of two real variables. We will also revisit the complex exponential function, now from the viewpoint of complex-valued mappings, and use it to describe the important exponential form of a complex number.

This section lays the groundwork for everything that follows. Once we understand what complex functions are and how they behave, we will be ready to study limits, continuity, differentiability, and ultimately the remarkable class of holomorphic functions.

We specialize to the setting of complex analysis.

Definition 2.1.1. A *complex function* is a function $f : D \to \mathbb{C}$ where the domain D is a subset of $\mathbb{C}$. Equivalently, a complex function is a function whose inputs and outputs are complex numbers.

A complex function is also called a complex-valued function of a complex variable. If z is the input and w is the output, then we often write

$$w = f(z)$$

Remark 2.1.2. Throughout these notes, z will usually denote the complex input variable and w will often denote the complex output variable. Thus, if

$$w = f(z)$$

then both z and w are complex numbers.

Example 2.1.3. Define

$$f(z) = z^2 - (2 + i)z$$

This defines a complex function because for each $z \in \mathbb{C}$, the expression on the right size produces exactly one complex number. Let us evaluate f at a few points

- At $z = i$,
$$f(i) = i^2 - (2 + i)i = -1 - (2i - i) = -2i$$
- At $z = 1 + i$,
$$f(1 + i) = (1 + i)^2 - (2 + i)(1 + i) = 2i - (1 + 3i) = -1 - i$$

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Example 2.1.4. Define

$$g(z) = z + 2\mathrm{Re}(z)$$

This is also a complex function. Let us evaluate at a few points.

- At $z = i$, since $\mathrm{Re}(i) = 0$,

$$g(i) = i + 2(0) = i$$

- At $z = 2 - 3i$, since $\mathrm{Re}(2 - 3i) = 2$, we get

$$g(2 - 3i) = (2 - 3i) + 2(2) = 6 - 3i$$

Often, a formula does not explicitly specify its domain. In such cases, we take the domain to be the set of complex numbers for which the formula makes sense.

Definition 2.1.5. The *natural domain* of a complex function is the set of complex numbers z for which the formula defining the function is meaningful.

Example 2.1.6. Consider $f(z) = z^2 - (2 + i)z$. This expression is defined for every complex number z, so

$$\mathrm{Dom}(f) = \mathbb{C}$$

Likewise, for $g(z) = z + 2\mathrm{Re}(z)$ the formula is also defined for all $z \in \mathbb{C}$, so

$$\mathrm{Dom}(g) = \mathbb{C}$$

Example 2.1.7. Consider the function

$$h(z) = \frac{z + 1}{z^2 + 1}$$

This expression is undefined whenever the denominator is zero. So we solve

$$z^2 + 1 = 0$$

Thus,

$$z^2 = -1$$

which gives $z = \pm i$. Hence, the natural domain is

$$\mathrm{Dom}(h) = \mathbb{C} \setminus \{i, -i\}$$

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A complex number can always be written in the form

$$z = x + iy$$

where $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$. Likewise, if

$$w = f(z)$$

then w can also be written as

$$w = u + iv$$

where $u = \mathrm{Re}(w)$ and $v = \mathrm{Im}(w)$. This leads to one of the most important representations in complex analysis.

Theorem 2.1.8. *Every complex function $f(z)$ can be written in the form*

$$f(z) = u(x, y) + iv(x, y)$$

where $z = x + iy$ and u, v are real-valued functions of the real variables x and y.

Definition 2.1.9. The functions $u = \mathrm{Re}(f(z))$ and $v = \mathrm{Im}(f(z))$ are called the *real part* and *imaginary part of f*, respectively.

Proof. Let $z = x + iy$ be any complex number in the domain of f, and let $w = f(z)$. Since w is a complex number, it has a unique decomposition into real and imaginary parts

$$w = u + iv$$

for some real numbers u and v.

Since w depends on z, and z depends on the real variables x and y, the quantities u and v may be viewed as real-valued functions of x and y. Thus, there exist real functions $u(x, y)$ and $v(x, y)$ such that

$$f(z) = f(x + iy) = u(x, y) + iv(x, y)$$

This proves the result. □

If

$$f(z) = u(x, y) + iv(x, y)$$

we often write

$$u(x, y) = \mathrm{Re}(f(z)), \quad v(x, y) = \mathrm{Im}(f(z))$$

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Remark 2.1.10. This theorem shows that the study of complex functions is closely related to the study of functions of two real variables.

In fact, much of complex analysis begins by expressing a complex function in terms of its real and imaginary parts and then studying how those two functions interact.

Remark 2.1.11. Since a complex-valued function

$$f(z) = u(x, y) + iv(x, y)$$

has both a two-dimensional input and a two-dimensional output, its full graph lies in $\mathbb{R}^4$ and cannot be drawn directly. A practical visualization method is therefore to plot the associated real-valued functions

$$u(x, y), \quad v(x, y), \quad |f(z)|, \quad \arg(f(z))$$

over the (x, y)-plane. This provides a useful geometric understanding of the behaviour of the function.

Example 2.1.12. Find the real and imaginary parts of $f(z) = z^2$.

Solution. Let $z = x + iy$. Then

$$f(z) = (x + iy)^2 = x^2 + 2ixy + i^2y^2 = x^2 - y^2 + 2ixy$$

Hence, $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

Example 2.1.13. Find the real and imaginary parts of

$$f(z) = z^2 - (2 + i)z$$

Solution. Let $z = x + iy$. Then

$$\begin{aligned} f(z) &= (x + iy)^2 - (2 + i)(x + iy) \\ &= x^2 - y^2 + 2ixy - ((2x - y) + i(x + 2y)) \\ &= (x^2 - y^2 - 2x + y) + i(2xy - x - 2y) \end{aligned}$$

Hence, the real and imaginary parts are

$$u(x, y) = x^2 - y^2 - 2x + y, \quad v(x, y) = 2xy - x - y$$

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Example 2.1.14. Find the real and imaginary parts of

$$g(z) = z + 2\mathrm{Re}(z)$$

Solution. Let $z = x + iy$. Since $\mathrm{Re}(z) = x$, then

$$g(z) = x + iy + 2x = 3x + iy$$

Hence,

$$u(x, y) = 3x, \quad v(x, y) = y$$

Theorem 2.1.8 shows that every complex function determines two real-valued functions u and v. Conversely, any pair of real-valued functions can be used to define a complex function.

Theorem 2.1.15. *Let $u(x, y)$ and $v(x, y)$ be real-valued functions defined on a set $D \subseteq \mathbb{R}^2$. Then the rule*

$$f(x + iy) = u(x, y) + iv(x, y)$$

defines a complex function on the corresponding subset of $\mathbb{C}$.

Proof. For each point $(x, y) \in D$, the values $u(x, y)$ and $v(x, y)$ are real numbers. Therefore,

$$u(x, y) + iv(x, y)$$

is a well-defined complex number.

Hence, the rule

$$z = x + iy \mapsto u(x, y) + iv(x, y)$$

assigns to each complex number in the domain exactly one complex value. Therefore, it defines a complex function. □

Example 2.1.16. Define a function by

$$u(x, y) = xy^2, \quad v(x, y) = x^2 - 4y^3$$

Then

$$f(z) = u(x, y) + iv(x, y) = xy^2 + i(x^2 - 4y^2)$$

defines a complex function. Find $f(3 + 2i)$.

Solution. Here, $x = 3$ and $y = 2$, so $u(3, 2) = 12$ and $v(3, 2) = -23$. Therefore,

$$f(3 + 2i) = 12 - 23i$$

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We briefly mentioned about the complex exponential in Section 1.3. Recall for $z = x + iy$, the complex exponential function is defined by

$$e^z = e^x(\cos(y) + i\sin(y))$$

Remark 2.1.17. This extends the usual real exponential function, because if $z = x$ is real, then

$$e^z = e^x(\cos(0) + i\sin(0)) = e^x$$

Thus, the complex exponential agrees with the ordinary exponential on the real axis.

Theorem 2.1.18. *If $z = x + iy$, then the real and imaginary parts of e^z are*

$$\mathrm{Re}(e^z) = e^x\cos(y), \quad \mathrm{Im}(e^z) = e^x\sin(y)$$

Proof. By definition,

$$e^z = e^x(\cos(y) + i\sin(y))$$

Distributing e^x gives

$$e^x\cos(y) + ie^x\sin(y)$$

Therefore, the real and imaginary parts are

$$u(x, y) = e^x\cos(y), \quad v(x, y) = e^x\sin(y)$$

□

Example 2.1.19. Evaluate e^z at the following points

(1) $z = 0$
(2) $z = i$
(3) $z = 2 + \pi i$

Solution.

(1) At $z = 0$, we have $x = 0$ and $y = 0$, so

$$e^0 = e^0(\cos(0) + i\sin(0)) = 1(1 + 0i) = 1$$

(2) At $z = i$, we have $x = 0$ and $y = 1$, so

$$e^i = e^0(\cos(1) + i\sin(1)) = \cos(1) + i\sin(1) \approx 0.5403 + 0.8415i$$

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(3) At $z = 2 + \pi i$, we have $x = 2$ and $y = \pi$, so

$$e^{2+\pi i} = e^2(\cos(\pi) + i\sin(\pi)) = e^2(-1 + 0i) = -e^2$$

The complex exponential satisfies many of the same algebraic rules as the real exponential.

Theorem 2.1.20. *For all $z, w \in \mathbb{C}$ and all integers n,*
(1) $e^0 = 1$
(2) $e^z e^w = e^{z+w}$
(3) $\frac{e^z}{e^w} = e^{z-w}$
(4) $(e^z)^n = e^{nz}$

Proof. The second statement was proved in Proposition 1.3.4. The others will follow similarly. □

Remark 2.1.21. Unlike the real exponential function, the complex exponential function is periodic. Indeed,

$$e^{z+2\pi i} = e^z$$

This means that the complex exponential has period $2\pi i$.

Theorem 2.1.22. *For all $z \in \mathbb{C}$,*

$$e^{z+2\pi i} = e^z$$

Proof. Let $z = x + iy$. Then

$$z + 2\pi i = x + i(y + 2\pi)$$

So by the definition of the complex exponential, we have

$$e^{z+2\pi i} = e^x(\cos(y + 2\pi) + i\sin(y + 2\pi))$$

Since $\cos(y + 2\pi) = \cos(y)$ and $\sin(y + 2\pi) = \sin(y)$, it follows that

$$e^{z+2\pi i} = e^x(\cos(y) + i\sin(y)) = e^z$$

Therefore, e^z is $2\pi i$-periodic. □

Recall from Section 1.3 that any nonzero complex number can be written in polar form as

$$z = r(\cos(\theta) + i\sin(\theta))$$

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where $r = |z|$ and $\theta = \arg(z)$.

Using the complex exponential, we also saw that the complex number can be written as

$$z = re^{i\theta}$$

Sometimes it is more natural to express a complex function in terms of polar coordinates rather than Cartesian coordinates. If

$$z = r(\cos(\theta) + i\sin(\theta)) = re^{i\theta}$$

then a function $f(z)$ may be rewritten in terms of r and θ.

Definition 2.1.23. If $z = re^{i\theta}$ and

$$f(z) = u(r, \theta) + iv(r, \theta)$$

then $u(r, \theta)$ and $v(r, \theta)$ are called the *real* and *imaginary parts of* f in polar coordinates.

Remark 2.1.24. These are generally different functions from the Cartesian-coordinate functions $u(x, y)$ and $v(x, y)$, even though they describe the same complex funtion. That is,

$$u(r, \theta) \neq u(x, y), \quad v(r, \theta) \neq v(x, y)$$

as formulas, because they are expressed in different variables.

Example 2.1.25. Express $f(z) = z^2$ in polar coordinates and determine its real and imaginary parts.

Solution. Write

$$z = re^{i\theta} = r(\cos(\theta) + i\sin(\theta))$$

Then

$$f(z) = z^2 = (re^{i\theta})^2 = r^2e^{2i\theta}$$

Using Euler's formula,

$$r^2e^{2i\theta} = r^2(\cos(2\theta) + i\sin(2\theta))$$

Therefore,

$$f(z) = r^2\cos(2\theta) + ir^2\sin(2\theta)$$

Hence,

$$u(r, \theta) = r^2\cos(2\theta), \quad v(r, \theta) = r^2\sin(2\theta)$$

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Example 2.1.26. Let $f(z) = r^3\cos(\theta) + i(2r\sin(\theta))$ where $z = re^{i\theta}$. Evaluate $f(2i)$.

Solution. First write $2i$ in polar form $2i = 2e^{\frac{i\pi}{2}}$. So $r = 2$ and $\theta = \frac{\pi}{2}$. Substitute into the formula

$$f(2i) = 2^3\cos\left(\frac{\pi}{2}\right) + i\left(2\cdot 2\sin\left(\frac{\pi}{2}\right)\right) = 4i$$

2.2 Limits and Continuity

After introducing complex functions in Section 2.1, the next natural question is how such functions behave when the input variable approaches a point in the complex plane. In real calculus, the notions of limit and continuity are fundamental: they underlie the definitions of derivative and integral, and they describe how functions behave locally. The same is true in complex analysis.

At first sight, the definition of a limit for a complex function looks very similar to the real-variable definition. However, there is an essential difference. A real variable approaches a point along the real line, so there are only two basic directions of approach: from the left and from the right. By contrast, a complex variable approaches a point in the plane, and therefore it may do so along infinitely many different paths. For this reason, the existence of a complex limit is a much stronger condition: the function must approach the same value no matter how the point is approached in the complex plane.

This section develops the basic theory of limits and continuity for complex functions. We begin with the precise epsilon-delta definition of a complex limit and show how to determine when a limit does or does not exist. We then relate complex limits to limits of the real and imaginary parts, which provides an effective computational tool. Next, we define continuity for complex functions and prove its basic properties. In particular, we show that polynomial functions are continuous everywhere and rational functions are continuous on their domains. We also discuss an important boundedness property of continuous functions on closed and bounded regions.

Finally, this section introduces the idea of a branch of a multi-valued function. This concept becomes necessary when dealing with functions such as the square root or the argument, whose values cannot be made single-valued and continuous on all of $\mathbb{C}$. The language of branches and branch cuts will play an important role later in complex analysis.

Before defining complex limits, we recall the definition of a limit for a real-valued function of a real variable.

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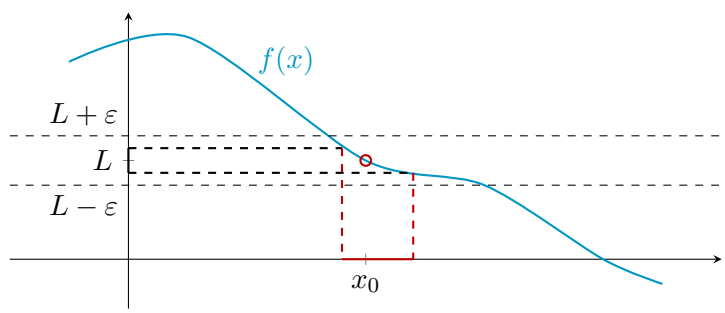
Definition 2.2.1. Let f be a real-valued function defined on a deleted neighbourhood of $x_0 \in \mathbb{R}$. We say that

$$\lim_{x \to x_0} f(x) = L$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $0 < |x - x_0| < \delta$, then $|f(x) - L| < \varepsilon$.

Remark 2.2.2. The condition $0 < |x - x_0| < \delta$ means that x is close to x_0, but not equal to x_0. The inequality $|f(x) - L| < \varepsilon$ means that the function values lie within ε of the limit L.



In the complex plane, distance is measured using the modulus. Thus, the complex version of the ε-δ definition is obtained by replacing absolute values on $\mathbb{R}$ with moduli on $\mathbb{C}$.

Definition 2.2.3. Suppose a complex function f is defined in a deleted neighbourhood of $z_0 \in \mathbb{C}$, and let $L \in \mathbb{C}$. We say that

$$\lim_{z \to z_0} fz) = L$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $0 < |z - z_0| < \delta$, then $|f(z) - L| < \varepsilon$.

Remark 2.2.4. This definition is formally identical to the real one, but conceptually it is stronger. In the real case, a point approaches x_0 along a line. In the complex case, a point z_0 in the plane, and hence, it can approach along infinitely many different curves or paths.

Therefore, for the limit

$$\lim_{z \to z_0} f(z)$$

to exist, the value $f(z)$ must approach the same complex number L along every possible path approaching z_0.

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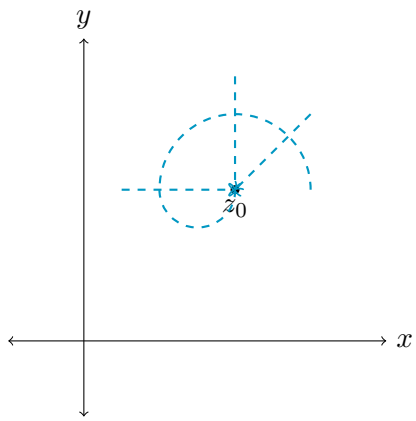


Definition 2.2.5. Let $z_0 \in \mathbb{C}$ and let $\delta > 0$. The set

$$\{z \in \mathbb{C} : 0 < |z - z_0| < \delta\}$$

is called the *deleted δ-neighbourhood* of z_0.

Definition 2.2.6. Let $L \in \mathbb{C}$ and let $\varepsilon > 0$. The set

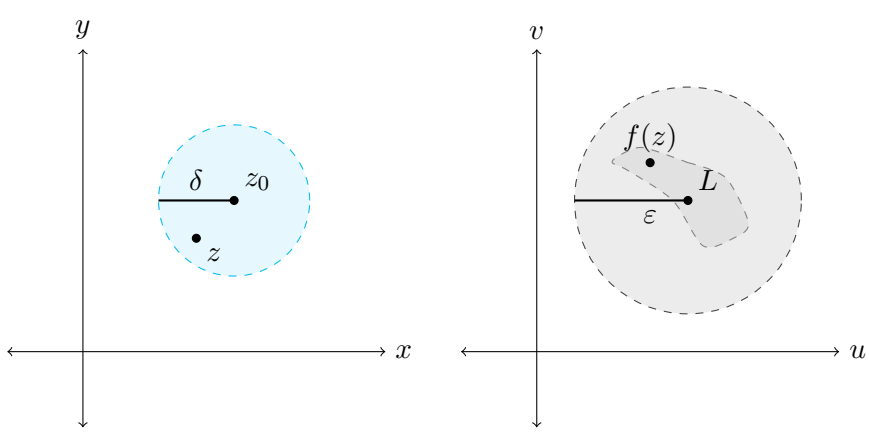
$$\{w \in \mathbb{C} : |w - L| < \varepsilon\}$$

is called the *deleted ε-neighbourhood* of L.

The condition $0 < |z - z_0| < \delta$ means that z lies inside a circle of radius δ centered at z_0, but with the center removed. The condition $|f(z) - L| < \varepsilon$ means that $f(z)$ lies inside the circle of radius ε centered at L. Thus,

$$\lim_{z \to z_0} f(z) = L$$

means that points sufficiently close to z_0 are mapped by f into points arbitrarily close to L.



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Because z may approach z_0 along many paths, one of the standard ways to prove that a limit does not exist is to exhibit two paths that lead to different limiting values.

Proposition 2.2.7. *If a complex function f approaches two different complex numbers $L_1 \neq L_2$ along tw different paths through z_0, then*

$$\lim_{z \to z_0} f(z)$$

does not exist.

Proof. If the limit existed and were equal to some L, then by Definition 2.2.3, the value of $f(z)$ would have to approach the same number L along every path approaching z_0. If two paths yield different values $L_1 \neq L_2$, this is impossible. Hence, the limit does not exist. □

Example 2.2.8. Show that

$$\lim_{z \to 0} \frac{z}{\bar{z}}$$

does not exist.

Solution. We examine two different approaches to 0.

(1) Let $z = x + 0i$ with $x \to 0$. Then $\bar{z} = x$, so

$$\frac{z}{\bar{z}} = \frac{x}{x} = 1$$

with $x \neq 0$. Hence,

$$\lim_{\substack{z \to 0 \\ z \text{ along real axis}}} \frac{z}{\bar{z}} = 1$$

(2) Let $z = iy$ with $y \to 0$. Then $\bar{z} = -y$ and so

$$\frac{z}{\bar{z}} = \frac{iy}{-iy} = -1$$

with $y \neq 0$. Hence,

$$\lim_{\substack{z \to 0 \\ z \text{ along imaginary axis}}} \frac{z}{\bar{z}} = -1$$

Since these two values are not equal, the limit does not exist.

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Example 2.2.9. Prove that

$$\lim_{z\to 1+i}(2+i)z = 1+3i$$

Solution. Let $\varepsilon > 0$ be arbitrary. We seek a $\delta > 0$ such that if $0 < |z-(1+i)| < \delta$, then $|(2+i)z-(1+3i)| < \varepsilon$.

Observe that

$$(2+i)(1+i) = 2+2i+i+i^2 = 1+3i$$

Therefore

$$|(2+i)z-(1+3i)| = |(2+i)z-(2+i)(1+i)| = |2+i||z-(1+i)|$$

Since

$$|2+i| = \sqrt{2^2+1^2} = \sqrt{5}$$

we obtain

$$|(2+i)z-(1+3i)| = \sqrt{5}|z-(1+i)|$$

Now choose

$$\delta = \frac{\varepsilon}{\sqrt{5}}$$

so that if $0 < |z-(1+i)| < \delta$, then

$$|(2+i)z-(1+3i)| = \sqrt{5}|z-(1+i)| < \sqrt{5}\cdot\frac{\varepsilon}{\sqrt{5}} = \varepsilon$$

Hence,

$$\lim_{z\to 1+i}(2+i)z = 1+3i$$

If $f(z) = u(x,y)+iv(x,y)$ and $z = x+iy$, then complex limits are naturally related to limits of the real-valued functions u and v. We first recall the real multivariable definition.

Definition 2.2.10. Let F be a real-valued function of two variables. We say that

$$\lim_{(x,y)\to(x_0,y_0)} F(x,y) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < \sqrt{(x-x_0)^2+(y-y_0)^2} < \delta$, then

$$|F(x,y)-L| < \varepsilon.$$

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Remark 2.2.11. This definition is analogous to the complex limit definition because

$$|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2}$$

whenever $z = x + iy$ and $z_0 = x_0 + iy_0$.

Theorem 2.2.12. *Suppose $f(z) = u(x, y) + iv(x, y)$ with $z = x + iy$. Let $z_0 = x_0 + iy_0$ and $L = u_0 + iv_0$. Then*

$$\lim_{z \to z_0} f(z) = L \iff \lim_{(x,y)\to(x_0,y_0)} u(x, y) = u_0, \quad \lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0$$

Proof. First assume that $\lim_{z\to z_0} f(z) = L$. Then

$$f(z) - L = (u(x, y) - u_0) + i(v(x, y) - v_0)$$

Hence

$$|u(x, y) - u_0| \leq |f(z) - L|, \quad |v(x, y) - v_0| \leq |f(z) - L|$$

For any $\varepsilon > 0$, we choose $\delta > 0$ such that whenever $0 < |z - z_0| < \delta$, $|f(z) - L| < \varepsilon$. Then whenever $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$,

$$|u(x, y) - u_0| < \varepsilon, \quad |v(x, y) - v_0| < \varepsilon$$

Therefore,

$$\lim_{(x,y)\to(x_0,y_0)} u(x, y) = u_0, \quad \lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0$$

Conversely, let assume that $\lim_{(x,y)\to(x_0,y_0)} u(x, y) = u_0$ and $\lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0$. Let $\varepsilon > 0$ be arbitrary. Since $\lim_{(x,y)\to(x_0,y_0)} u(x, y) = u_0$, there exists $\delta_1 > 0$ such that if $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta_1$,

$$|u(x, y) - u_0| < \frac{\varepsilon}{2}$$

Similarly, since $\lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0$, there exists $\delta_2 > 0$ such that if $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta_2$,

$$|v(x, y) - v_0| < \frac{\varepsilon}{2}$$

Now let $\delta = \min\{\delta_1, \delta_2\}$. Then if $0 < |z - z_0| < \delta$, we have

$$|u(x, y) - u_0| < \frac{\varepsilon}{2}, \quad |v(x, y) - v_0| < \frac{\varepsilon}{2}$$

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Therefore,

$$|f(z) - L| = |(u - u_0) + i(v - v_0)| \leq |u - u_0| + |v - v_0| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Hence,

$$\lim_{z \to z_0} f(z) = L.$$

□

Example 2.2.13. Use Theorem 2.2.12 to compute

$$\lim_{z \to 1+i} (z^2 + i)$$

Solution. Let $z = x + iy$. Then

$$z^2 + i = (x + iy)^2 + i = x^2 - y^2 + i(2xy - 1)$$

So now,

$$u(x, y) = x^2 - y^2, \quad v(x, y) = 2xy + 1$$

Since $z \to 1 + i$ means $(x, y) \to (1, 1)$, we compute

$$u_0 = \lim_{(x,y) \to (1,1)} u(x, y) = \lim_{(x,y) \to (1,1)} (x^2 - y^2) = 0$$
$$v_0 = \lim_{(x,y) \to (1,1)} v(x, y) = \lim_{(x,y) \to (1,1)} (2xy + 1) = 3$$

Therefore,

$$\lim_{z \to 1+i} (z^2 + i) = 3i$$

Theorem 2.2.14. *Suppose* $\lim_{z \to z_0} f(z) = L$ *and* $\lim_{z \to z_0} g(z) = M$. *Then*

(1) $\lim_{z \to z_0} c = c$
(2) $\lim_{z \to z_0} z = z_0$
(3) For any complex constant c, $\lim_{z \to z_0} cf(z) = cL$
(4) $\lim_{z \to z_0} (f(z) + g(z)) = L + M$
(5) $\lim_{z \to z_0} f(z)g(z) = LM$
(6) If $M \neq 0$, *then* $\lim_{z \to z_0} \frac{f(z)}{g(z)} = \frac{L}{M}$

Proof. The first two limits are trivial.

Write $f(z) = u(x, y) + iv(x, y)$ and $g(x, y) = a(x, y) + ib(x, y)$. By Theorem 2.2.12, the real and imaginary parts of f tend to those of L, and the real and imaginary parts of g tend to those of M.

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The results then follow from the corresponding properties of real limits of two variables applied to the real and imaginary parts. For instance, the product

$$f(z)g(z) = (u + iv)(a + ib) = (ua - vb) + i(ub + va)$$

and the real-variable limit law imply that the real and imaginary parts tend to those of LM. The quotient follows similarly, provided that $M \neq 0$. □

Example 2.2.15. Compute

$$\lim_{z \to i} \frac{(3+i)z^4 - z^2 + 2z}{z+1}$$

Solution. Using the algebraic limit laws,

$$\lim_{z \to i} z = i, \quad \lim_{z \to i} z^2 = i^2 = -1, \quad \lim_{z \to i} z^4 = i^4 = 1$$

Therefore,

$$\lim_{z \to i} \frac{(3+i)z^4 - z^2 + 2z}{z+1} = \frac{(3+i) - (-1) + 2i}{i+1} = \frac{4+3i}{1+i} = \frac{7}{2} - \frac{1}{2}i$$

Example 2.2.16. Compute

$$\lim_{z \to 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i}$$

Solution. First observe that direct substitution gives

$$\lim_{z \to 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} = \frac{0}{0}$$

so the expression has the indeterminate form $\frac{0}{0}$.

Factor the numerator

$$z^2 - 2z + 4 = (z - 1 - \sqrt{3}i)(z - 1 + \sqrt{3}i)$$

For $z \neq z + 1 + \sqrt{3}i$, we may cancel

$$\frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} = \frac{(z - 1 - \sqrt{3}i)(z - 1 + \sqrt{3}i)}{z - 1 - \sqrt{3}i} = z - 1 + \sqrt{3}i$$

Therefore,

$$\lim_{z \to 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} = \lim_{z \to 1+\sqrt{3}i} (z - 1 + \sqrt{3}i) = 2\sqrt{3}i$$

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Definition 2.2.17. A complex function f is *continuous at the point* z_0 if

$$\lim_{z \to z_0} f(z) = f(z_0)$$

That is, f is continuous at z_0 if and only if
(1) $\lim_{z \to z_0} f(z)$ exists
(2) $f(z_0)$ is defined, and
(3) $\lim_{z \to z_0} f(z) = f(z_0)$
If a function f is not continuous at z_0, then f is said to be *discontinuous at* z_0.

Example 2.2.18. Show that $f(z) = z^2 - iz + 2$ is continuous at $z_0 = 1 - i$.

Solution. Using the limit laws,

$$\begin{aligned}\lim_{z \to 1-i} (z^2 - iz + 2) &= (1-i)^2 - i(1-i) + 2 \\ &= -2i - i - 1 + 2 \\ &= 1 - 3i\end{aligned}$$

On the other hand,

$$f(1-i) = (1-i)^2 - i(1-i) + 2 = 1 - 3i$$

Therefore,

$$\lim_{z \to 1-i} f(z) = f(1-i)$$

so f is continuous at $z_0 = 1 - i$.

Theorem 2.2.19. *Suppose $f(z) = u(x,y) + iv(x,y)$, $z = x + iy$, and $z_0 = x_0 + iy_0$. Then f is continuous at z_0 if and only if both u and v are continuous at (x_0, y_0).*

Proof. Assume that f is continuous at z_0. Then

$$\lim_{z \to z_0} f(z) = f(z_0) = u(x_0, y_0) + iv(x_0, y_0)$$

By Theorem 2.2.12,

$$\lim_{(x,y) \to (x_0,y_0)} u(x,y) = u(x_0, y_0), \qquad \lim_{(x,y) \to (x_0,y_0)} v(x,y) = v(x_0, y_0)$$

Thus, u and v are continuous at (x_0, y_0).

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Conversely, if u and v are continuous at (x_0, y_0), then

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u(x_0,y_0), \qquad \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v(x_0,y_0)$$

By Theorem 2.2.12,

$$\lim_{z\to z_0} f(z) = u(x_0,y_0) + iv(x_0,y_0) = f(z_0)$$

Hence, f is continuous at z_0. □

Example 2.2.20. Show that $f(z) = \bar{z}$ is continuous on $\mathbb{C}$.

Solution. If $z = x + iy$, then $\bar{z} = x - iy$. Thus,

$$u(x,y) = x, \quad v(x,y) = -y$$

Both u and v are polynomial functions in x and y, hence continuous everywhere on $\mathbb{R}^2$. By Theorem 2.2.19, $f(z)$ is continuous on $\mathbb{C}$.

Theorem 2.2.21. *If f and g are continuous at z_0, then the following functions are also continuous at z_0*

(1) cf, where c is a complex constant
(2) $f + g$
(3) fg
(4) $\frac{f}{g}$, provided $g(z_0) \neq 0$.

Proof. We prove the product case. Since f and g are continuous at z_0,

$$\lim_{z\to z_0} f(z) = f(z_0), \quad \lim_{z\to z_0} g(z) = g(z_0)$$

By Theorem 2.2.19,

$$\lim_{z\to z_0} f(z)g(z) = f(z_0)g(z_0)$$

This is the same statement that fg is continuous at z_0. The other parts are proved similarly, using the algebraic limit laws. □

Theorem 2.2.22. *Polynomial functions are continuous on the entire complex plane $\mathbb{C}$.*

Proof. A polynomial has the form

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

The identity $z \mapsto z$ is continuous, and every constant function is continuous. By repeated use of Theorem 2.2.21, powers of z, scalar multiples, and finite sums of continuous functions are continuous. Hence, p is continuous on all of $\mathbb{C}$. □

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Corollary 2.2.23. *Rational functions are continuous on their domains. That is, if $r(z) = \frac{p(z)}{q(z)}$ with p and q polynomials, then r is continuous at every point z_0 for which $q(z_0) \neq 0$.*

Proof. By Theorem 2.2.22, both p and q are continuous on $\mathbb{C}$. If $q(z_0) \neq 0$, then Theorem 2.2.19 shows that $\frac{p}{q}$ is continuous on z_0. □

An important property of continuous functions in real analysis is that continuous functions on closed and bounded sets are bounded. The same idea applies in the complex setting.

Definition 2.2.24. A complex function f is *bounded* on a set $R \subseteq \mathbb{C}$ if there exists a real constant $M > 0$ such that

$$|f(z)| \leq M$$

for all $z \in R$.

Theorem 2.2.25. *If a complex function f is continuous on a closed and bounded region R, then f is bounded on R.*

Proof. Write $f(z) = u(x, y) + iv(x, y)$. Since f is continuous on R, Theorem 2.2.19 implies that u and v are continuous on the corresponding subset of $\mathbb{R}^2$. Since R is continuous and bounded, the real-variable theorem on boundedness of continuous functions show that u and v are bounded on R. Thus, there exist constants $M_1, M_2 > 0$ such that

$$|u(x, y)| \leq M_1, \quad |v(x, y)| \leq M_2$$

for all $z = x + iy \in R$. Hence,

$$|f(z)| + |u + iv| \leq |u| + |v| \leq M_1 + M_2$$

So f is bounded on R. □

Certain complex expressions naturally give rise to more than one value. For example, the argument of a nonzero complex number is not unique.

To study such objects analytically, we often choose one continuous determination of the function on a suitable domain.

Definition 2.2.26. Let F be a multi-valued complex function. A *branch* of F is a single-valued function f defined on some domain D such that

(1) for every $z \in D$, the value $f(z)$ is one of the values of $F(z)$, and
(2) f is continuous on D.

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A *branch cut* is a curve or set removed from the domain of a multi-valued function in order to obtain a branch that is single-valued and continuous on the remaining domain.

Example 2.2.27. Consider the principal square root $f(z) = z^{\frac{1}{2}}$. Show that it is discontinuous at $z_0 = -1$.

Solution. To test continuity at -1, we examine two different paths approaching -1 on the unit circle. Write $z = e^{i\theta}$. Then for the principal square root,

$$z^{\frac{1}{2}} = e^{\frac{i\theta}{2}}$$

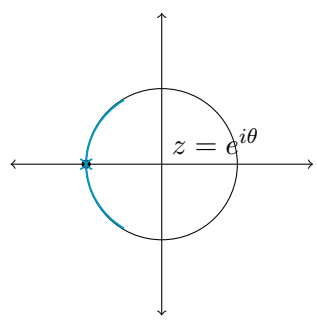
- If $\theta \to \pi^-$, then

$$\lim_{z\to -1} z^{\frac{1}{2}} = \lim_{\theta\to\pi^-} e^{\frac{i\theta}{2}} = e^{\frac{i\pi}{2}} = i$$

- If $\theta \to \pi^+$, then

$$\lim_{z\to -1} z^{\frac{1}{2}} = \lim_{\theta\to\pi^+} e^{-\frac{i\theta}{2}} = e^{-\frac{i\pi}{2}} = -i$$

Since these two limits are different, the limit of $z^{\frac{1}{2}}$ as $z \to -1$ does not exist. Hence, $f(z) = z^{\frac{1}{2}}$ is discontinuous at $z_0 = -1$.



Definition 2.2.28. A standard branch of the square root obtained by restricting to the argument $-\pi < \arg(z) < \pi$. On this domain, we define

$$f_1(z) = \sqrt{r}e^{\frac{i\theta}{2}}, \quad z = re^{i\theta}, \quad -\pi < \theta < \pi$$

This is called the *principal branch* of the square root.

Example 2.2.29. Show that the principal branch

$$f_1(z) = \sqrt{r}e^{\frac{i\theta}{2}}, \quad z = re^{i\theta}, \quad -\pi < \theta < \pi$$

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is a branch of the multi-valued function $z^{\frac{1}{2}}$.

Solution. For each nonzero complex number $z = re^{i\theta}$ with $-\pi < \theta < \pi$, the value $f_1(z) = \sqrt{r}e^{\frac{i\theta}{2}}$ is one of the square roots of z, since

$$\left(\sqrt{r}e^{\frac{i\theta}{2}}\right)^2 = re^{i\theta} = z$$

Thus, $f_1(z)$ is indeed one of the possible values of $z^{\frac{1}{2}}$. Moreover, on the domain

$$D = \{z \in \mathbb{C} : z \neq 0, -\pi < \arg(z) < \pi\}$$

the functions $r = |z|$ and $\theta = \arg(z)$ can be chosen continuously. Therefore, f_1 is continuous on D¿ Hence, f_1 is a branch of the multi-valued square root function.

Definition 2.2.30. A point z_0 is called a *branch point* of a multi-valued function if, when one travels around a sufficiently small closed curve encircling z_0, the values of the function do not return to their original values.

Remark 2.2.31. For the multi-valued square root function $z^{\frac{1}{2}}$, the point $z = 0$ is a branch point. Indeed, if $z = re^{i\theta}$, then

$$z^{\frac{1}{2}} = \sqrt{r}e^{\frac{i\theta}{2}}$$

As θ increases by 2π, the value of $z^{\frac{1}{2}}$ changes sign

$$\sqrt{r}e^{\frac{i(\theta+2\pi)}{2}} = \sqrt{r}e^{\frac{i\theta}{2}}e^{i\pi} = -\sqrt{r}e^{\frac{i\theta}{2}}$$

Thus, a full turn around the origin does not return the square root to its initial value.

Remark 2.2.32. In real analysis, continuity is often visualized by saying that the graph of a function has no breaks or holes. Since complex functions generally cannot be graphed in the same simple way, continuity in complex analysis is studied primarily through algebraic and analytic definitions rather than geometric graphs.

Remark 2.2.33. The existence of a complex limit is harder to verify than the existence of a real limit because one must consider every possible path of approach.

Remark 2.2.34. On the other hand, once the basic limit laws are established, many limits of polynomial and rational functions can be computed directly, just as in elementary real calculus.

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2.3 Differentiability and Holomorphicity

In the previous section, we studied limits and continuity of complex functions. These ideas now allow us to introduce one of the central concepts of complex analysis: the derivative of a complex function. At first sight, the definition of the derivative in the complex setting looks exactly like the one from real calculus:

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

However, although the formula is familiar, its meaning is much more restrictive. In real calculus, the increment approaches zero along the real line, so there are only two directions of approach. In complex analysis, the increment Δz approaches 0 in the plane, and hence from infinitely many possible directions. For the derivative to exist, the difference quotient must approach the same limit no matter how Δz tends to 0.

This stronger requirement is what makes complex differentiability so remarkable. Many functions that are perfectly well behaved from the point of view of real-variable calculus fail to be differentiable as complex functions. At the same time, the functions that do satisfy this condition form a highly structured and powerful class. These are the holomorphic functions, which lie at the heart of complex analysis.

In this section, we first define the derivative of a complex function and illustrate how it is computed. We then establish the familiar algebraic rules of differentiation, including the sum, product, quotient, chain, and power rules. After this, we examine examples showing that some seemingly simple functions are not complex differentiable. This helps clarify how demanding the complex derivative really is.

We then introduce the notion of a function being holomorphic at a point and holomorphic on a domain, and explain how these ideas are stronger than differentiability at a single point. We prove that polynomial functions are holomorphic everywhere and that rational functions are holomorphic on any domain that avoids their singularities. Finally, we establish one of the most important basic facts in the subject: differentiability implies continuity.

Let $z = x + iy$ and $z_0 = x_0 + iy_0$. Then the change in the input variable is

$$\Delta z = z - z_0 = \Delta x + i\Delta y$$

If $w = f(z)$, then the corresponding change in the function value is

$$\Delta w = f(z_0 + \Delta z) - f(z_0)$$

The derivative is defined as the limit of the difference quotient $\frac{\Delta w}{\Delta z}$ as $\Delta z \to 0$.

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Definition 2.3.1. Suppose the complex function f is defined in a neighbourhood of z_0. The *derivative* of f at z_0, denoted by $f'(z_0)$ is defined by

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

provided this limit exists.

If the limit exists, then we say that f is differentiable at z_0.

Remark 2.3.2. Other common notations for the derivative of $w = f(z)$ are w' and $\frac{dw}{dz}$. If the derivative is taken at a specific point z_0, one may write

$$\left.\frac{dw}{dz}\right|_{z=z_0}$$

Example 2.3.3. Use Definition 2.3.1 to find the derivative of $f(z) = z^2 - 5z$.

Solution. To compute the derivative at a general point, we write

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Indeed,

$$\begin{aligned} f'(z) &= \lim_{\Delta z \to 0} \frac{(z + \Delta z)^2 - 5(z + \Delta z) - (z^2 - 5z)}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{z^2 + 2z\Delta z + \Delta z^2 - 5z - 5\Delta z - z^2 + 5z}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{2z\Delta z + \Delta z^2 - 5\Delta z}{\Delta z} \\ &= \lim_{\Delta z \to 0} (2z + \Delta z - 5) \end{aligned}$$

Taking the limit as $\Delta z \to 0$, we obtain

$$f'(z) = 2z - 5$$

The familiar differentiation rules from real calculus also hold for complex functions.

Theorem 2.3.4. *Suppose f and g are differentiable at a point z, and let c be a complex constant. Then*

(1) Constant rules

$$\frac{d}{dz}(c) = 0, \quad \frac{d}{dz}(cf(z)) = c'f(z)$$

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(2) *Sum rule*

$$\frac{d}{dz}(f(z)+g(z)) = f'(z)+g'(z)$$

(3) *Product rule*

$$\frac{d}{dz}(f(z)g(z)) = f'(z)g(z)+f(z)g'(z)$$

(4) *Quotient rule*

$$\frac{d}{dz}\left(\frac{f(z)}{g(z)}\right) = \frac{f'(z)g(z)-f(z)g'(z)}{g^2(z)}, \quad g(z)\neq 0$$

(5) *Chain rule*

$$\frac{d}{dz}(f(g(z))) = f'(g(z))g'(z)$$

Remark 2.3.5. As in real calculus, these rules make it possible to compute derivatives of many functions without returning each time to the limit definition.

Theorem 2.3.6 (**Power Rule for Integer Powers**). *If n is an integer, then*

$$\frac{d}{dz}(z^n) = nz^{n-1}$$

Corollary 2.3.7 (**Power Rule for Functions**). *If g is differentiable and n is an integer, then*

$$\frac{d}{dz}(g^n(z)) = ng^{n-1}(z)g'(z)$$

Example 2.3.8. Differentiate each of the following functions.

(1) $f(z) = 3z^4 - 5z^3 + 2z$
(2) $f(z) = \frac{z^2}{4z+1}$
(3) $f(z) = (iz^2+3z)^5$

Solution.

(1) Using the power rule and the sum rule

$$f'(z) = 3\cdot 4z^3 - 5\cdot 3z^2 + 2 = 12z^3 - 15z^2 + 2$$

(2) Apply the quotient rule

$$f'(z) = \frac{(4z+1)(2z)-z^2(4)}{(4z+1)^2} = \frac{8z^2+2z-4z}{(4z^2+1)} = \frac{4z^2+2z}{(4z+1)^2}$$

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(3) Use the power rule for functions with $g(z) = iz^2 + 3z$. Then $g'(z) = 2iz + 3$, so
$$f'(z) = 5(iz^2 + 3z)^4(2iz + 3)$$

For a function to be differentiable at a point z_0, the limit
$$\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$
must exist and must be the same no matter how Δz approaches 0.

This is far more restrictive than real calculus, because in the complex plane, Δz may approach 0 from infinitely many directions.

Example 2.3.9. Show that the function
$$f(z) = x + 4iy$$
is not differentiable at any point z.

Solution. Let $z = x + iy$ and $\Delta z = \Delta x + i\Delta y$. Then
$$f(z + \Delta z) = (x + \Delta x) + 4i(y + \Delta y) = x + \Delta x + 4iy + 4i\Delta y$$
Therefore,
$$f(z + \Delta z) - f(z) = \Delta x + 4i\Delta y$$
So the difference quotient is
$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{\Delta x + 4i\Delta y}{\Delta x + i\Delta y}$$
We now compare two different ways of letting $\Delta z \to 0$.

- Approach parallel to the x-axis: Take $\Delta y = 0$ so that $\Delta z = \Delta x$. Then
$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{\Delta x}{\Delta x} = 1$$
Hence, along this path,
$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = 1$$
- Approach parallel to the y-axis: Take $\Delta x = 0$ so that $\Delta z = i\Delta y$. Then
$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{4i\Delta y}{i\Delta y} = 4$$
Hence, along this path
$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = 4$$

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Since these two values are different, the limit does not exist. Therefore, f is nowhere differentiable.

Remark 2.3.10. This example shows that a function may have simple real and imaginary parts and still fail to be differentiable as a complex function.

In particular, the basic power rule does not apply to functions involving $\bar{z}$ in a naive way, because such functions often fail to be differentiable.

Differentiability at a single point is important, but in complex analysis the functions of real interest are those that are differentiable on an open set.

Definition 2.3.11. A complex function f is said to be *holomorphic at a point* z_0 if it is differentiable at z_0 and at every point in some neighbourhood of z_0.

A function f is *holomorphic in a domain* D if it is holomorphic at every point of D.

Definition 2.3.12. A function that is holomorphic at every point of the complex plane $\mathbb{C}$ is called an *entire function*.

Remark 2.3.13. Holomorphicity at a point is *not* a pointwise property only. It is a *neighbourhood property*. To say that f is holomorphic at z_0 means not merely that $f'(z_0)$ exists, but that f is differentiable in some open neighbourhood of z_0.

Example 2.3.14. Consider $f(z) = |z|^2$. This function is differentiable at $z = 0$, but it is not differentiable everywhere. Hence, it is not holomorphic at 0.

To see this, although the derivative exists at the single point $z = 0$, there is no neighbourhood fo 0 throughout which f is differentiable. Therefore, f is not holomorphic at 0.

Remark 2.3.15. In contrast, the polynomial $f(z) = z^2$ is differentiable at every point in $\mathbb{C}$, and hence is entire.

The standard algebraic rules allow us to identify many familiar holomorphic functions.

Theorem 2.3.16. *(1) A polynomial function $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, for $n \geq 0$ is an entire function.*

(2) A rational function $f(z) = \frac{p(z)}{q(z)}$, where p and q are polynomial functions, is holomorphic in any domain D that contains no point z_0 such that $q(z_0) = 0$.

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Proof. (1) A polynomial is built from constants and powers of z using addition and multiplication. Since

$$\frac{d}{dz}(z^n) = nz^{n-1}$$

for every nonnegative integer n, each term $a_k z^k$ is differentiable everywhere. By repeated use of the sum rule, the whole polynomial is differentiable everywhere on $\mathbb{C}$. Hence, it is entire.

(2) Since polynomials are entire, both p and q are differentiable everywhere. Wherever $q(z) \neq 0$, the quotient rule shows that f is differentiable. Thus, f is holomorphic on any domain that excluded the zeros of the denominator.

□

Definition 2.3.17. A point z_0 at which a complex function f fails to be holomorphic is called a *singular point* of f.

Example 2.3.18. Consider

$$f(z) = \frac{4z}{z^2 - 2z + 2}$$

The denominator satisfies

$$z^2 - 2z + 2 = 0$$

whenever $z = 1 + i$ or $z = 1 - i$. Thus, f is not holomorphic at these two points. Hence, $1 + i$ and $1 - i$ are singular points of f.

Theorem 2.3.19. *If f and g are holomorphic in a domain D, then the following are also holomorphic in D*

(1) $f + g$
(2) fg
(3) $\frac{f}{g}$*, provided* $g(z) \neq 0$ *for all* $z \in D$.

Proof. Since f and g are holomorphic in D, they are differentiable at every point of D. Applying the corresponding differentiation rules pointwise shows that the sum, product, and quotient are also differentiable at each point of D wherever defined. Therefore, they are holomorphic in D. □

Since $\Delta z = z - z_0$, we may rewrite the derivative formula in a more familiar form.

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Theorem 2.3.20. *If f is differentiable at z_0, then*

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

Proof. Set $z = z_0 + \Delta z$. Then as $\Delta z \to 0$, we have $z \to z_0$, and

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \frac{f(z) - f(z_0)}{z - z_0}$$

Thus, the two limits are identical. □

One of the most important foundational facts is that complex differentiability implies continuity.

Theorem 2.3.21. *If f is differentiable at a point z_0, then f is continuous at z_0.*

Proof. Assume f is differentiable at z_0. Then the limit

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists. Now write

$$f(z) - f(z_0) = \frac{f(z) - f(z_0)}{z - z_0}(z - z_0)$$

with $z \neq z_0$. Taking limits as $z \to z_0$, and using the limit laws, we obtain

$$\lim_{z \to z_0} (f(z) - f(z_0)) = \left(\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \right) \left(\lim_{z \to z_0} (z - z_0) \right)$$

That is,

$$\lim_{z \to z_0} (f(z) - f(z_0)) = f'(z_0) \cdot 0 = 0$$

Therefore,

$$\lim_{z \to z_0} f(z) = f(z_0)$$

so f is continuous at z_0. □

Remark 2.3.22. The converse is false: continuity does not imply differentiability. Example 2.3.9 already illustrates this point. The function $f(z) = x + 4iy$ is continuous everywhere, but nowhere differentiable.

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2.4 The Cauchy–Riemann Equations

In Section 2.3, we introduced the complex derivative and saw that differentiability in the complex sense is far more restrictive than differentiability in ordinary real calculus. A complex derivative is defined by the same formal difference quotient as in real analysis, but because the variable z approaches a point through infinitely many directions in the complex plane, the existence of that limit imposes strong conditions on the function.

If a complex function is written in the form $f(z) = u(x, y) + iv(x, y)$ with $z = x + iy$, then the behaviour of f is determined by the two real-valued functions u and v. It is therefore natural to ask: what conditions must u and v satisfy in order for f to be differentiable? The answer is given by one of the most important results in complex analysis: the Cauchy–Riemann equations.

These equations provide a precise relationship between the first-order partial derivatives of the real and imaginary parts of a complex function. They arise as a necessary condition for complex differentiability, and under suitable continuity assumptions on the partial derivatives, they also become a sufficient condition. Thus, the Cauchy–Riemann equations serve as the fundamental bridge between complex differentiability and multivariable calculus.

In this section, we derive the Cauchy–Riemann equations from the definition of the derivative, first as a necessary condition for differentiability. We then show how they can be used to rule out holomorphicity, and afterwards we establish a sufficient criterion for holomorphicity when the relevant partial derivatives are continuous. This gives us a practical and powerful test for deciding when a function is holomorphic.

We also derive a useful formula for the derivative in terms of the partial derivatives of u and v, examine examples of functions that satisfy the Cauchy–Riemann equations only on certain sets, and conclude with a polar-coordinate version of the equations. These results deepen our understanding of holomorphic functions and prepare us for later sections, where the consequences of holomorphicity become even more striking.

Suppose $f(z) = u(x, y) + iv(x, y)$, $z = x + iy$. We want to determine what conditions on u and v are forced by the assumption that f is differentiable.

Theorem 2.4.1 (**Cauchy–Riemann Equations**). *Suppose $f(z) = u(x, y) + iv(x, y)$ is differentiable at the point $z = x + iy$. Then the first-order partial derivatives of u and v exist at (x, y) and satisfy the Cauchy–Riemann equations*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

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Proof. Since f is differentiable at z, its derivative is

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Write $\Delta z = \Delta x + i\Delta y$. Since $f(z) = u(x, y) + iv(x, y)$, we have

$$f(z + \Delta z) = u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y)$$

Therefore,

$$f'(z) = \lim_{\Delta z \to 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}$$

We now evaluate this limit in two special ways.

(1) Let $\Delta z \to 0$ along a horizontal line. Take $\Delta y = 0$. Then $\Delta z = \Delta x$, so

$$\begin{aligned} f'(z) &= \lim_{\Delta x \to 0} \frac{u(x + \Delta x, y) - u(x, y) + i(v(x + \Delta x, y) - v(x, y))}{\Delta x} \\ &= \lim_{\Delta x \to 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v(x + \Delta x, y)}{\Delta x} \\ &= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \end{aligned}$$

(2) Let $\Delta z \to 0$ along a vertical line. Take $\Delta x = 0$. Then $\Delta z = i\Delta y$, so

$$\begin{aligned} f'(z) &= \lim_{\Delta y \to 0} \frac{u(x, y + \Delta y) - u(x, y) + i(v(x, y + \Delta y) - v(x, y))}{i\Delta y} \\ &= -i \lim_{\Delta y \to 0} \frac{u(x, y + \Delta y) - u(x, y)}{\Delta y} + \lim_{\Delta y \to 0} \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} \\ &= \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y} \end{aligned}$$

Since both expressions represent the same derivative $f'(z)$, we equate the real and imaginary parts to get

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

Equivalently,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

This proves the result. □

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Remark 2.4.2. The theorem says that the Cauchy–Riemann equations are a necessary condition for differentiability. That is,

- if f is differentiable, then the equations must hold.
- If the equations fail, then f cannot be differentiable.

However, the equations alone are not yet sufficient without additional hypotheses.

Corollary 2.4.3. *If $f(z) = u(x, y) + iv(x, y)$ is holomorphic in a domain D, then the Cauchy–Riemann equations hold at every point of D.*

Proof. If f is holomorphic in D, then it is differentiable at every point of D. Applying Theorem 2.4.1 at each point yields the result. □

Example 2.4.4. Verify that the polynomial $f(z) = z^2 + z$ satisfies the Cauchy–Riemann equations everywhere.

Solution. Write $z = x + iy$. Then

$$\begin{aligned} f(z) &= z^2 + z \\ &= (x + iy)^2 + (x + iy) \\ &= x^2 - y^2 + x + i(2xy + y) \end{aligned}$$

Therefore,

$$u(x, y) = x^2 - y^2 + x, \quad v(x, y) = 2xy + y$$

Now compute the first-order partial derivatives

$$u_x = 2x + 1, \quad u_y = -2y, \quad v_x = 2y, \quad v_y = 2x + 1$$

Hence,

$$u_x = v_y, \quad u_y = -v_x$$

Therefore, the Cauchy–Riemann equations are satisfied at every point in the plane.

The contrapositive of Theorem 2.4.1 is extremely useful.

Proposition 2.4.5. *If the Cauchy–Riemann equations are not satisfied at a point z_0, then the function $f(z) = u(x, y) + iv(x, y)$ is not differentiable at z_0. If the equations fail at every point of a domain D, then f cannot be holomorphic in D.*

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Example 2.4.6. Show that

$$f(z) = 2x^2 + y + i(y^2 - x)$$

is not holomorphic at any point.

Solution. We identify

$$u(x, y) = 2x^2 + y, \quad v(x, y) = y^2 - x$$

Then

$$u_x = 4x, \quad u_y = 1, \quad v_x = -1, \quad v_y = 2y$$

Now the second Cauchy–Riemann equation gives

$$u_y = -v_x$$

But the first Cauchy–Riemann equation requires

$$u_x = v_y$$

that is,

$$4x = 2y \implies y = 2x$$

Therefore, the Cauchy–Riemann equations do not hold throughout any neighbourhood of any point. In particular, there is no open disk on which the equations hold everywhere. Hence, f is nowhere differentiable.

Remark 2.4.7. This example shows an important subtlety: a function may satisfy the Cauchy–Riemann equations at some points, or even along a curve, without being holomorphic there. Holomorphicity is a neighbourhood property.

The Cauchy–Riemann equations alone do not automatically guarantee differentiability. But if the relevant first-order partial derivatives are continuous, then they do.

Theorem 2.4.8. *Suppose the real-valued functions $u(x, y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in a domain D. If they satisfy the Cauchy–Riemann equations*

$$u_x = v_y, \quad u_y = -v_x$$

at every point of D, then the complex function

$$f(z) = u(x, y) + iv(x, y)$$

is holomorphic in D.

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Remark 2.4.9. This theorem gives a practical test for holomorphicity

(1) Write $f(z) = u(x, y) + iv(x, y)$.
(2) Compute the first-order partial derivatives.
(3) Check that these partial derivatives are continuous.
(4) Verify the Cauchy–Riemann equations.

If all of this holds on a domain D, then f is holomorphic on D.

Example 2.4.10. Consider

$$f(z) = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2}$$

Show that f is holomorphic in any domain that does not contain the point $z = 0$.

Solution. We identify

$$u(x, y) = \frac{x}{x^2 + y^2}, \quad v(x, y) = -\frac{y}{x^2 + y^2}$$

These functions are defined and continuous whenever $x^2 + y^2 = 0$, that is, for all $z \neq 0$.

Now compute the partial derivatives,

$$u_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad u_y = \frac{-2xy}{(x^2 + y^2)^2}, \quad v_x = \frac{2xy}{(x^2 + y^2)^2}, \quad v_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

Hence

$$u_x = v_y, \quad u_y = -v_x$$

for every point with $x^2 + y^2 \neq 0$.

Since u, v, and their first-order partial derivatives are continuous away from $(0, 0)$, Theorem 2.4.8 applies. Therefore, $f(z)$ is holomorphic in any domain that does not contain $z = 0$.

The computation in the proof of Theorem 2.4.1 showed that, whenever f is differentiable,

$$f'(z) = u_x + iv_x, \quad f'(z) = v_y - iu_x$$

Using the Cauchy–Riemann equations, these expressions agree.

Theorem 2.4.11. *Suppose the hypotheses of Theorem 2.4.8 hold. Then*

$$f'(z) = u_x + iv_x = v_y - iu_y$$

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Proof. Since f is holomorphic by Theorem 2.4.8, it is differentiable. From the derivation in Theorem 2.4.1, we know that

$$f'(z) = u_x + iv_x$$

when the limit is taken along a horizontal line, and

$$f'(z) = v_y - iu_x$$

when taken along a vertical line. Since the derivative exists, these two quantities are equal. □

Example 2.4.12. Use the derivative formula to compute the derivative of $f(z) = z^2$.

Solution. Write $z = x + iy$. Then

$$f(z) = x^2 - y^2 + i(2xy)$$

So then $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$. Then

$$u_x = 2x, \quad v_x = 2y$$

Hence,

$$f'(z) = u_x + iv_x = 2x + i(2y) = 2(x + iy) = 2z$$

Thus, $f'(z) = 2z$.

Example 2.4.13. Consider again $f(z) = 2x^2 + y + i(y^2 - x)$. Show that although f is nowhere holomorphic, it is differentiable at the points on the line $y = 2x$.

Solution. From Example 2.4.6,

$$u(x, y) = 2x^2 + y, \quad v(x, y) = y^2 - x$$

with

$$u_x = 4x, \quad u_y = 1, \quad v_x = -1, \quad v_y = 2y$$

These partial derivatives are continuous everywhere. The Cauchy–Riemann equations hold exactly when

$$u_x = v_y, \quad u_y = -v_x$$

The second equation holds, everywhere, while the first gives

$$4x = 2y \implies y = 2x$$

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Therefore, by Theorem 2.4.8, the function is differentiable at points on the line $y = 2x$. Now along this line,

$$f'(z) = u_x + iv_x = 4x - i$$

Since $y = 2x$, this may also be written as

$$f'(z) = 2y - i$$

Thus, f is differentiable at points on the line $y = 2x$, but nowhere holomorphic, since there is no neighbourhood on which it is differentiable at every point.

Theorem 2.4.14. *Suppose $f(z) = u(x, y) + iv(x, y)$ is holomorphic in a domain D.*

(1) If $|f(z)|$ is constant in D, then f is constant in D.
(2) If $f'(z) = 0$ for all $z \in D$, then f is constant in D.

Remark 2.4.15. This is distinctly complex phenomenon. Later, it will become a recurring theme that holomorphic functions are far more rigid than ordinary differentiable real functions.

Sometimes, it is more convenient to express a function in polar form. If $z = re^{i\theta}$, then a complex function may be written as

$$f(z) = u(r, \theta) + iv(r, \theta)$$

In polar coordinates, the Cauchy–Riemann equations take a different form.

Theorem 2.4.16 (Cauchy–Riemann Equations: Polar Coordinates). *Suppose $f(z) = u(r, \theta) + iv(r, \theta)$ is holomorphic. Then*

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$$

These are the polar coordinate versions of the Cartesian equations

$$u_x = v_y, \quad u_y = -v_x$$

Theorem 2.4.17. *Under the same hypotheses,*

$$f'(z) = e^{-i\theta}(u_r + iv_r) = \frac{1}{r}e^{-i\theta}(v_\theta - iu_\theta).$$

Remark 2.4.18. The polar form is especially useful for functions naturally expressed in terms of modulus and argument, such as powers, logarithms, and trigonometric expressions in polar coordinates.

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One of the most important applications of the Cauchy–Riemann equations is to the complex exponential function. Recall that

$$e^z = e^x(\cos(y) + i\sin(y))$$

Thus,

$$u(x, y) = e^x \cos(y), \quad v(x, y) = e^x \sin(y)$$

Then

$$u_x = e^x \cos(y), \quad u_y = -e^x \sin(y), \quad v_x = e^x \sin(y), \quad v_y = e^x \cos(y)$$

So $u_x = v_y$ and $u_y = -v_x$. Hence, the Cauchy–Riemann equations hold everywhere, and therefore, e^z is holomorphic everywhere.

Moreover,

$$f'(z) = u_x + iv_x = e^x \cos(y) + ie^x \sin(y) = e^z$$

So the complex exponential satisfies

$$\frac{d}{dz}(e^z) = e^z$$

just as in real calculus.

2.5 Harmonic Functions

In Section 2.4 we saw that if a complex function

$$f(z) = u(x, y) + iv(x, y)$$

is holomorphic, then its real and imaginary parts are not independent; rather they are linked by the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x$$

These equations already show that holomorphicity imposes strong first-order conditions on the functions u and v. In this section, we push this idea one step further and examine the consequences for their second-order partial derivatives.

A remarkable fact emerges: the real and imaginary parts of an holomorphic function satisfy the partial differential equation

$$u_{xx} + u_{yy} = 0, \quad v_{xx} + v_{yy} = 0$$

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This equation is known as the *Laplace's equation*, and any function satisfying it is called a *harmonic function*. Harmonic functions play a central role not only in complex analysis, but also in physics, engineering, and applied mathematics, especially in the study of heat flow, gravitation, electrostatics, and fluid motion.

Thus, holomorphic functions automatically produce harmonic functions. Conversely, under suitable conditions, a harmonic function can be used to construct a holomorphic function by finding another harmonic function that satisfies the Cauchy–Riemann equations. This second function is called a *harmonic conjugate*. In this way, harmonic functions and holomorphic functions are deeply connected.

In this section, we define harmonic functions, prove that the real and imaginary parts of an holomorphic function are harmonic, and introduce the notion of a harmonic conjugate. We then work through examples of finding harmonic conjugates and forming corresponding holomorphic functions. Finally, we discuss the significance of these ideas and their relation to Laplace's equation in real analysis and mathematical physics.

In Section 2.4, we saw that if $f(z) = u(x, y) + iv(x, y)$ is holomorphic, then the Cauchy–Riemann equations hold

$$u_x = v_y, \quad u_y = -v_x$$

If we differentiate these relations once more, we are led naturally to the second order partial differential equation

$$\phi_{xx} + \phi_{yy} = 0$$

This equation is called *Laplace's equation* in two variables. The operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

is called the *Laplacian*. Thus, Laplace's equation may be written compactly as $\nabla^2\phi = 0$.

Definition 2.5.1. A real-valued function $\phi(x, y)$ of two real variables is said to be *harmonic in a domain D* if

(1) ϕ has continuous first- and second-order partial derivatives in D, and
(2) ϕ satisfies Laplace's equation in D

$$\phi_{xx} + \phi_{yy} = 0$$

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Remark 2.5.2. Thus, to show that a real-valued function $\phi(x, y)$ is harmonic, it is enough to compute its second-order partial derivatives and verify that their sum is zero.

Remark 2.5.3. Harmonic functions arise naturally in many applications, especially in the study of temperatures, gravitational potentials, electric potentials, and steady-satte physical systems.

One of the most important facts in this section is the following theorem.

Theorem 2.5.4. *Suppose the complex function $f(z) = u(x, y) + iv(x, y)$ is holomorphic in D. Then both $u(x, y)$ and $v(x, y)$ are harmonic in D.*

Proof. Assume that $f(z) = u(x, y) + iv(x, y)$ is holomorphic in D. Then from the Cauchy–Riemann equations, we have

$$u_x = v_y, \quad u_y = -v_x$$

throughout D. Differentiate the first equation with respect to x to get

$$u_{xx} = v_{yx}$$

and differentiate the second equation with respect to y to get

$$u_{yy} = -v_{xy}$$

Adding these two equations give

$$u_{xx} + u_{yy} = v_{yx} - v_{xy}$$

Since the mixed partial derivatives are equal under the continuity assumptions, $v_{yx} = v_{xy}$, and so we obtain

$$u_{xx} + u_{yy} = 0$$

Hence, u is harmonic in D.

Similarly, differentiate the first equation with respect to y to get

$$u_{xy} = v_{yy}$$

and differentiate the second equation with respect to x to get

$$u_{yx} = -v_{xx}$$

Subtracting these relations, or equivalently using $u_{xy} = u_{yx}$, we obtain

$$v_{xx} + v_{yy} = 0$$

Hence, v is harmonic in D.

Therefore, both the real and imaginary parts of a holomorphic function are harmonic. □

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Corollary 2.5.5. *If $f(z) = u(x, y) + iv(x, y)$ is entire, then both u and v are harmonic on the entire plane $\mathbb{C}$.*

Example 2.5.6. Show that the real and imaginary parts of $f(z) = z^2$ are harmonic.

Solution. We write $z = x + iy$. Then

$$f(z) = z^2 = (x + iy)^2 = x^2 - y^2 + i(2xy)$$

So then

$$u(x, y) = x^2 - y^2, \quad v(x, y) = 2xy$$

Now compute the second-order partial derivatives of u

$$u_x = 2x, \quad u_{xx} = 2, \quad u_y = -2y, \quad u_{yy} = -2$$

Therefore,

$$u_{xx} + u_{yy} = 2 + (-2) = 0$$

So u is harmonic.

Now compute the second-order partial derivatives of v

$$v_x = 2y, \quad v_{xx} = 0, \quad v_y = 2x, \quad v_{yy} = 0$$

Hence

$$v_{xx} + v_{yy} = 0 + 0 = 0$$

So v is also harmonic.

Therefore, both the real and imaginary parts of z^2 are harmonic.

We now reverse the viewpoint. Instead of starting with a holomorphic function and concluding that its real and imaginary parts are harmonic, suppose we begin with a harmonic function $u(x, y)$. Can we find another harmonic function $v(x, y)$ such that

$$f(z) = u(x, y) + iv(x, y)$$

is holomorphic? This leads to the following definition.

Definition 2.5.7. Suppose $u(x, y)$ is harmonic in a domain D. If there exists a real-valued function $v(x, y)$ such that $f(z) = u(x, y) + iv(x, y)$ is holomorphic in D, then v is called a *harmonic conjugate* of u in D.

Remark 2.5.8. If v is a harmonic conjugate of u, then u is also a harmonic conjugate of $-v$, since the Cauchy–Riemann equations are symmetric in this way.

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Remark 2.5.9. To find a harmonic conjugate v of a given harmonic function u, we use the Cauchy–Riemann equations

$$v_y = u_x, \quad v_x = -u_y$$

We then integrate one of these equations and use the other to determine the "constant of integration."

Example 2.5.10. Verify that $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic, and then find a harmonic conjugate $v(x, y)$.

Solution. Compute the first- and second-order partial derivatives

$$u_x = 3x^2 - 3y^2, \quad u_{xx} = 6x, \quad u_y = -6xy - 5, \quad u_{yy} = -6x$$

Hence

$$u_{xx} + u_{yy} = 6x - 6x = 0$$

Therefore, u is harmonic.

If v is a harmonic conjugate of u, then $v_y = u_x = 3x^2 - 3y^2$ and $v_x = -u_y = 6xy + 5$. Integrate $v_y = 3x^2 - 3y^2$ with respect to y to get

$$v(x, y) = 3x^2y - y^3 + h(x)$$

where $h(x)$ is a function of x only. Now differentiate this expression with respect to x

$$v_x = 6xy + h'(x)$$

But from the Cauchy–Riemann equations we also need $v_x = 6xy + 5$. Therefore, $h'(x) = 5$, so $h(x) = 5x + C$, where C is a real constant.

So the harmonic conjugate is

$$v(x, y) = 3x^2y - y^3 + 5x + C$$

The resulting holomorphic function is

$$f(z) = u(x, y) + iv(x, y) = x^3 - 3xy^2 - 5y + i(3x^2y - y^3 + 5x + C)$$

Remark 2.5.11. If $v(x, y)$ is a harmonic conjugate of $u(x, y)$, then so is $v(x, y) + C$ for any real constant C, since adding a constant does not affect the partial derivatives. Thus, harmonic conjugates are unique only up to an additive real constant.

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Example 2.5.12. Find a harmonic conjugate of $u(x, y) = x^2 - y^2$.

Solution. First compute

$$u_x = 2x, \quad u_y = -2y$$

Then the Cauchy–Riemann equations require

$$v_y = u_x = 2x, \quad v_x = -u_y = 2y$$

Integrate $v_y = 2x$ with respect to y, we have

$$v(x, y) = 2xy + h(x)$$

Differentiate with respect to x to get

$$v_x = 2y + h'(x)$$

Since we need $v_x = 2y$, it follows that $h'(x) = 0$, so $h(x) = C$, where C is constant. Hence, a harmonic conjugate is

$$v(x, y) = 2xy + C$$

The associated holomorphic function is

$$f(z) = x^2 - y^2 + i(2xy) = z^2$$

up to an imaginary constant.

Remark 2.5.13. Theorem 2.5.4 gives a powerful connection between complex analysis and partial differential equations:

- analytic functions produce harmonic functions,
- harmonic functions can often be completed to holomorphic functions by finding harmonic conjugates.

This means that many problems involving Laplace's equation can be studied using holomorphic functions.

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Chapter 3

Elementary Functions

Having established the foundational ideas of limits, continuity, differentiability, holomorphicity, the Cauchy–Riemann equations, and harmonic functions, we are now ready to study some of the most important examples of holomorphic functions in complex analysis: the elementary functions. These functions extend familiar real-variable functions into the complex plane, but in doing so they exhibit several new and distinctly complex phenomena that have no true analogue in real calculus.

In real analysis, functions such as the exponential, logarithm, sine, cosine, and powers are usually introduced as single-valued and globally well-behaved. In the complex setting, however, the situation becomes significantly richer. The complex exponential function retains many of its familiar algebraic and holomorphic properties, but it also becomes periodic, revealing a deep connection between exponential and trigonometric behavior through Euler's formula. By contrast, the complex logarithm is no longer globally single-valued, and its study leads naturally to the ideas of branches, branch cuts, and multi-valued functions, which are central themes in complex analysis.

This chapter develops the theory of these elementary functions systematically. We begin with the complex exponential and logarithmic functions, which form the basis for much of the subject. The exponential function is entire and never vanishes, while the logarithm serves as a local inverse of the exponential wherever a suitable branch can be chosen. From there, we define complex powers, which generalize expressions of the form z^α and reveal why even seemingly simple operations such as taking roots or fractional powers require careful attention in the complex plane. Finally, we study the trigonometric and hyperbolic functions, which can be elegantly defined in terms of exponentials and whose properties become especially transparent

from the complex viewpoint.

Throughout this chapter, we will see that elementary functions in complex analysis are not merely extensions of their real counterparts, but rather gateways to deeper ideas involving periodicity, holomorphic continuation, conformal structure, and the geometry of the complex plane. These functions will play a central role in later topics such as contour integration, residues, and conformal mappings.

3.1 Exponential and Logarithmic Functions

The exponential and logarithmic functions are among the most important elementary functions in both real and complex analysis. In the real-variable setting, these functions are familiar and well behaved: the exponential function is everywhere defined and one-to-one, while the logarithm serves as its inverse on the positive real axis. In this complex setting, however, these functions exhibit new and much richer behaviour.

The complex exponential function extends Euler's formula, as we have seen, and plays a fundamental role in connecting algebraic, geometric, and trigonometric aspects of complex numbers. Unlike the real exponential function, the complex exponential is periodic, and this periodicity leads to one of the most important differences between real and complex analysis: the inverse function, the logarithm, is no longer single-valued.

This section develops the complex exponential and logarithmic functions in detail. We begin by defining the exponential function and proving that it is entire. After that, we introduce the complex logarithm as a multi-valued inverse of the exponential function, and then define its principal branch, which provides a single-valued logarithmic function on an appropriate domain.

Recall the definition of the complex exponential

Definition 3.1.1. Let $z = x + iy \in \mathbb{C}$. The *complex exponential function* is defined by

$$e^z = e^{x+iy} = e^x(\cos(y) + i\sin(y))$$

Theorem 3.1.2. *The exponential function e^z is entire, and*

$$\frac{d}{dz}(e^z) = e^z$$

Proof. Write $e^z = e^x(\cos(y) + i\sin(y))$. Thus, the real and imaginary parts are

$$u(x, y) = e^x\cos(y), \quad v(x, y) = e^x\sin(y)$$

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Compute the partial derivatives

$$u_x = e^x \cos(y), \quad u_y = -e^x \sin(y), \quad v_x = e^x \sin(y), \quad v_y = e^x \cos(y)$$

Hence, $u_x = v_y$ and $u_y = -v_x$. So the Cauchy–Riemann equations are satisfied everywhere in $\mathbb{C}$, and all partial derivatives are continuous. Therefore, e^z is holomorphic on all of $\mathbb{C}$, and its derivative is

$$\frac{d}{dz}(e^z) = u_x + iv_x = e^x \cos(y) + ie^x \sin(y) = e^x(\cos(y) + i\sin(y)) = e^z$$

□

Example 3.1.3. Differentiate the following functions.
(1) $f(z) = iz^4(z^2 - e^z)$
(2) $f(z) = e^{z^2-(1+i)z+3}$

Solution.
(1) Using the product rule and the derivative of e^z, we obtain

$$\begin{aligned} f'(z) &= i[4z^3(z^2 - e^z) + z^4(2z - e^z)] \\ &= 4iz^5 - 4iz^3e^z + 2iz^5 - iz^4e^z \\ &= 6iz^5 - 4iz^3e^z - iz^4e^z \end{aligned}$$

(2) Using the chain rule,

$$f'(z) = e^{z^2-(1+i)z+3}(2z - (1+i))$$

Suppose $w = e^z = e^x(\cos(y) + i\sin(y))$. Compare this with polar form $w = r(\cos(\theta) + i\sin(\theta))$, we obtain immediately

Proposition 3.1.4. *For every $z = x + iy \in \mathbb{C}$,*
(1) $|e^z| = e^x$
(2) $\arg(e^z) = y + 2\pi n$ *for* $n \in \mathbb{Z}$.

Proof. Since $e^z = e^x(\cos(y) + i\sin(y))$, the modulus is the coefficient of the trigonometric factor, namely, $|e^z| = e^x$, and the angle is determined by y up to integer multiples of 2π, so $\arg(e^z) = y + 2\pi n$, for $n \in \mathbb{Z}$. □

Corollary 3.1.5. *For every $z \in \mathbb{C}$, $e^z \neq 0$.*

Proof. From Proposition 3.1.4, $|e^z| = e^x > 0$, so e^z can never equal 0. □

Proposition 3.1.6. *For every $z \in \mathbb{C}$, $\overline{e^z} = e^{\bar{z}}$.*

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Proof. Let $z = x + iy$. Then $e^z = e^x(\cos(y) + i\sin(y))$. Taking conjugates give

$$\overline{e^z} = e^x(\cos(y) - i\sin(y))$$

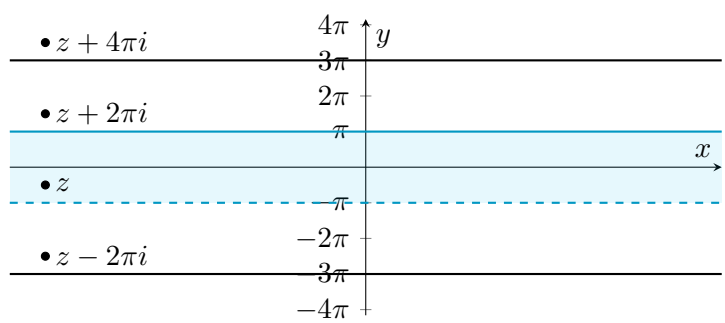
Since $\cos(-y) = \cos(y)$ and $\sin(-y) = -\sin(y)$, we have

$$e^x(\cos(y) - i\sin(y)) = e^x(\cos(-y) + i\sin(-y)) = e^{x-iy} = e^{\bar{z}}$$

□

Since the exponential function repeats every $2\pi i$, it is useful to restrict its domain to one horizontal strip of height 2π.

Definition 3.1.7. The strip $-\infty < x < \infty$, $-\pi < y \leq \pi$ is called a *fundamental region* of the complex exponential function.



In real analysis, the natural logarithm $\log(x)$ is defined as the inverse of the exponential function e^x for $x > 0$. In the complex setting, however, the exponential function is not one-to-one, so its inverse cannot be single-valued. This leads to the complex logarithm, which is necessarily a multi-valued function.

Suppose $e^w = z$, where $z \neq 0$. Write $w = u + iv$ so that

$$e^w = e^{u+iv} = e^u(\cos(v) + i\sin(v))$$

If $e^w = z$, then comparing the modulus and argument gives $e^u = |z|$ and $v = \arg(z)$. Taking the real logarithm of the first equation yields $u = \log|z|$. Therefore,

$$w = \log|z| + i\arg(z)$$

Since the argument is multi-valued, this gives infinitely many values of w.

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Definition 3.1.8. For $z \neq 0$, the *complex logarithm* is the multi-valued function defined by

$$\ln(z) = \log|z| + i\arg(z)$$

Since $\arg(z) = \theta + 2\pi n$ where $n \in \mathbb{Z}$, we may also write

$$\ln(z) = \log|z| + i(\theta + 2\pi n)$$

Example 3.1.9. Find all complex solutions to

(1) $e^w = i$
(2) $e^w = 1 + i$
(3) $e^w = -2$.

Solution.

(1) Since $|i| = 1$ and $\arg(i) = \frac{\pi}{2} + 2\pi n$, we obtain

$$w = \ln(i) = \log(1) + i\left(\frac{\pi}{2} + 2\pi n\right) = i\left(\frac{\pi}{2} + 2\pi n\right)$$

(2) Since $|1 + i| = \sqrt{2}$ and $\arg(1 + i) = \frac{\pi}{4} + 2\pi n$, we obtain

$$w = \ln(1 + i) = \log(\sqrt{2}) + i\left(\frac{\pi}{4} + 2\pi n\right) = \frac{1}{2}\log(2) + i\left(\frac{\pi}{4} + 2\pi n\right)$$

(3) Since $|-2| = 2$ and $\arg(-2) = \pi + 2\pi n$, we obtain

$$w = \ln(-2) = \log(2) + i(\pi + 2\pi n)$$

Theorem 3.1.10. *If $z, w \neq 0$ and $n \in \mathbb{Z}$, the following hold.*

(1) $\ln(zw) = \ln(z) + \ln(w)$
(2) $\ln\left(\frac{z}{w}\right) = \ln(z) - \ln(w)$
(3) $\ln(z^n) = n\ln(z)$

Proof. We will show that (1) holds. Note by definition,

$$\ln(z) = \log|z| + i\arg(z), \quad \ln(w) = \log|w| + i\arg(w)$$

Then adding the two equations give

$$\ln(z) + \ln(w) = \log|z| + \log|w| + i(\arg(z) + \arg(w))$$

Using properties of the real logarithm,

$$\log|z| + \log|w| = \log|zw|$$

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and using Corollary 1.2.5,

$$\arg(z) + \arg(w) = \arg(zw)$$

Therefore,

$$\ln(z) + \ln(w) = \log|zw| + i\arg(zw) = \ln(zw)$$

□

Remark 3.1.11. These identities hold for the multi-valued logarithm $\ln(z)$, but they do now always hold for the principal logarithm $\mathrm{Ln}(z)$. For example,

$$\mathrm{Ln}(zw) \neq \mathrm{Ln}(z) + \mathrm{Ln}(w)$$

in general.

Because $\ln(z)$ is multi-valued, it is often convenient to choose a single value by restricting the argument.

Definition 3.1.12. The *principal value* of the logarithm is defined by

$$\mathrm{Ln}(z) = \log|z| + i\,\mathrm{Arg}(z)$$

where the *principal argument* satisfies

$$-\pi < \mathrm{Arg}(z) \leq \pi$$

Thus, $\mathrm{Ln}(z)$ is a single-valued function.

Example 3.1.13. Compute the principal values.

(1) $\mathrm{Ln}(i)$
(2) $\mathrm{Ln}(1+i)$
(3) $\mathrm{Ln}(-2)$

Solution.

(1) Since $|i| = 1$ and $\mathrm{Arg}(z) = \frac{\pi}{2}$, we obtain

$$\mathrm{Ln}(i) = \log(1) + i\frac{\pi}{2} = \frac{\pi i}{2}$$

(2) Since $|1+i| = \sqrt{2}$ and $\mathrm{Arg}(1+i) = \frac{\pi}{4}$, we obtain

$$\mathrm{Ln}(1+i) = \frac{1}{2}\log(2) + \frac{\pi i}{4}$$

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(3) Since $|-2| = 2$ and $\operatorname{Arg}(-2) = \pi$,

$$\operatorname{Ln}(-2) = \log(2) + \pi i$$

Theorem 3.1.14. *For every $z \neq 0$,*

$$e^{\ln(z)} = z$$

Moreover, if $z = x + iy$ lies in the strip $-\infty < x < \infty$, $-\pi < y \leq \pi$, then

$$\operatorname{Ln}(e^z) = z$$

Proof. If $\ln(z) = \log|z| + i\arg(z)$, then

$$e^{\ln(z)} = e^{\log|z| + i\arg(z)} = e^{\log|z|}(\cos(\arg(z)) + i\sin(\arg(z))$$

Since $e^{\log|z|} = |z|$, we obtain

$$e^{\ln(z)} = |z|(\cos(\arg(z)) + i\sin(\arg(z))) = z$$

Thus, $e^{\ln(z)} = z$.

For the second identity, if $z = x + iy$ lies in the fundamental strip, then $|e^z| = e^x$ and $\operatorname{Arg}(e^z) = y$. Hence,

$$\operatorname{Ln}(e^z) = \log(e^x) + iy = x + iy = z$$

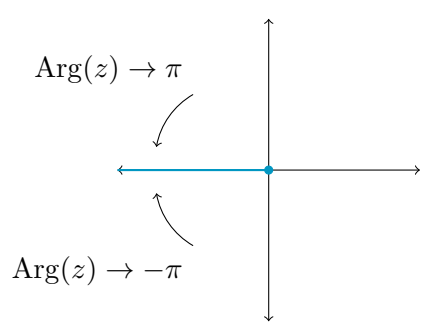
□

The principal logarithm is not defined at $z = 0$, and it is discontinuous across the negative real axis because the principal argument jumps there. This leads to the need for a branch cut.

Definition 3.1.15. The principal branch of the logarithm is defined on the domain

$$\mathbb{C} \setminus (-\infty, 0]$$

This is the complex plane with the nonpositive real axis removed.



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Theorem 3.1.16. *The principal branch* $\mathrm{Ln}(z) = \log|z| + i\,\mathrm{Arg}(z)$ *is holomorphic on* $\mathbb{C} \setminus (-\infty, 0]$ *and*

$$\frac{d}{dz}(\mathrm{Ln}(z)) = \frac{1}{z}$$

Proof. Write $z = re^{i\theta}$, $r > 0$, and $-\pi < \theta < \pi$. Then

$$\mathrm{Ln}(z) = \log(r) + i\theta$$

Thus, in polar form

$$u(r, \theta) = \log(r), \quad v(r, \theta) = \theta$$

Compute the partial derivatives

$$u_r = \frac{1}{r}, \quad u_\theta = 0, \quad v_r = 0, \quad v_\theta = 1$$

The Cauchy–Riemann equations in polar form are

$$u_r = \frac{1}{r} v_\theta, \quad v_r = -\frac{1}{r} u_\theta$$

Indeed,

$$\frac{1}{r} v_\theta = \frac{1}{r} = u_r, \quad -\frac{1}{r} u_\theta = 0 = v_r$$

So $\mathrm{Ln}(z)$ is holomorphic on its domain. Furthermore, using the derivative formula in polar form $f'(z) = e^{-i\theta}(u_r + iv_r)$, we have

$$\mathrm{Ln}'(z) = e^{-i\theta}\left(\frac{1}{r} + i0\right) = \frac{e^{-i\theta}}{r} = \frac{1}{re^{i\theta}} = \frac{1}{z}$$

□

Example 3.1.17. Find the derivatives of the following functions on appropriate domains.

(1) $f(z) = z\,\mathrm{Ln}(z)$
(2) $f(z) = \mathrm{Ln}(z + 1)$

Solution.

(1) Using the product rule

$$f'(z) = 1 \cdot \mathrm{Ln}(z) + z \cdot \frac{1}{z} = \mathrm{Ln}(z) + 1$$

for $z \in \mathbb{C} \setminus (-\infty, 0]$.

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(2) By the chain rule,

$$f'(z) = \frac{1}{z+1}$$

This is valid wherever $z+1$ does not lie on the branch cut of Ln, that is, whenever $z \notin (-\infty, -1]$. Therefore,

$$\frac{d}{dz}(\mathrm{Ln}(z+1)) = \frac{1}{z+1}$$

for $z \in \mathbb{C} \setminus (-\infty, -1]$.

The principal branch is only one possible branch of the logarithm.

Definition 3.1.18. For any fixed $\theta_0 \in \mathbb{R}$, the function

$$f_{\theta_0}(z) = \log(r) + i\theta, \quad \theta_0 < \theta < \theta_0 + 2\pi$$

defines a branch of the logarithm on a suitable slit pllane. Each such branch satisfies

$$f'_{\theta_0}(z) = \frac{1}{z}$$

3.2 Complex Powers

In the real-variable setting, powers such as x^n, $\sqrt[n]{x}$ and more generally x^α are usually introduced in stages, beginning with integer exponents, and then extending to rational and real exponents. For positive real numbers, these notions behave in a familiar and largely unambiguously way. In the complex setting, however, the situation is more subtle.

For integer exponents, the expression z^n is simply repeated multiplication and causes no difficulty. But when we try to define expressions such as z^α for arbitrary complex exponents α, we must use the complex logarithm. Since the complex logarithm is multi-valued, it follows that complex powers are in general multi-valued as well. Thus, unlike in real analysis, a complex power may have infinitely many values.

This section develops the theory of complex powers using the exponential and logarithmic functions from the previous section. We begin by defining z^α in terms of the complex logarithm and then examine its algebraic properties. We show that when the exponent is an integer, the usual single-valued notion of powers is recovered. We then introduce the principal value of a complex power by using the principal branch of the logarithm, and study the holomorphicity of this principal branch. Along the way, we see that many

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familiar rules of exponents continue to hold, but others fail or require care because of the multi-valued nature of the logarithm.

Recall from Section 3.1 that for $z \neq 0$, the complex logarithm is the multi-valued function

$$\ln(z) = \log|z| + i\arg(z)$$

Since the logarithm is multi-valued, any definition involving $\ln(z)$ will generally also be multi-valued.

Definition 3.2.1. Let $\alpha \in \mathbb{C}$ and let $z \in \mathbb{C}$ with $z \neq 0$. The *complex power* z^α is defined by

$$z^\alpha = e^{\alpha \ln(z)}$$

Since $\ln(z)$ is multi-valued, the expression z^α is in general a multi-valued function.

Remark 3.2.2. If $\ln(z) = \log|z| + i(\theta + 2\pi n)$ for $n \in \mathbb{Z}$, then

$$z^\alpha = e^{\alpha \log|z|} E^{i\alpha(\theta+2\pi n)}$$

Thus, a single complex number z may produce infinitely many values of z^α.

Theorem 3.2.3. *If $n \in \mathbb{Z}$, then for $z \neq 0$, the complex power z^n agrees with the usual integer power and is single-valued.*

Proof. Let $n \in \mathbb{Z}$. By Definition 3.2.1,

$$z^n = e^{n\ln(z)}$$

Now write

$$\ln(z) = \log|z| + i\arg(z)$$

Then

$$z^n = e^{n(\log|z| + i\arg(z))} = e^{n\log|z|} e^{in\arg(z)}$$

Let $\theta = \mathrm{Arg}(z)$ be one choice of argument, so that every argument of z is of the form

$$\arg(z) = \theta + 2\pi k$$

for $k \in \mathbb{Z}$. Hence,

$$e^{in\arg(z)} = e^{in(\theta+2\pi k)} = e^{in\theta} e^{i2\pi nk}$$

Since $n, k \in \mathbb{Z}$, we have $e^{i2\pi nk} = 1$. Therefore,

$$e^{in\arg(z)} = e^{in\theta}$$

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which is independent of the choice of k. Thus,

$$z^n = e^{n \log |z|} e^{in\theta}$$

Since $e^{n \log |z|} = |z|^n$, this is precisely the usual single-valued integer power of z. Hence z^n agrees with the standard meaning of integer powers.

$$z^n = e^{n \log |z|} e^{in \operatorname{Arg}(z)}$$

and this is single-valued. □

Remark 3.2.4. Although integer powers are single-valued, the expression $z^\alpha = e^{\alpha \ln(z)}$ is generally multi-valued when α is not an integer.

Example 3.2.5. Find all values of i^{2i}.

Solution. We use the definition $z^\alpha = e^{\alpha \ln(z)}$. Here, $z = i$ and $\alpha = 2i$. Since $|i| = 1$, $\arg(i) = \frac{\pi}{2} + 2\pi n$ for $n \in \mathbb{Z}$, we have

$$\ln(i) = \log(1) + i\left(\frac{\pi}{2} + 2\pi n\right) = i\left(\frac{\pi}{2} + 2\pi n\right)$$

Therefore,

$$i^{2i} = e^{2i \ln(i)} = e^{2i \cdot i\left(\frac{\pi}{2} + 2\pi n\right)} = e^{-2\left(\frac{\pi}{2} + 2\pi n\right)} = e^{-(4n+1)\pi}$$

Example 3.2.6. Find all values of $(1+i)^i$.

Solution. Again, $(1+i)^i = e^{i \ln(1+i)}$. Now, $|1+i| = \sqrt{2}$ and $\arg(1+i) = \frac{\pi}{4} + 2\pi n$ for $n \in \mathbb{Z}$. Hence

$$\log(1+i) = \log(\sqrt{2}) + i\left(\frac{\pi}{4} + 2\pi n\right) = \frac{1}{2}\log(2) + i\left(\frac{\pi}{2} + 2\pi n\right)$$

Therefore,

$$(1+i)^i = e^{i\left(\frac{1}{2}\log(2) + i\left(\frac{\pi}{4} + 2\pi n\right)\right)} = e^{\frac{i \log(2)}{2}} e^{-\left(\frac{\pi}{4} + 2\pi n\right)}$$

Theorem 3.2.7. *Let $\alpha_1, \alpha_2 \in \mathbb{C}$, and let $z \neq 0$. Then the complex powers satisfy*

(1) $z^{\alpha_1} z^{\alpha_2} = z^{\alpha_1 + \alpha_2}$
(2) $\frac{z^{\alpha_1}}{z^{\alpha_2}} = z^{\alpha_1 - \alpha_2}$.
(3) For every integer n, $(z^\alpha)^n = z^{n\alpha}$.

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Proof. From the definition, $z^{\alpha_1} = e^{\alpha_1 \ln(z)}$ and $z^{\alpha_2} = e^{\alpha_2 \ln(z)}$. Hence, by Theorem 2.1.20,

$$z^{\alpha_1} z^{\alpha_2} = e^{\alpha_1 \ln(z)} e^{\alpha_2 \ln(z)} = e^{(\alpha_1+\alpha_2)\ln(z)} = z^{\alpha_1+\alpha_2}$$

Similarly, using Theorem 2.1.20

$$\frac{z^{\alpha_1}}{z^{\alpha_2}} = e^{\alpha_1 \ln(z)} e^{-\alpha_2 \ln(z)} = e^{(\alpha_1-\alpha_2)\ln(z)} = z^{\alpha_1-\alpha_2}$$

Finally, if $n \in \mathbb{Z}$, then

$$(z^\alpha)^n = (e^{\alpha \ln(z)})^n = e^{n\alpha \ln(z)} = z^{n\alpha}$$

□

Remark 3.2.8. Some exponent laws from real analysis do not automatically remain valid in the complex setting without care, especially when principal values are used. The multi-valued nature of $\ln(z)$ must always be kept in mind.

Since the complex logarithm is multi-valued, the power z^α is generally multi-valued. To obtain a single-valued function, we use the principal branch of the logarithm.

Recall that the principal logarithm is

$$\operatorname{Ln}(z) = \log|z| + i \operatorname{Arg}(z)$$

where $-\pi < \operatorname{Arg}(z) \leq \pi$.

Definition 3.2.9. Let $\alpha \in \mathbb{C}$ and let $z \neq 0$. The *principal value* of the complex power z^α is defined by

$$z^\alpha = e^{\alpha \operatorname{Ln}(z)}$$

When necessary, this may be called the *principal branch of the power function.*

Remark 3.2.10. In context, the notation z^α is often used both for the multi-valued power and for its principal value. The intended meaning is usually clear from context.

Example 3.2.11. Find the principal value of $(-3)^{\frac{i}{\pi}}$.

Solution. We use the principal logarithm

$$(-3)^{\frac{i}{\pi}} = e^{\frac{i}{\pi} \operatorname{Ln}(-3)}$$

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Now, $|-3| = 3$ and $\operatorname{Arg}(-3) = \pi$, so

$$\operatorname{Ln}(-3) = \log(3) + i\pi$$

Hence,

$$(-3)^{\frac{i}{\pi}} = e^{\frac{i}{\pi}(\log(3)+i\pi)} = e^{-1+\frac{i\log(3)}{\pi}}$$

Using Definition 3.1.1,

$$(-3)^{\frac{i}{\pi}} = e^{-1}\left(\cos\left(\frac{\log(3)}{\pi}\right) + i\sin\left(\frac{\log(3)}{\pi}\right)\right)$$

Example 3.2.12. Find the principal value of $(2i)^{1-i}$.

Solution. We compute

$$(2i)^{1-i} = e^{(1-i)\operatorname{Ln}(2i)}$$

Since $|2i| = 2$ and $\operatorname{Arg}(2i) = \frac{\pi}{2}$, we have

$$\operatorname{Ln}(2i) = \log(2) + \frac{\pi i}{2}$$

Therefore,

$$(2i)^{1-i} = e^{(1-i)(\log(2)+\frac{\pi i}{2})}$$

Expanding the exponent

$$(1-i)\left(\log(2) + \frac{\pi i}{2}\right) = \log(2) + \frac{\pi}{2} + i\left(\frac{\pi}{2} - \log(2)\right)$$

Hence,

$$(2i)^{1-i} = e^{\log(2)+\frac{\pi}{2}}\left(\cos\left(\frac{\pi}{2} - \log(2)\right) + i\sin\left(\frac{\pi}{2} - \log(2)\right)\right)$$

Since $\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$ and $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$, we have

$$(2i)^{1-i} = 2e^{\frac{\pi}{2}}(\sin(\log(2)) + i\cos(\log(2))$$

In general, the multi-valued power function cannot be treated as a single holomorphic function on all of $\mathbb{C} \setminus \{0\}$. However, its principal branch is holomorphic on a slit plane.

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Theorem 3.2.13. *Let $\alpha \in \mathbb{C}$ be fixed, and define*

$$f(z) = e^{\alpha \operatorname{Ln}(z)}$$

where $|z| > 0$ and $-\pi < \arg(z) < \pi$. Then f is a branch of the multi-valued power function z^α. This branch is called the principal branch of z^α and its branch cut is the nonpositive real axis.

Proof. The principal logarithm $\operatorname{Ln}(z)$ is holomorphic on the domain $|z| > 0$ $-\pi < \arg(z) < \pi$. The exponential function is entire. Therefore, the composition $f(z) = e^{\alpha \operatorname{Ln}(z)}$ is holomorphic on the same domain.

Since $\operatorname{Ln}(z)$ is one of the values of $\ln(z)$, it follows that $f(z)$ gives one single-valued branch of the multi-valued function $z^\alpha = e^{\alpha \ln(z)}$. Hence, f is a branch of z^α, namely the principal branch. □

Theorem 3.2.14. *Let $f(z) = z^\alpha = e^{\alpha \operatorname{Ln}(z)}$, $|z| > 0$, and $-\pi < \arg(z) < \pi$. Then f is differentiable on this domain and*

$$\frac{d}{dz} z^\alpha = \alpha z^{\alpha - 1}$$

Proof. Using the chain rule,

$$f'(z) = \frac{d}{dz}\left(e^{\alpha \operatorname{Ln}(z)}\right) = e^{\alpha \operatorname{Ln}(z)} \frac{d}{dz}(\alpha \operatorname{Ln}(z)) = \alpha e^{\alpha \operatorname{Ln}(z)} \frac{d}{dz}(\operatorname{Ln}(z))$$

By Theorem 3.1.16,

$$\frac{d}{dz}(\operatorname{Ln}(z)) = \frac{1}{z}$$

Therefore,

$$f'(z) = \alpha e^{\alpha \operatorname{Ln}(z)} \frac{1}{z} = \alpha z^\alpha \frac{1}{z} = \alpha z^{\alpha - 1}$$

□

Example 3.2.15. Find the derivative of the principal value z^i at the point $1 + i$.

Solution. Since $1 + i$ lies in the domain of $|z| > 0$ and $-\pi < \arg(z) < \pi$, the derivative formula applies

$$\frac{d}{dz} z^i = i z^{i-1}$$

Hence,

$$\left.\frac{d}{dz} z^i\right|_{z=1+i} = i(1+i)^{i-1} = i(1+i)^i (1+i)^{-1}$$

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From Example 3.2.6, the principal value of $(1+i)^i$ corresponds to $n = 0$, namely

$$(1+i)^i = e^{-\frac{\pi}{4}+\frac{i\log(2)}{2}}$$

Therefore,

$$\left.\frac{d}{dz}\right|_{z=1+i} = \frac{1+i}{2}e^{-\frac{\pi}{4}+\frac{i\log(2)}{2}}$$

Just as the logarithm has many branches, so too does the power function. If we define the branch of the logarithm by

$$f_{\theta_0} = \log(r) + i\theta$$

where $\theta_0 < \theta < \theta_0 + 2\pi$, then the corresponding branch of the power function is

$$z^\alpha = e^{\alpha f_{\theta_0}(Z)}$$

For example, using $-\frac{\pi}{4} < \theta < \frac{7\pi}{4}$ produces a branch cut along the ray $\arg(z) = -\frac{\pi}{4}$.

3.3 Complex Trigonometric Functions

In real analysis, the trigonometric functions $\sin(x)$ and $\cos(x)$ are introduced geometrically or through differential equations, and they play a central role in geometry, calculus, and applied mathematics. In complex analysis, these functions can be defined in a remarkably elegant way using the complex exponential function. This point of view not only extends the familiar sine and cosine functions from real numbers to complex numbers, but also reveals new and distinctly complex behavior.

The complex sine and cosine functions agree with their real counterparts when the input is real, so they are true extensions of the ordinary trigonometric functions. At the same time, they inherit many important algebraic and differential properties from the exponential function. In particular, they are entire functions, satisfy the familiar identities from trigonometry, and have derivatives analogous to those in real calculus. However, unlike the real sine and cosine functions, their behavior in the complex plane is much richer: they are not bounded, and their mapping properties are far more intricate.

Having established the foundational ideas of limits, continuity, differentiability, holomorphicity, the Cauchy–Riemann equations, and harmonic functions, we are now ready to study some of the most important examples of holomorphic functions in complex analysis: the elementary functions. These

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functions extend familiar real-variable functions into the complex plane, but in doing so they exhibit several new and distinctly complex phenomena that have no true analogue in real calculus.

We begin with Euler's formulas for a real variable x

$$e^{ix} = \cos(x) + i\sin(x), \quad e^{-ix} = \cos(x) - i\sin(x)$$

Adding these two equations, we get

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

and by subtracting them, we get

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

These formulas suggest the following definitions for complex arguments.

Definition 3.3.1. For $z \in \mathbb{C}$, define the complex sine and cosine by

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos(z) = \frac{e^{iz} + e^{-iz}}{2}$$

Remark 3.3.2. If $z = x \in \mathbb{R}$, then these definitions agree with the usual sine and cosine functions. Hence, the complex trigonometric functions are genuine extensions of the real trigonometric functions.

Definition 3.3.3. Whenever the denominators are nonzero, define

$$\tan(z) = \frac{\sin(z)}{\cos(z)}, \quad \cot(z) = \frac{\cos(z)}{\sin(z)}, \quad \sec(z) = \frac{1}{\cos(z)}, \quad \csc(z) = \frac{1}{\sin(z)}$$

These agree with the corresponding real trigonometric functions when z is real.

Example 3.3.4. Express each of the following in the form $a + ib$.

(1) $\cos(i)$
(2) $\sin(2 + i)$
(3) $\tan(\pi - 2i)$

Solution.

(1) Using the definition of $\cos(z)$,

$$\cos(i) = \frac{e^{i(i)} + e^{-i(i)}}{2} = \frac{e^{-1} + e}{2}$$

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(2) Using the definition of $\sin(z)$,

$$\sin(2+i) = \frac{e^{i(i+1)} - e^{-i(2+i)}}{2i} = \frac{e^{-1+2i} - e^{1-2i}}{2i}$$

Writing each exponential in trigonometric form

$$\begin{aligned} e^{-1+2i} &= e^{-1}(\cos(2) + i\sin(2)) \\ e^{1-2i} &= e(\cos(2) - i\sin(2)) \end{aligned}$$

Therefore,

$$\sin(2+i) = \frac{e^{-1}(\cos(2) + i\sin(2)) - e(\cos(2) - i\sin(2))}{2i}$$

(3) Using the definition of $\tan(z)$, togetehr with the exponential definitions of sine and cosine, we have

$$\tan(\pi - 2i) = \frac{\frac{e^{i(\pi-2i)} - e^{-i(\pi-2i)}}{2i}}{\frac{e^{i(\pi-2i)} + e^{-i(\pi-2i)}}{2}} = \frac{e^{i(\pi-2i)} - e^{-i(\pi-2i)}}{i(e^{i(\pi-2i)} + e^{-i(\pi-2i)})} = \frac{e^{-2} - e^{-2}}{i(e^{2} + e^{-2})}$$

Since $\frac{1}{i} = -i$, this becomes

$$\tan(\pi - 2i) = -i\frac{e^{2} - e^{-2}}{e^{2} + e^{-2}}$$

Most of the familiar trigonometric identities from real analysis remain valid for complex arguments.

Theorem 3.3.5. *For all $z, w \in \mathbb{C}$, the following identities hold*

(1) $\sin(-z) = -\sin(z)$
(2) $\cos(-z) = \cos(z)$
(3) $\cos^2(z) + \sin^2(z) = 1$
(4) $\sin(z \pm w) = \sin(z)\cos(w) \pm \cos(z)\sin(w)$
(5) $\cos(z \pm w) = \cos(z)\cos(w) \mp \sin(z)\sin(w)$
(6) $\sin(2z) = 2\sin(z)\cos(z)$
(7) $\cos(2z) = \cos^2(z) - \sin^2(z)$

Proof. We will prove (1) and (3). By definition,

$$\sin(-z) = \frac{e^{i(-z)} - e^{-i(-z)}}{2i} = \frac{e^{-iz} - e^{iz}}{2i} = -\frac{e^{iz} - e^{-iz}}{2i} = -\sin(z)$$

which proves (1).

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To see that (3) holds, using the definitions of complex sine and cosine, we have

$$\cos^2(z) = \left(\frac{e^{iz} + e^{-iz}}{2}\right)^2 = \frac{e^{2iz} + 2 + e^{-2iz}}{4}$$
$$\sin^2(z) = \left(\frac{e^{iz} - e^{-iz}}{2i}\right)^2 = -\frac{e^{2iz} - 2 + e^{-2iz}}{4}$$

Adding gives

$$\cos^2(z) + \sin^2(z) = \frac{e^{2iz} + 2 + e^{-2iz}}{4} + \left(-\frac{e^{2iz} - 2 + e^{-2iz}}{4}\right) = 1$$

proving (3). □

Remark 3.3.6. Although many familiar identities remain valid, some inequalities from real trigonometry do not. For example, it is not true that $|\sin(z)| \leq 1$ or $|\cos(z)| \leq 1$ for complex z. Indeed,

$$|\cos(i)| = \frac{e^{-1} + e}{2} > 1$$

Thus, the complex sine and cosine functions are not bounded by 1 in modulus.

Theorem 3.3.7. *For all $z \in \mathbb{C}$,*

$$\sin(z + 2\pi) = \sin(z), \quad \cos(z + 2\pi) = \cos(z)$$

Thus, both $\sin(z)$ and $\cos(z)$ are periodic with real period 2π.

Proof. We provide the proof for sine. By definition,

$$\sin(z+2\pi) = \frac{e^{i(z+2\pi)} - e^{-i(z+2\pi)}}{2i} = \frac{e^{iz}e^{i2\pi} - e^{-iz}e^{-i2\pi}}{2i} = \frac{e^{iz} - e^{-i}}{2i} = \sin(z)$$

□

Corollary 3.3.8. *The functions $\tan(z)$ and $\cot(z)$ are periodic with period π wherever they are defined.*

Proof. Since $\sin(z + \pi) = -\sin(z)$ and $\cos(z + \pi) = -\cos(z)$, we have

$$\tan(z + \pi) = \frac{\sin(z + \pi)}{\cos(z + \pi)} = \frac{-\sin(z)}{-\cos(z)} = \tan(z)$$

A similar argument shows that $\cot(z + \pi) = \cot(z)$. □

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Example 3.3.9. Find all complex solutions of $\sin(z) = 5$.

Solution. By definition,

$$\frac{e^{iz} - e^{-iz}}{2i} = 5$$

Multiplying both sides by $2ie^{iz}$, we get

$$e^{2iz} - 1 = 10ie^{iz}$$

and so

$$e^{2iz} - 10ie^{iz} - 1 = 0$$

Let $w = e^{iz}$. Then the equation becomes the quadratic

$$w^2 - 1 - iw - 1 = 0$$

Using the quadratic formula

$$w = \frac{10i \pm \sqrt{(-10i)^2 - 4(1)(-1)}}{2} = \frac{10i \pm 4\sqrt{6}i}{2} = (5 \pm 2\sqrt{6})i$$

Thus, we must solve the two exponential equations

$$e^{iz} = (5 + 2\sqrt{6})i, \quad e^{-iz} = (5 - 2\sqrt{6})i$$

Taking logarithms, we have the following cases.

(1) Consider $e^{iz} = (5 + 2\sqrt{6})i$. Since

$$|(5 + 2\sqrt{6})i| = 5 + 2\sqrt{6}, \quad \arg((5 + 2\sqrt{6})i) = \frac{\pi}{2} + 2\pi n$$

then

$$iz = \log(5 + 2\sqrt{6}) + i\left(\frac{\pi}{2} + 2\pi n\right)$$

Multiplying both sides by $-i$ give

$$z = -i\log(5 + 2\sqrt{6}) + \frac{(4n+1)\pi}{2}$$

for $n \in \mathbb{Z}$.

(2) Similarly, consider $e^{-iz} = (5 - 2\sqrt{6})i$. Since

$$|(5 - 2\sqrt{6})i| = 5 - 2\sqrt{6}, \quad \arg((5 - 2\sqrt{6})i) = \frac{\pi}{2} + 2\pi n$$

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Then

$$iz = \log(5 - 2\sqrt{6}) + i\left(\frac{\pi}{2} + 2\pi n\right)$$

and so

$$z = \frac{(4n+1)\pi}{2} - i\log(5 - 2\sqrt{6})$$

for $n \in \mathbb{Z}$.

Let $z = x + iy$. We now express $\sin(z)$ and $\cos(z)$ in terms of their real and imaginary parts.

Theorem 3.3.10. *If $z = x + iy$, then*
(1) $\sin(z) = \sin(x)\cosh(y) + i\cos(x)\sinh(y)$
(2) $\cos(z) = \cos(x)\cosh(y) - i\sin(x)\sinh(y)$

Proof. Using the definition of sine,

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

If $z = x + iy$, then $iz = ix - y$ and $-iz = -ix + y$. Then

$$e^{iz} = e^{-y}(\cos(x) + i\sin(x)), \quad e^{-iz} = e^{y}(\cos(x) - i\sin(x))$$

Substituting gives

$$\sin(z) = \frac{e^{-y}(\cos(x) + i\sin(x)) - e^{y}(\cos(x) - i\sin(x))}{2i}$$

Grouping real and imaginary parts and using

$$\cosh(y) = \frac{e^{y} + e^{-y}}{2}, \quad \sinh(y) = \frac{e^{y} - e^{-y}}{2}$$

we obtain

$$\sin(z) = \sin(x)\cosh(y) + i\cos(x)\sinh(y)$$

The proof of (2) is similar. □

Theorem 3.3.11. *If $z = x + iy$, then*
(1) $|\sin(z)| = \sqrt{\sin^2(x) + \sinh^2(y)}$
(2) $|\cos(z)| = \sqrt{\cos^2(x) + \sinh^2(y)}$

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Proof. Using Theorem 3.3.10,

$$|\sin(z)|^2 = \sin^2(x)\cosh^2(y) + \cos^2(x)\sinh^2(y)$$

Using the identity $\cosh^2(y) = 1 + \sinh^2(y)$, we get

$$\begin{aligned}|\sin(z)|^2 &= \sin^2(x)(1+\sinh^2(y)) + \cos^2(x)\sinh^2(y)\\ &= \sin^2(x) + (\sin^2(x)+\cos^2(x))\sinh^2(y)\\ &= \sin^2(x) + \sinh^2(y)\end{aligned}$$

Therefore,

$$|\sin(z)| = \sqrt{\sin^2(x) + \sinh^2(y)}$$

The proof of (2) is similar. □

Remark 3.3.12. These formulas show immediately that the complex sine and cosine functions are not bounded on $\mathbb{C}$. Indeed, as $|y| \to \infty$, we have $\sinh(y) \to \infty$, so both $|\sin(x+iy)|$ and $|\cos(x+iy)|$ can be arbitrarily large. This is one of the main differences between real and complex trigonometric functions.

Theorem 3.3.13. *(1) The zeros of* $\sin(z)$ *are exactly the real numbers* $z = n\pi$ *for* $n \in \mathbb{Z}$.

(2) The zeros of $\cos(z)$ *are exactly the real numbers* $z = \frac{(2n+1)\pi}{2}$ *for* $n \in \mathbb{Z}$.

Proof. Let $z = x + iy$. By the modulus formula, we have

$$|\sin(z)| = \sqrt{\sin^2(x) + \sinh^2(y)}$$

Therefore, $\sin(z) = 0$ if and only if $|\sin(z)| = 0$ if and only if $\sin^2(x) + \sinh^2(y) = 0$. But both terms are nonnegative real numbers, so their sum is zero if and only if each term is zero. Hence, $\sin(x) = 0$ and $\sinh(y) = 0$. Now, $\sin(x) = 0$ if and only if $x = n\pi$ for $n \in \mathbb{Z}$ and $\sinh(y) = 0$ if and only if $y = 0$. Therefore, the only zeros of $\sin(z)$ are the real multiples of π.

The proof of (2) is similar. □

Theorem 3.3.14. *The functions* $\sin(z)$ *and* $\cos(z)$ *are entire, and their derivatives are given by*

$$\frac{d}{dz}\sin(z) = \cos(z), \quad \frac{d}{dz}\cos(z) = -\sin(z)$$

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Proof. Since the exponential function is entire and compositions of entire functions are entire, it follows that $\sin(z)$ and $\cos(z)$ are entire.

Differentiating using the chain rule,

$$\frac{d}{dz}\sin(z) = \frac{1}{2i}(ie^{iz}-(-i)e^{-iz}) = \frac{1}{2i}(ie^{iz}+ie^{-iz}) = \frac{i}{2i}(e^{iz}+e^{-iz}) = \frac{e^{iz}+e^{-iz}}{2} = \cos(z)$$

Similarly,

$$\frac{d}{dz}\cos(z) = \frac{1}{2}(ie^{iz} - ie^{-iz}) = i\frac{e^{iz}-e^{-iz}}{2} = -\frac{e^{iz}-e^{-iz}}{2i} = -\sin(z)$$

Therefore, both $\sin(z)$ and $\cos(z)$ are entire. □

Corollary 3.3.15. *Wherever they are defined*

$$\frac{d}{dz}\tan(z) = \sec^2(z), \quad \frac{d}{dz}\cot(z) = -\csc^2(z),$$
$$\frac{d}{dz}\sec(z) = \sec(z)\tan(z), \quad \frac{d}{dz}\csc(z) = -\csc(z)\cot(z)$$

Proof. These follow from the quotient rule and the derivatives of $\sin(z)$ and $\cos(z)$, as exactly in real-variable calculus. □

Remark 3.3.16. Since $\tan(z)$ and $\sec(z)$ involve the division by $\cos(z)$, they are undefined at the zeros of $\cos(z)$, namely at $z = \frac{(2n+1)\pi}{2}$ for $n \in \mathbb{Z}$. Similarly, $\cot(z)$ and $\csc(z)$ are undefined at the zeros of $\sin(z)$, namely at $z = n\pi$, for $n \in \mathbb{Z}$.

3.4 Complex Hyperbolic Functions

In real-variable calculus, the hyperbolic sine and cosine functions are defined in terms of the exponential function by

$$\sinh(x) = \frac{e^x - e^{-x}}{2}, \quad \cosh(x) = \frac{e^x + e^{-x}}{2}$$

These functions arise naturally in many areas of mathematics, including differential equations, geometry, and mathematical physics. In complex analysis, these definitions extend directly to complex inputs by replacing the real exponential function with the complex exponential function. This produces the complex hyperbolic functions, which are holomorphic analogues of their real counterparts.

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These functions arise naturally in many areas of mathematics, including differential equations, geometry, and mathematical physics. In complex analysis, these definitions extend directly to complex inputs by replacing the real exponential function with the complex exponential function. This produces the complex hyperbolic functions, which are holomorphic analogues of their real counterparts.

In this section, we define the complex hyperbolic functions and the associated functions $\tanh(z)$, $\coth(z)$, $\operatorname{sech}(z)$, and $\operatorname{csch}(z)$. We derive their basic differential properties, establish their relations with the complex trigonometric functions, and prove the standard hyperbolic identities. These results show that, while the complex hyperbolic functions retain many of the algebraic features of their real counterparts, they also display several phenomena unique to the complex setting.

Recall that for a real variable x,

$$\sinh(x) = \frac{e^x - e^{-x}}{2}, \quad \cosh(z) = \frac{e^z + e^{-z}}{2}$$

This suggests the following complex definitions.

Definition 3.4.1. For $z \in \mathbb{C}$, the *complex hyperbolic sine* and *complex hyperbolic cosine* are defined by

$$\sinh(z) = \frac{e^z - e^{-z}}{2}, \quad \cosh(z) = \frac{e^z + e^{-z}}{2}$$

Remark 3.4.2. Since the complex exponential function agrees with the real exponential function on real input, it follows immediately that for real x,

$$\sinh(x) = \frac{e^x - e^{-x}}{2}, \quad \cosh(x) = \frac{e^x + e^{-x}}{2}$$

are exactly the usual real hyperbolic functions. Thus, the complex hyperbolic sine and cosine functions are genuine extensions of the corresponding real functions.

Definition 3.4.3. Wherever the denominators are nonzero, define

$$\tanh(z) = \frac{\sinh(z)}{\cosh(z)}, \quad \coth(z) = \frac{\cosh(z)}{\sinh(z)}, \quad \operatorname{sech}(z) = \frac{1}{\cosh(z)}, \quad \operatorname{csch}(z) = \frac{1}{\sinh(z)}$$

These are called the *complex hyperbolic tangent*, *hyperbolic cotangent*, *hyperbolic secant*, and *hyperbolic cosecant*, respectively.

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Since the functions e^z and e^{-z} are entire, it is immediate that $\sinh(z)$ and $\cosh(z)$ are entire as well.

Theorem 3.4.4. *For all $z \in \mathbb{C}$,*

$$\frac{d}{dz}\sinh(z) = \cosh(z), \quad \frac{d}{dz}\cosh(z) = \sinh(z)$$

In particular, $\sinh(z)$ *and* $\cosh(z)$ *are entire functions.*

Proof. From the definition of hyperbolic sine, differentiating term by term gives

$$\frac{d}{dz}\sinh(z) = \frac{1}{2}\left(\frac{d}{dz}e^z - \frac{d}{dz}e^{-z}\right) = \frac{1}{2}(e^z + e^{-z}) = \cosh(z)$$

The proof for hyperbolic cosine is similar. □

Corollary 3.4.5. *Wherever these functions are defined,*

$$\frac{d}{dz}\tanh(z) = \operatorname{sech}^2(z), \quad \frac{d}{dz}\coth(z) = -\operatorname{csch}^2(z),$$
$$\frac{d}{dz}\operatorname{sech}(z) = -\operatorname{sech}(z)\tanh(z), \quad \frac{d}{dz}\operatorname{csch}(z) = -\operatorname{csch}(z)\coth(z)$$

Proof. These rules follow Definition 3.4.3, Theorem 3.4.4, and the quotient rule. □

One of the most beautiful features of complex hyperbolic functions is their direct relationship with the complex trigonometric derivatives.

Theorem 3.4.6. *For all $z \in \mathbb{C}$,*
(1) $\sinh(z) = -i\sinh(iz)$
(2) $\cos(z) = \cosh(iz)$
(3) $\sinh(z) = -i\sin(iz)$
(4) $\cosh(z) = \cos(iz)$

Proof. We prove (1). From Definition 3.4.1, we have

$$\sinh(iz) = \frac{e^{iz} - e^{-iz}}{2i}$$

By the definition of $\sin(z)$, we have $\sinh(iz) = i\sin(z)$. Proof of (2) is similar.

Replacing z by iz in these identities yield (3) and (4). □

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Example 3.4.7. Show that $\tan(iz) = i\tanh(z)$.

Solution. Using the identities from Theorem 3.4.6,

$$\tan(iz) = \frac{\sin(iz)}{\cos(iz)} = \frac{i\sinh(z)}{\cosh(z)} = i\tanh(z)$$

as desired.

Using the identities from complex trigonometry, we can also derive the standard hyperbolic identities.

Theorem 3.4.8. *For all $z, w \in \mathbb{C}$,*

(1) $\sinh(-z) = -\sinh(z)$
(2) $\cosh(-z) = \cosh(z)$
(3) $\cosh^2(z) - \sinh^2(z) = 1$
(4) $\sinh(z \pm w) = \sinh(z)\cosh(w) \pm \cosh(z)\sinh(w)$
(5) $\cosh(z \pm w) = \cosh(z)\cosh(w) \pm \sinh(z)\sinh(w)$

Proof. To see that (1) is true, by definition, we have

$$\sinh(-z) = \frac{e^{-z} - e^{z}}{2} = -\frac{e^{z} - e^{-z}}{2} = -\sinh(z)$$

The proof of (2) is similar.

To see that (3) is true, using the definitions of hyperbolic sine and cosine,

$$\cosh^2(z) = \left(\frac{e^{z} + e^{-z}}{2}\right)^2 = \frac{e^{2z} + 2 + e^{-2z}}{4},$$
$$\sinh^2(z) = \left(\frac{e^{z} - e^{-z}}{2}\right)^2 = \frac{e^{2z} - 2 + e^{-2z}}{4}$$

Subtracting gives

$$\cosh^2(z) - \sinh^2(z) = \frac{e^{2z} + 2 + e^{-2z}}{4} - \frac{e^{2z} - 2 + e^{-2z}}{4} = 1$$

To see that (5) is true, by Theorem 3.4.6, we have

$$\cosh(z + w) = \cos(i(z + w)) = \cos(iz + iw)$$

Using the addition formula for cosine

$$\cos(iz + iw) = \cos(iz)\cos(iw) - \sin(iz)\sin(iw)$$

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Again, by Theorem 3.4.6

$$\cos(iz) = \cosh(z), \quad \cos(iw) = \cosh(w), \quad \sin(iz) = i\sinh(z), \quad \sin(iw) = i\sinh(w)$$

Therefore,

$$\cosh(z+w) = \cosh(z)\cosh(w) - i\sinh(z)i\sinh(w) = \cosh(z)\cosh(w) + \sinh(z)\sinh(w)$$

The proof of (4) is similar. □

The complex hyperbolic functions differ from their real counterparts in several important ways.

Remark 3.4.9. In real analysis, the functions $\sinh(x)$ and $\cosh(x)$ are not periodic. In contrast, the complex hyperbolic functions are periodic in the imaginary direction. Indeed, from the definitions

$$\sinh(z + 2\pi i) = \frac{e^{z+2\pi i} - e^{-(z+2\pi i)}}{2} = \frac{e^z - e^{-z}}{2} = \sinh(z)$$

and similarly, $\cosh(z + 2\pi i) = \cosh(z)$. Thus, both $\sinh(z)$ and $\cosh(z)$ have period $2\pi i$.

Remark 3.4.10. In real analysis, $\cosh(z)$ has no real zeros and $\sinh(z)$ has exactly one real zero at $x = 0$. In contrast, the complex functions $\sinh(z)$ and $\cosh(z)$ have infinitely many zeros in $\mathbb{C}$. This follows from the trigonometric relations in Theorem 3.4.6 together with the known zeros of $\sin(z)$ and $\cos(z)$.

Theorem 3.4.11. *(1) The zeros of* $\sinh(z)$ *are* $z = n\pi i$ *for* $n \in \mathbb{Z}$.
(2) The zeros of $\cosh(z)$ *are* $z = \frac{(2n+1)\pi i}{2}$ *for* $n \in \mathbb{Z}$.

Proof. From Theorem 3.4.6, we have $\sinh(z) = -i\sin(iz)$. Therefore, $\sinh(z) = 0$ if and only if $\sin(iz) = 0$. By Theorem 3.3.13, $iz = n\pi$ if and only if $z = -in\pi$. Since n ranges over all integers, this is equivalent to $z = n\pi i$ for $n \in \mathbb{Z}$, so the zeros of $\sinh(z)$ are exactly the numbers $n\pi i$. The proof of (2) is similar. □

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Chapter 4

Integration in the Complex Plane

Integration is one of the central ideas in analysis, and in complex analysis it becomes even more powerful than in the real-variable setting. In real calculus, integration is often introduced as accumulation along an interval or as the signed area under a curve. In the complex plane, however, integration is performed along paths or contours, and this geometric viewpoint leads to some of the most remarkable results in mathematics.

This chapter develops the theory of integration for complex-valued functions of a complex variable. Although the definition of a complex integral resembles the familiar notion of a line integral from multivariable calculus, the consequences are dramatically stronger. In particular, when a function is holomorphic, its integrals exhibit extraordinary rigidity: many contour integrals vanish, integrals often depend only on endpoints, and the values of a holomorphic function inside a region can be completely recovered from its values along the boundary.

We begin with a brief review of real integrals and line integrals, emphasizing the parametrized viewpoint that will be essential for complex integration. This section serves as a bridge between single-variable and multivariable integration and the new complex setting.

We define complex integrals along curves in the complex plane and establish their basic properties. We will see how to parametrize contours, compute integrals explicitly, and interpret these integrals in terms of their real and imaginary parts.

The heart of the chapter the Cauchy–Goursat Theorem, one of the foundational results of complex analysis. This theorem shows that the

integral of a holomorphic function around a closed contour is zero under suitable conditions. Unlike in real-variable calculus, this powerful vanishing result is not an isolated fact but a gateway to an entire theory.

We explore path independence, which reveals that under appropriate hypotheses, the value of a complex integral depends only on the endpoints of the path and not on the particular route taken. This connects naturally to the existence of antiderivatives and gives a complex analogue of the Fundamental Theorem of Calculus.

Finally, we arrive at the celebrated Cauchy Integral Formulas, which show that the values of a holomorphic function—and even all of its derivatives—are determined entirely by contour integrals. These formulas are among the most important results in the subject and form the basis for many later developments, including power series expansions, Liouville's Theorem, the Maximum Modulus Principle, and residue theory.

4.1 Review of Real and Line Integration

Before defining integration of complex-valued functions along curves in the complex plane, it is helpful to review the real-variable and multivariable integration ideas on which the theory is built. In ordinary calculus, the definite integral is introduced as a limit of Riemann sums and is often interpreted geometrically as signed area. In multivariable calculus, this viewpoint is extended to line integrals, where one integrates a function along a curve rather than over an interval.

These ideas are the natural foundation for complex integration. A complex contour integral is, in essence, a special type of line integral in the plane, but one whose algebraic and geometric consequences are much richer. Thus, this section reviews the definitions and computational methods for definite integrals and line integrals in the plane, together with the terminology of curves that will be used throughout the chapter.

We begin with the definite integral of a real-valued function on an interval, emphasizing its definition as a limit of Riemann sums. We then introduce curves in the plane, together with notions such as smoothness, piecewise smoothness, simplicity, and orientation. After that, we define the three most common types of line integrals in the plane:

- with respect to x,
- with respect to y,
- and with respect to arc length s.

Finally, we discuss practical methods of evaluation, both for parametri-

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cally defined curves and for curves given explicitly as graphs $y = f(x)$. These ideas will be used directly in the next section when we define and study complex integrals.

We begin by recalling the definition of the define integral from single-variable calculus.

Definition 4.1.1. Let $f : [a, b] \to \mathbb{R}$ be a function. A function $F : [a, b] \to \mathbb{R}$ is called an *antiderivative of* f on $[a, b]$ if $F'(x) = f(x)$ for all $x \in [a, b]$.

Theorem 4.1.2 (**Fundamental Theorem of Calculus, Part II**). *If f is continuous on $[a, b]$ and F is an antiderivative of f, then*

$$\int_a^b f(x)\ dx = F(b) - F(a)$$

Remark 4.1.3. The formula above is a method of evaluation, not the actual definition of the definite integral. The true definition is given in terms of limits of sums.

Definition 4.1.4. A *partition* $\mathcal{P}$ of the interval $[a, b]$ is a finite collection of points

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

The interval $[a, b]$ is thereby divided into subintervals $[x_{k-1}, x_k]$ for $k = 1, 2, ..., n$.

Definition 4.1.5. For a partition $\mathcal{P}$, the *length* of the kth subinterval is

$$\Delta x_k = x_k - x_{k-1}$$

Definition 4.1.6. The *norm* of a partition $\mathcal{P}$, denoted $\|\mathcal{P}\|$ is defined by

$$\|\mathcal{P}\| = \max_{1 \le k \le n} \Delta x_k$$

Thus, $\|\mathcal{P}\|$ is the length of the longest subinterval.

Definition 4.1.7. Let $f : [a, b] \to \mathbb{R}$ be a function and let $\mathcal{P}$ be a partition of $[a, b]$. Choose a sample point $x_k^* \in [x_{k-1}, x_k]$ for each $k = 1, 2, ..., n$. Then the corresponding *Riemann sum* is

$$\sum_{k=1}^{n} f(x_k^*)\Delta x_k$$

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Definition 4.1.8. Let $f : [a, b] \to \mathbb{R}$. We say that f *is integrable* on $[a, b]$ if the limit

$$\lim_{\|\mathcal{P}\| \to 0} \sum_{k=1}^{n} f(x_k^*) \Delta x_k$$

exists and is independent of the choice of partition and sample points. In this case, the *definite integral* on $[a, b]$ is defined by

$$\int_a^b f(x)\ dx = \lim_{\|\mathcal{P}\| \to 0} \sum_{k=1}^{n} f(x_k^*) \Delta x_k$$

Theorem 4.1.9. *Every continuous function $f : [a, b] \to \mathbb{R}$ is integrable on $[a, b]$.*

We will assume this fact throughout.

Example 4.1.10. Evaluate

$$\int_{-1}^{2} x^2\ dx$$

Solution. An antiderivative of x^2 is

$$F(x) = \frac{x^3}{3}$$

Hence, by the Fundamental Theorem of Calculus (Theorem 4.1.2)

$$\int_{-1}^{2} x^2\ dx = \left[\frac{x^3}{3}\right]_{-1}^{2} = \frac{8}{3} - \left(-\frac{1}{3}\right) = 3$$

The definite integral accumulates values of a function over an interval of the real line. A natural generalization is to accumulate values of a function along a curve in the plane. To do this carefully, we first need some terminology about curves.

Definition 4.1.11. A *curve* C in the plane is given by a pair of parametric equations $x = x(t)$, $y = y(t)$, $a \leq t \leq b$, where $x(t)$ and $y(t)$ are real-valued functions. The point $(x(t), y(t))$ is the point on the curve corresponding to the parameter value t.

Definition 4.1.12. If C is a curve defined by $x = x(t)$ and $y = y(t)$ on $a \leq t \leq b$, then the *initial point* of C is $A = (x(a), y(a))$, and the *terminal point* of C is $B = (x(b), y(b))$.

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Definition 4.1.13. If C is a curve defined by $x = x(t)$ and $y = y(t)$ on $a \leq t \leq b$ is called *smooth* if

(1) $x'(t)$ and $y'(t)$ are continuous on $[a, b]$, and
(2) $x'(t)$ and $y'(t)$ are not simultaneously zero for all $t \in (a, b)$.

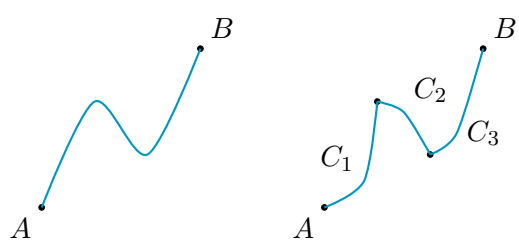
In other words, $(x'(t), y'(t)) \neq (0, 0)$ for all $t \in (a, b)$.

This condition ensures that the curve has a well-defined tangent direction and no "stopping point" in the interior.

Definition 4.1.14. A curve C is called *piecewise smooth* if it consists of finitely many smooth curves joined end to end. That is, there exist smooth curves C_1, C_2, ..., C_m such that the terminal point of C_k is the initial point of C_{k+1} for each k.

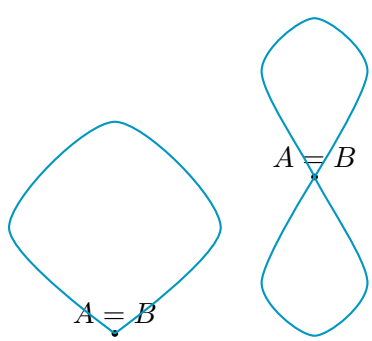
Definition 4.1.15. A curve C is called *simple* if it does not cross itself, except possibly at its endpoints.

Thus, distinct parameter values correspond to distinct points of the curve, except possibly when the initial and terminal points coincide.



Definition 4.1.16. A curve C is called *closed* if its initial point and terminal points are the same.

Definition 4.1.17. A curve C is called a *simple closed curve* if it is both *simple*, and *closed*.



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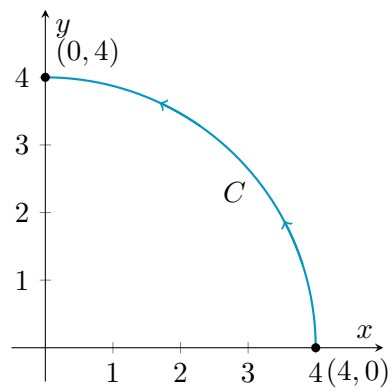
A simple closed curve encloses a region of the plane and plays a fundamental role in complex analysis.

Example 4.1.18. Consider the parametrized curve $x(t) = 4\cos(t)$ and $y(t) = 4\sin(t)$ for $0 \le t \le \frac{\pi}{2}$. Then the initial point is $(4, 0)$ and the terminal point is $(0, 4)$. The curve is the quarter-circle of radius 4 in the first quadrant.

Since $x'(t) = -4\sin(t)$ and $y'(t) = 4\cos(t)$ and these are continuous and never simultaneously zero on $\left(0, \frac{\pi}{2}\right)$, the curve is smooth.



Let $G(x, y)$ be a real-valued function defined on a region of the plane, and let C be a smooth or piecewise smooth curve. There are three standard types of integrals.

(1) $\displaystyle\int_C G(x, y)\ dx$

(2) $\displaystyle\int_C G(x, y)\ dy$

(3) $\displaystyle\int_C G(x, y)\ ds$

where ds denotes differential arc length. These are all defined as limits of sums, analogous to the Riemann integral.

Definition 4.1.19. Suppose C is the curve defined by the parametric equations $x = x(t)$ and $y = y(t)$ on $a \le t \le b$. Let $\mathcal{P}$ be a partition of the parameter interval $[a, b]$. This partition induces a subdivision of the curve C into subarcs. For each subarc, define $\Delta x_k = x(t_k) - x(t_{k-1})$ and $\Delta y_k = y(t_k) - y(t_{k-1})$, and let Δs_k denote the arc length of the kth subarc. Choose a sample point (x_k^*, y_k^*) on each subarc.

(1) The *line integral of G along C with respect to x* is

$$\int_C G(x, y)\ dx = \lim_{\|\mathcal{P}\| \to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*) \Delta x_k$$

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provided the limit exists.

(2) The *line integral of G along C with respect to y* is

$$\int_C G(x,y)\ dy = \lim_{\|\mathcal{P}\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\Delta y_k$$

provided the limit exists.

(3) The *line integral of G along C with respect to arc length* is

$$\int_C G(x,y)\ ds = \lim_{\|\mathcal{P}\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\Delta s_k$$

provided the limit exists.

Theorem 4.1.20. *If G is continuous on a smooth curve C, then all three integrals*

$$\int_C G(x,y)\ dx, \quad \int_C G(x,y)\ dy, \quad \int_C G(x,y)\ ds$$

exist.

We assume this result throughout.

The abstract definitions above are important conceptually, but in practice we evaluate line integrals by converting them into ordinary definite integrals.

Suppose C is the curve defined by the parametric curves $x = x(t)$ and $y = y(t)$ on the interval $a \le t \le b$. Assume that C is a smooth parametrization of the curve. Then $dx = x'(t)\ dt$ and $dy = y'(t)\ dt$. Also, the differential arc length is

$$ds = \sqrt{x'(t)^2 + y'(t)^2}\ dt$$

This yields the following theorem.

Theorem 4.1.21. *Suppose C is the curve defined by the parametric curves $x = x(t)$ and $y = y(t)$ on the interval $a \le t \le b$. Assume that C is a smooth parametrization of the curve. Then*

(1) $\displaystyle\int_C G(x,y)\ dx = \int_a^b G(x(t), y(t))x'(t)\ dt$

(2) $\displaystyle\int_C G(x,y)\ dy = \int_a^b G(x(t), y(t))y'(t)\ dt$

(3) $\displaystyle\int_C G(x,y)\ ds = \int_a^b G(x(t), y(t))\sqrt{x'(t)^2 + y'(t)^2}\ dt$

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Proof. We prove the formula for (1); the others are analogous.

By definition, we have

$$\int_C G(x,y)\ dx = \lim_{\|\mathcal{P}\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\Delta x_k$$

Now if (x_k^*, y_k^*) corresponds to some $t_k^* \in [t_{k-1}, t_k]$, then $x_k^* = x(t_k^*)$, $y_k^* = y(t_k^*)$, and $\Delta x_k = x(t_k) - x(t_{k-1})$.

By the Mean Value Theorem, this difference behaves like $x'(t_k^*)\Delta t_k$, so the sum becomes

$$\sum_{k=1}^{n} G(x(t_k^*), y(t_k^*))x'(t_k^*)\Delta t_k$$

which is precisely a Riemann sum $\int_a^b G(x(t), y(t))x'(t)\ dt$. Hence,

$$\int_C G(x,y)\ dx = \int_a^b G(x(t), y(t))x'(t)\ dt$$

□

Example 4.1.22. Evaluate the following line integrals along the quarter circle defined in Example 4.1.10 for the function $G(x,y) = xy^2$

(1) $\int_C xy^2\ dx$
(2) $\int_C xy^2\ dy$
(3) $\int_C xy^2\ ds$

Solution. We first compute $x'(t) = -4\sin(t)$ and $y'(t) = 4\cos(t)$. Also,

$$ds = \sqrt{(-4\sin(t))^2 + (4\cos(t))^2} = \sqrt{16\sin^2(t) + 16\cos^2(t)} = 4$$

Furthermore,

$$G(x(t), y(t)) = (4\cos(t))(4\sin(t))^2 = 4\cos(t)\cdot 16\sin^2(t) = 64\cos(t)\sin^2(t)$$

(1) By Theorem 4.1.20,

$$\begin{aligned}\int_C xy^2\ dx &= \int_0^{\frac{\pi}{2}} 64\sin^2(t)\cos(t)(-4\sin(t))\ dt \\ &= -256\int_0^{\frac{\pi}{2}} \sin^3(t)\cos(t)\ dt\end{aligned}$$

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Let $u = \sin(t)$. Then $du = \cos(t)\ dt$. Then

$$\begin{aligned}\int_C xy^2\ dx &= -256 \int_0^1 u^3\ du \\ &= -256 \left[\frac{u^4}{4}\right]_0^1 \\ &= -256 \cdot \frac{1}{4} \\ &= -64\end{aligned}$$

(2) Again, by Theorem 4.1.20,

$$\begin{aligned}\int_C xy^2\ dy &= \int_0^{\frac{\pi}{2}} 64 \sin^2(t)\cos(t)(4\cos(t))\ dt \\ &= 256 \int_0^{\frac{\pi}{2}} \sin^2(t)\cos^2(t)\ dt\end{aligned}$$

Using the identity

$$\sin^2(t)\cos^2(t) = \frac{1}{2}\sin^2(2t) = \frac{1}{4}(1 - \cos(4t))$$

so

$$\begin{aligned}\int_C xy^2\ dy &= 32 \int_0^{\frac{\pi}{2}} (1 - \cos(4t))\ dt \\ &= 32 \left[t - \frac{1}{4}\sin(4t)\right]_0^{\frac{\pi}{2}} \\ &= 32 \cdot \frac{\pi}{2} = 16\end{aligned}$$

(3) By Theorem 4.1.20,

$$\begin{aligned}\int_C xy^2\ ds &= \int_0^{\frac{\pi}{2}} 64 \sin^2(t)\cos(t)(4)\ dt \\ &= 256 \int_0^{\frac{\pi}{2}} \sin^2(t)\cos(t)\ dt\end{aligned}$$

Let $u = \sin(t)$ so then $du = \cos(t)\ dt$. Then

$$\int_C xy^2 = 256 \int_0^1 u^2\ du$$

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$$= 256 \left[\frac{u^3}{3} \right]_0^1$$
$$= \frac{256}{3}$$

Sometimes, the curve C is given explicitly as the graph of a function $y = f(x)$ with $a \leq x \leq b$. In that case, it is natural to use x itself as the parameter. Then $y = f(x)$ and $dy = f'(x)\ dx$. Also, the differential arc length is

$$ds = \sqrt{1 + f'(x)^2}\ dx$$

This gives the following formulas.

Theorem 4.1.23. *Let C be the graph of a smooth function $y = f(x)$ on $a \leq x \leq b$. Then for a continuous function $G(x, y)$,*

(1) $\displaystyle\int_C G(x, y)\ dx = \int_a^b G(x, f(x))\ dx$

(2) $\displaystyle\int_C G(x, y)\ dy = \int_a^b G(x, f(x))f'(x)\ dx$

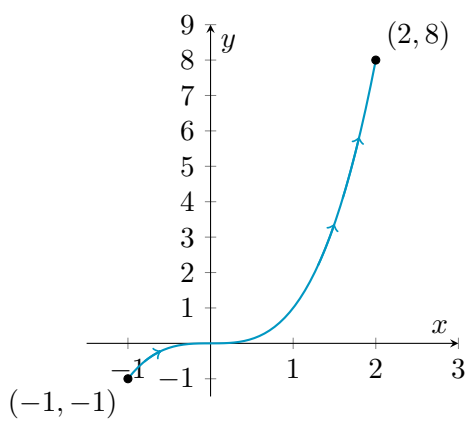
(3) $\displaystyle\int_C G(x, y)\ ds = \int_a^b G(x, f(x))\sqrt{1 + f'(x)^2}\ dx.$

Proof. Take the parametrization $x = t$, $y = f(t)$, $a \leq t \leq b$. Then $x'(t) = 1$ and $y'(t) = f'(t)$ and $ds = \sqrt{1 + f'(x)^2}\ dt$. Substituting these into the formulas of Theorem 4.1.20 gives the result immediately. □

Example 4.1.24. Evaluate

$$\int_C xy\ dx + x^2\ dy$$

where C is the graph of $y = x^3$ on $-1 \leq x \leq 2$.



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Solution. Since $y = x^3$, we have $dy = 3x^2\ dx$. Thus,

$$\begin{aligned}\int_C xy\ dx + x^2\ dy &= \int_{-1}^{2} x(x^3)\ dx + \int_{-1}^{2} x^2(3x^2)\ dx \\ &= \int_{-1}^{2} x^4\ dx + \int_{-1}^{2} 3x^4\ dx \\ &= \int_{-1}^{2} 4x^4\ dx \\ &= 4\left[\frac{x^5}{5}\right]_{-1}^{2} \\ &= \frac{132}{5}\end{aligned}$$

Many curves arise as the union of simpler smooth pieces.

Definition 4.1.25. If a piecewise smooth curve C is the union of smooth subcurves $C_1, C_2, ..., C_m$, traversed consecutively, then line integrals over C are defined by summing the integrals over each piece

(1) $\displaystyle\int_C G(x,y)\ dx = \sum_{j=1}^{m} \int_{C_j} G(x,y)\ dx$

(2) $\displaystyle\int_C G(x,y)\ dy = \sum_{j=1}^{m} \int_{C_j} G(x,y)\ dy$

(3) $\displaystyle\int_C G(x,y)\ ds = \sum_{j=1}^{m} \int_{C_j} G(x,y)\ ds$

Example 4.1.26. Evaluate

$$\oint_C y^2\ dx - x^2\ dy$$

where C is the closed curve consisting of the following three pieces:

(1) C_1: from $(0,0)$ to $(2,0)$ along the x-axis.
(2) C_2: from $(2,0)$ to $(2,4)$ vertically.
(3) C_3: from $(2,4)$ back to $(0,0)$ along the parabola $y = x^2$.

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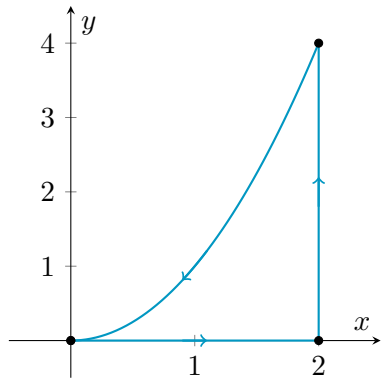

Solution. Since $C = C_1 \cup C_2 \cup C_3$, then

$$\oint_C y^2\ dx - x^2\ dy = \int_{C_1} y^2\ dx - x^2\ dy + \int_{C_2} y^2\ dx - x^2\ dy + \int_{C_3} y^2\ dx - x^2\ dy$$

On C_1, we have $y = 0$, $0 \leq x \leq 2$, and $dy = 0$, so

$$\int_{C_1} y^2\ dx - x^2\ dy = \int_0^2 0\ dx - \int_0^2 x^2(0) = 0$$

On C_2, we have $x = 2$, y goes from 0 to 4, and $dx = 0$. Thus,

$$\int_{C_2} y^2\ dx - x^2\ dy = \int_0^4 y^2(0) - 4\ dy = -\int_0^4 4\ dy = -16$$

On C_3, the curve is $y = x^2$, but it is traversed from $(2, 4)$ to $(0, 0)$, so x goes from 2 to 0. Also, $dy = 2x\ dx$. Hence,

$$\begin{aligned}\int_{C_3} y^2\ dx - x^2\ dy &= \int_2^0 (x^2)^2\ dx - \int_2^0 x^2(2x)\ dx \\ &= \int_2^0 (x^4 - 2x^3)\ dx \\ &= \left[\frac{x^5}{5} - \frac{x^4}{2}\right]_2^0 \\ &= \frac{8}{5}\end{aligned}$$

Therefore,

$$\oint_C y^2\ dx - x^2\ dy = 0 + (-16) + \frac{8}{5} = -\frac{72}{5}$$

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When integrating along a curve, the direction in which the curve is traversed matters.

Definition 4.1.27. An *orientation* of a curve C is a choice of direction along the curve. If a curve is traversed from its initial point A to its terminal point B, we call this the *positive orientation*.

The same geometric curve traversed in the opposite direction is denoted by $-C$, and is called the curve with opposite orientation.

Theorem 4.1.28. *Let C be an oriented smooth or piecewise smooth curve. Then*

(1) $\displaystyle\int_{-C} G(x,y)\ dx = -\int_C G(x,y)\ dx$

(2) $\displaystyle\int_{-C} G(x,y)\ dy = -\int_C G(x,y)\ dy,$

(3) $\displaystyle\int_{-C} P(x,y)\ dx + Q(x,y)\ dy = -\int_C P(x,y)\ dx + Q(x,y)\ dy$

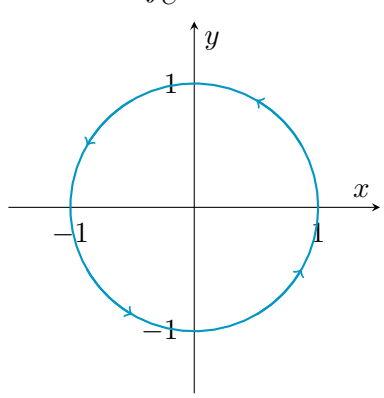
Proof. Suppose C is parametrized by $x = x(t)$, $y = y(t)$ for $a \le t \le b$. Then $-C$ is parametrized by reversing the parameter $x = x(t)$, $y = y(t)$, and $b \ge t \ge a$. Equivalently, set $u = a + b - t$. Then $du = -dt$, so the line integral over $-C$ becomes the negative of the corresponding integral over C. For example

$$\int_{-C} G(x,y)\,dx = \int_b^a G(x(t),y(t))x'(t)\,dt = -\int_a^b G(x(t),y(t))x'(t)\,dt = -\int_C G(x,y)\,dx$$

The other formulas are proven similarly. □

Example 4.1.29. Let C be the unit circle $x = \cos(t)$, $y = \sin(t)$, $0 \le t \le 2\pi$. Evaluate

$$\oint_C x\ dx$$



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Solution. Since $dx = -\sin(t)\ dt$, we have

$$\begin{aligned}\oint_C x\ dx &= \int_0^{2\pi} \cos(t)(-\sin(t))\ dt \\ &= -\int_0^{2\pi} \cos(t)\sin(t)\ dt \\ &= -\frac{1}{2}\int_0^{2\pi} \sin(2t)\ dt \\ &= 0\end{aligned}$$

It is common to write

$$\int_C P(x,y)\ dx + Q(x,y)\ dy$$

more compactly as

$$\int_C P\ dx + Q\ dy$$

If C is a closed, curve, one often writes

$$\oint_C P\ dx + Q\ dy$$

to emphasize that the path is closed.

4.2 Complex Integrals

In the previous section, we reviewed real integrals and line integrals in the plane. Those ideas now serve as the foundation for one of the central objects in complex analysis: the complex integral, also called a contour integral. Just as a real line integral accumulates values of a function along a curve in the plane, a complex integral accumulates values of a complex-valued function along a curve in the complex plane.

At first glance, the definition of a complex integral looks very similar to the definition of a real line integral. However, because both the function and the variable are complex-valued, contour integrals combine geometric information about the path with algebraic structure coming from complex multiplication. This added structure is what gives complex integration its extraordinary power.

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In this section, we first revisit curves in the complex plane and describe how a parametrized curve can be written in the form

$$z(t) = x(t) + iy(t)$$

We then define the complex integration of a function along a contour C as a limit of sums, in direct analogy with the Riemann integral. After that, we derive the practical formula

$$\int_C f(z)\ dz = \int_a^b f(z(t))z'(t)\ dt$$

which is the main tool for evaluating contour integrals in practice.

We also study the basic algebraic properties of contour integrals, including linearity, additivity over piecewise smooth curves, and the effect of reversing orientation. Finally, we prove an important estimate, the ML-inequality, which gives an upper bound for the magnitude of a contour integral in terms of the size of the integrand and the length of the contour.

We begin by translating the language of parametrized curves from $\mathbb{R}^2$ into the language of complex numbers.

Definition 4.2.1. Let $x = x(t)$ and $y = y(t)$ be defined on $a \leq t \leq b$, where $x(t)$ and $y(t)$ are real-valued continuous functions. Then the corresponding curve C in the complex plane is described by

$$z = z(t) = x(t) + iy(t)$$

for $a \leq t \leq b$. The function $z(t)$ is called a *parametrization* of the curve C.

Definition 4.2.2. If $z = z(t)$, $a \leq t \leq b$ parametrizes a curve, then the *initial point of* C is $z(a) = x(a) + iy(a)$, and the *terminal point of* C is $z(b) = x(b) + iy(b)$.

Definition 4.2.3. A curve C parametrized by $z(t) = x(t) + iy(t)$, $a \leq t \leq b$ is called *smooth* if $z'(t)$ is continuous on $[a, b]$ and $z'(t) \neq 0$ for all $t \in [a, b]$.

Since $z'(t) = x'(t) + iy'(t)$, this is equivalent to saying that $x'(t)$ and $y'(t)$ are continuous and not simultaneously zero.

Definition 4.2.4. A curve C is called *piecewise smooth* if it is the union of finitely many smooth curves joined end to end.

Definition 4.2.5. A curve C is called *simple* if $z(t_1) \neq z(t_2)$ whenever $t_1 \neq t_2$, except possibly when $t_1 = a$ and $t_2 = b$. Thus, a simple curve does not intersect itself, except possibly at its endpoints.

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Definition 4.2.6. A curve C is called *closed* if $z(a) = z(b)$.

Definition 4.2.7. A curve C is called a *simple closed curve* if it is both simple and closed.

Definition 4.2.8. A piecewise smooth curve is often called a *contour* or *path*.

Definition 4.2.9. Let $z = z(t)$, $a \leq t \leq b$ parametrize a contour C. The direction corresponding to increasing values of t is called the *positive orientation of* C.

If the same geometric curve is traversed in the opposite direction, it is denoted by $-C$ and is called the contour with *opposite orientation*.

For a simple closed curve, the positive orientation is usually taken to be the counterclockwise direction, so that the interior of the curve remains to the left as the curve is traversed.

Example 4.2.10. The unit circle may be parametrized by

$$z(t) = e^{it} = \cos(t) + i\sin(t)$$

for $0 \leq t \leq 2\pi$. This is a smooth simple closed curve with positive (counterclockwise) orientation.

We now define integration of a complex-valued function along a contour.

Definition 4.2.11. Let C be parametrized by $z = z(t)$ for $a \leq t \leq b$.

(1) A *partition* $\mathcal{P}$ of $[a, b]$ is a finite set of points

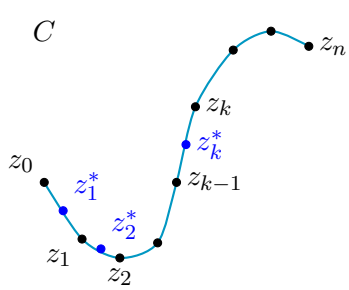
$$a = t_0 < t_1 < \cdots < t_n = b$$

This partition induces points on the contour $z_l = z(t_k)$ for $k = 0, 1, 2, ..., n$.

(2) We write $\Delta z_k = z_k - z_{k-1}$ for $k = 1, 2, ..., n$.

(3) The *norm* of the partition is

$$\|\mathcal{P}\| = \max_{1 \leq i \leq n} (t_k - t_{k-1})$$



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Definition 4.2.12. Let f be a complex-valued function defined on a contour C, where C is parametrized by $z = z(t)$ for $a \leq t \leq b$. Choose a point z_k^* on each subarc of C, determined by the partition. The *complex integral of f over C* is defined by

$$\int_C f(z)\ dz = \lim_{\|\mathcal{P}\| \to 0} \sum_{k=1}^{n} f(z_k^*) \Delta z_k$$

provided this limit exists. This is also called the *contour integral* of f over C.

Remark 4.2.13. This definition is formally analogous to the definition of a line integral in the plane. The difference is that both $f(z)$ and Δz_k are now complex numbers.

Before evaluating contour integrals, it is useful to recall how one integrates a complex-valued function of a real variable.

Definition 4.2.14. Let $f : [a, b] \to \mathbb{C}$ is called a *complex-valued function* of a real variable if for each $t \in [a, b]$,

$$f(t) = f_1(t) + i f_2(t)$$

where f_1 and f_2 are real-valued functions on $[a, b]$.

Definition 4.2.15. If f_1 and f_2 are continuous on $[a, b]$, then f is continuous on $[a, b]$, and we define

$$\int_a^b f(t)\ dt = \int_a^b f_1(t)\ dt + i \int_a^b f_2(t)\ dt$$

Example 4.2.16. Evaluate

$$\int_0^1 (2t + i)^2\ dt$$

Solution. First expand

$$(2t + i)^2 = 4t^2 + 4it + i^2 = 4t^2 - 1 + 4it$$

Therefore,

$$\int_0^1 (2t + i)^2\ dt = \int_0^1 (4t^2 - 1)\ dt + i \int_0^1 4t\ dt$$

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$$= \left[\frac{4^3}{3} - t\right]_0^1 + i\left[2t^2\right]_0^1$$
$$= \frac{1}{3} + 2i$$

Theorem 4.2.17. *Let f and g be continuous complex-valued functions on $[a, b]$ and let $\alpha \in \mathbb{C}$. Then*

(1) $\displaystyle\int_a^b \alpha f(t)\ dt = \alpha \int_a^b f(t)\ dt$

(2) $\displaystyle\int_a^b (f(t) + g(t))\ dt = \int_a^b f(t)\ dt + \int_a^b g(t)\ dt$

(3) If $c \in [a, b]$, then $\displaystyle\int_a^b f(t)\ dt = \int_a^c f(t)\ dt + \int_c^b f(t)\ dt$

(4) $\displaystyle\int_b^a f(t)\ dt = -\int_a^b f(t)\ dt$

Proof. Write $f = f_1 + if_2$ and $g = g_1 + ig_2$, where f_1, f_2, g_1, g_2 are real-valued continuous functions on $[a, b]$. Then each statement follows immediately from the corresponding property of real integrals applied to the real and imaginary parts separately. □

We now derive the practical formula for evaluating contour integrals. Suppose $f(z) = u(x, y) + iv(x, y)$ where u and v are real-valued functions, and suppose $z(t) = x(t) + iy(t)$ for $a \leq t \leq b$. Then

$$dz = dx + idy$$

Formally,

$$f(z)\ dz = (u + iv)(dx + idy) = (u\ dx - v\ dy) + i(v\ dx + u\ dy)$$

Thus, we are led to the identity

$$\int_C f(z)\ dz = \int_C (u\ dx - v\ dy) + i\int_C (v\ dx + u\ dy)$$

Theorem 4.2.18. *Let f be continuous on a smooth contour C given by $z = z(t)$ for $a \leq t \leq b$. Then*

$$\int_C f(z)\ dz = \int_a^b f(z(t))z'(t)\ dt$$

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Proof. Let $z(t) = x(t) + iy(t)$ and $f(z) = u(x, y) + iv(x, y)$. Then $z'(t) = x'(t) + iy'(t)$, so

$$f(z(t))z'(t) = (u + iv)(x' + iy') = (ux' - vy') + i(vx' + uy')$$

Therefore,

$$\int_a^b f(z(t))z'(t)\ dt = \int_a^b (ux' - vy')\ dt + i \int_a^b (vx' + uy')\ dt$$

Using $dx = x'(t)\ dt$ and $dy = y'(t)\ dt$, we have

$$\int_a^b f(z(t))z'(t)\ dt = \int_C (u\ dx - v\ dy) + i \int_C (v\ dx + u\ dy) = \int_C f(z)\ dz$$

This proves the formula. □

Remark 4.2.19. This theorem shows that contour integrals are evaluated in essentially the same way as line integrals: substitute the parametrization $z = z(t)$, compute $z'(t)$, and integrate with respect to t.

Example 4.2.20. Evaluate

$$\int_C \bar{z}\ dz$$

where C is the contour $x = 3t$, $y = t^2$, $-1 \leq t \leq 4$.

Solution. The parametrization is $z(t) = 3t + it^2$, so $\bar{z}(t) = 3t - it^2$ and $z'(t) = 3 + 2it$. By Theorem **??**,

$$\begin{aligned}
\int_C \bar{z}\ dz &= \int_{-1}^4 (3t - it^2)(3 + 2it)\ dt \\
&= \int_{-1}^4 (2t^3 + 9t)\ dt + i \int_{-1}^4 3t^2\ dt \\
&= \left[\frac{1}{2}t^4 + \frac{9}{2}t^2\right]_{-1}^4 + i\left[t^3\right]_{-1}^4 \\
&= 195 + 65i
\end{aligned}$$

Example 4.2.21. Evaluate

$$\oint_C \frac{1}{z}\ dz$$

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where C is the unit circle $z(t) = e^{it}$, $0 \leq t \leq 2\pi$.

Solution. Since $z(t) = e^{it}$ with $0 \leq t \leq 2\pi$, we have $z'(t) = ie^{it}$ and $\frac{1}{z(t)} = e^{-it}$. Thus,

$$\oint_C \frac{1}{z}\, dz = \int_0^{2\pi} e^{-it}(ie^{it})\, dt = \int_0^{\pi} i\, dt = 2\pi i$$

Example 4.2.22. Evaluate

$$\int_C (8x^2 - iy)\, dz$$

where C is the line segment $y = 5x$, $0 \leq x \leq 2$, or equivalently $z = x + 5ix$ for $0 \leq x \leq 2$.

Solution. Using x as the parameter, we write

$$z(x) = x + 5ix = (1 + 5i)x$$

so

$$dz = (1 + 5i)\, dx$$

Since $y = 5x$, then

$$8x^2 - iy = 8x^2 - 5ix$$

Therefore,

$$\begin{aligned}
\int_C (8x^2 - iy)\, dz &= \int_0^2 (8x^2 - 5ix)(1 + 5i)\, dx \\
&= (1 + 5i) \int_0^2 (8x^2 - 5ix)\, dx \\
&= (1 + 5i) \left[\frac{8}{3}x^3 - 5i\frac{x^2}{2}\right]_0^2 \\
&= (1 + 5i)\left(\frac{64}{3} - 10i\right) \\
&= \frac{214}{3} + \frac{290}{3}i
\end{aligned}$$

We now state some basic algebraic properties of contour integrals.

Theorem 4.2.23. *Suppose f and g are continuous in a domain D, and C is a smooth curve lying entirely in D. Then*

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(1) For any $\alpha \in \mathbb{C}$,

$$\int_C \alpha f(z)\ dz = \alpha \int_C f(z)\ dz$$

(2)

$$\int_C (f(z) + g(z))\ dz = \int_C f(z)\ dz + \int_C g(z)\ dz$$

(3) If C consists of two smooth curves C_1 and C_2 joined end to end, then

$$\int_C f(z)\ dz = \int_{C_1} f(z)\ dz + \int_{C_2} f(z)\ dz$$

(4) If $-C$ denotes the same curve with opposite orientation, then

$$\int_{-C} f(z)\ dz = -\int_C f(z)\ dz$$

These statements hold when C is piecewise smooth.

Proof. To see that (1) is true, using Theorem 4.2.18, we have

$$\int_C \alpha f(z)\ dz = \int_a^b \alpha f(z(t))z'(t)\ dt = \alpha \int_a^b f(z(t))z'(t)\ dt = \alpha \int_C f(z)\ dz$$

To see that (2) is true, using Theorem 4.2.18, we have

$$\begin{aligned}\int_C (f(z) + g(z))\ dz &= \int_a^b (f(z(t)) + g(z(t))z'(t)\ dt \\ &= \int_a^b f(z(t))z'(t)\ dt + \int_a^b g(z(t))z'(t)\ dt \\ &= \int_C f(z)\ dz + \int_C g(z)\ dz\end{aligned}$$

To see that (3) is true, suppose C is obtained by traversing C_1 and then C_2. If the first corresponds to $a \leq t \leq c$ and the second to $c \leq t \leq b$, then

$$\begin{aligned}\int_C f(z)\ dz &= \int_a^b f(z(t))z'(t)\ dt \\ &= \int_a^c f(z(t))z'(t)\ dt + \int_c^b f(z(t))z'(t)\ dt \\ &= \int_{C_1} f(z)\ dz + \int_{C_2} f(z)\ dz\end{aligned}$$

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To see that (4) is true, if C is parametrized by $z = z(t)$, $a \leq t \leq b$, then $-C$ is parametrized by the same curve with reversed parameter. Hence,

$$\int_{-C} f(z)\ dz = \int_b^a f(z(t))z'(t)\ dt = -\int_a^b f(z(t))z'(t)\ dt = -\int_C f(z)\ dz$$

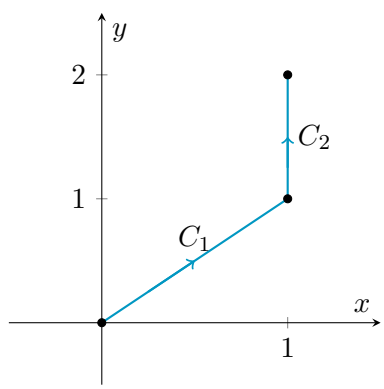
□

Example 4.2.24. Let C be the piecewise smooth contour consisting of

- C_1: the line segment from 0 to $1+i$ given by $y = x$.
- C_2: the vertical line segment from $1+i$ to $1+2i$ given by $x = 1$.

Evaluate

$$\int_C (x^2 + iy^2)\ dz$$



Solution. By additivity

$$\int_C (x^2 + iy^2)\ dz = \int_{C_1} (x^2 + iy^2)\ dz + \int_{C_2} (x^2 + iy^2)\ dz$$

We first integrate over C_1. Since $y = x$, use x as a parameter

$$z(x) = x + ix$$

for $0 \leq x \leq 1$, so $dz = (1+i)\ dx$, and so

$$x^2 + iy^2 = x^2 + ix^2 = (1+i)x^2$$

Hence,

$$\int_{C_1} (x^2 + iy^2)\ dz = \int_0^1 (1+i)x^2(1+i)\ dx$$

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$$
\begin{aligned}
&= (1+i)^2 \int_0^1 x^2 \, dx \\
&= 2i \left[\frac{x^3}{3} \right]_0^1 \\
&= \frac{2i}{3}
\end{aligned}
$$

Now we integrate over C_2. Since $x = 1$, use y as a parameter

$$z(y) = 1 + iy$$

for $1 \leq y \leq 2$, so $dz = i \, dy$. Also, $x^2 + iy^2 = 1 + iy^2$. Therefore

$$
\begin{aligned}
\int_{C_2} (x^2 + iy^2) \, dz &= \int_1^2 (1 + iy^2) i \, dy \\
&= \int_1^2 (i - y^2) \, dy \\
&= \left[iy - \frac{y^3}{3} \right]_1^2 \\
&= i - \frac{7}{3}
\end{aligned}
$$

Hence, combining gives

$$\int_C (x^2 + iy^2) \, dz = \frac{2i}{3} + \left(i - \frac{7}{3} \right) = -\frac{7}{3} + \frac{5i}{3}$$

We now prove an important estimate.

Definition 4.2.25. If a smooth contour C is parametrized by $z = z(t) = x(t) + iy(t)$ for $a \leq t \leq b$, then its length is

$$L = \int_a^b |z'(t)| \, dt$$

Since

$$|z'(t)| = \sqrt{x'(t)^2 + y'(t)^2}$$

this agrees with the usual arc length formula from calculus.

Theorem 4.2.26 (ML-Inequality). *Let f be continuous on a smooth contour C. If $|f(z)| \leq M$ for all $z \in C$ and L is the length of C, then*

$$\left| \int_C f(z) \, dz \right| \leq ML$$

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Proof. Let $z = z(t)$ for $a \leq t \leq b$ be a parametrization of C. Then by Theorem 4.2.18,

$$\int_C f(z)\ dz = \int_a^b f(z(t))z'(t)\ dt$$

Taking absolute values and applying the triangle inequality for integrals

$$\left|\int_C f(z)\ dz\right| \leq \int_a^b |f(z(t))||z'(t)|\ dt$$

Since $|f(z(t))| \leq M$ for all t, it follows that

$$\left|\int_C f(z)\ dz\right| \leq \int_a^b M|z'(t)|\ dt = M\int_a^b |z'(t)|\ dt = ML$$

This proves the inequality. □

Remark 4.2.27. This is often called the ML-inequality, where M is an upper bound for $|f(z)|$ on C, and L is the length of C. It is one of the most useful tools for estimating contour integrals.

Example 4.2.28. Find an upper bound for

$$\left|\oint_C \frac{e^z}{z+1}\ dz\right|$$

where C is the circle $|z| = 4$.

Solution. We apply the ML-inequality. First, the contour C is a circle of radius 4, so its length is

$$L = 2\pi(4) = 8\pi$$

Next, for $z \in C$, we have $|z| = 4$. By the lower triangle inequality, we have

$$|z+1| \geq ||z| - 1| = 4 - 1 = 3$$

Therefore,

$$\left|\frac{e^z}{z+1}\right| \leq \frac{|e^z|}{3}$$

Now,

$$|e^z| = e^x$$

where $x = \mathrm{Re}(z)$. On the unit circle $|z| = 4$, we have $x \leq 4$, so $|e^z| \leq e^4$. Hence,

$$\left|\frac{e^z}{z+1}\right| \leq \frac{e^4}{3}$$

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Thus, take $M = \frac{e^4}{3}$. By the ML-inequality

$$\left| \oint_C \frac{e^z}{z+1}\, dz \right| \leq ML = \frac{e^4}{3} \cdot 8\pi$$

Remark 4.2.29. A contour can have many different parametrizations. For example, the unit circle $|z| = 1$ with positive orientation may be parametrized by $z(t) = e^{it}$ for $0 \leq t \leq 2\pi$ or by $z(t) = e^{2\pi it}$ for $0 \leq t \leq 1$, or by other equivalent parametrizations.

What matters is that the parametrization traces out the same curve with the same orientation. The value of the contour integral does not depend on which such parametrization is chosen.

4.3 The Cauchy–Goursat Theorem

In the previous section, we introduced contour integrals and developed the basic tools needed to compute them. We now arrive at one of the most important turning points in complex analysis: the discovery that contour integrals of holomorphic functions behave in a remarkably rigid way. In real-variable calculus, line integrals generally depend strongly on the path taken. In complex analysis, however, holomorphicity imposes such strong structure that, under suitable conditions, the integral of a function around a closed contour is actually zero.

This fundamental fact is expressed by the Cauchy–Goursat Theorem. It is one of the central results of the subject and serves as the foundation for many deeper theorems, including path independence, Cauchy's Integral Formula, Liouville's Theorem, and the development of residue theory. The theorem shows that holomorphicity is not merely a local differentiability condition: it has powerful global consequences for integration.

We begin by introducing the distinction between simply connected and multiply connected domains, since the geometry of the domain plays an essential role in determining when contour integrals vanish. We then state and prove a version of Cauchy's theorem using Green's theorem and the Cauchy–Riemann equations. This proof requires slightly stronger hypotheses than the modern version. We then state the more standard Cauchy–Goursat Theorem, which removes the extra smoothness assumption on f'.

After establishing the theorem for simply connected domains, we extend the discussion to multiply connected domains, where the integral around one contour can be related to the integrals around inner boundary components. This leads naturally to the principle of deformation of contours, which allows

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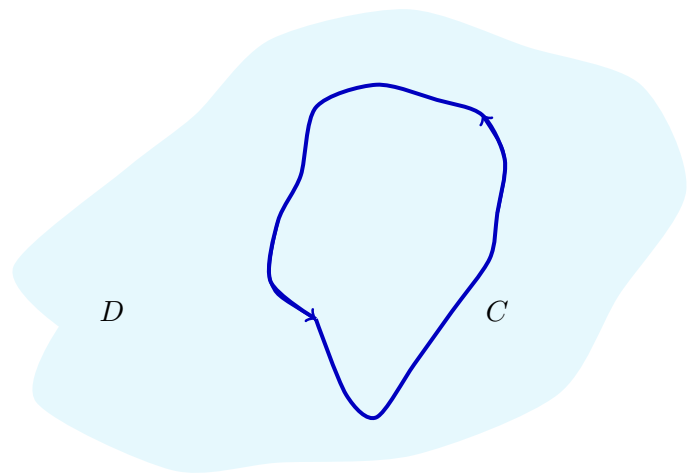
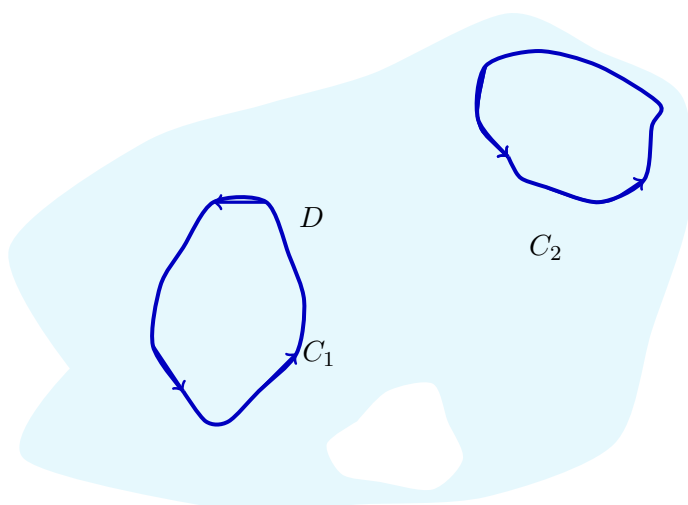
one to replace a complicated contour by a simpler one without changing the value of the integral, provided no singularities are crossed in the process.

Definition 4.3.1. A domain D is called *simply connected* if every simple contour C lying entirely in D can be shrunk continuously to a point without leaving D. A domain that is not simply connected is called *multiply connected*.





Intuitively, a simply connected domain has no holes. Examples of simply connected domain include

- the entire complex plane,
- any open disk,

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- any rectangle
- any open-half plane.

Example 4.3.2. The annulus $A(0,1,2) = \{z \in \mathbb{C} : 1 < |z| < 2\}$ is not simply connected, because a circle centered at the origin cannot be shrunk to a point without leaving the annulus.

We now state one of the central theorems of the subject.

Theorem 4.3.3 (**Cauchy's Theorem**)**.** *Suppose that f is holomorphic in a simply connected domain D, and suppose also that f' is continuous in D. Then for every simple closed contour C lying in D,*

$$\oint_C f(z)\ dz = 0$$

Remark 4.3.4. Historically, this theorem was proved by Cauchy under the assumption that f' is continuous. Later, Goursat showed that continuity of f' is actually not necessary. The modern version is therefore stronger and is usually called the *Cauchy–Goursat Theorem*.

To prove Theorem 4.3.3, we use Green's Theorem from multivariable calculus.

Theorem 4.3.5 (**Green's Theorem**)**.** *Let C be a positively oriented, piecewise smooth, simple closed curve bounding a region $R \subseteq \mathbb{R}^2$. Suppose P and Q have continuous first partial derivatives on an open set containing $R \cup C$. Then*

$$\oint_C P\ dx + Q\ dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\ dA$$

Proof of Theorem 4.3.3. We first prove the classical version assuming f' is continuous. Let

$$f(z) = u(x,y) + iv(x,y)$$

where u and v are real-valued functions. By Theorem 4.2.18, we have

$$\oint_C f(z)\ dz = \oint_C (u\ dx - v\ dy) + i \oint_C (v\ dx + u\ dy)$$

Let R be the region enclosed by C. We apply Green's Theorem (Theorem 4.3.5) separately to the two real integrals.

For the first integral, take $P = u$ and $Q = -v$. Then

$$\oint_C (u\ dx - v\ dy) = \iint_R (-v_x - u_y)\ dA$$

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For the second integral, take $P = v$ and $Q = u$. Then

$$\oint_C (v\ dx + u\ dy) = \iint_R (u_x - v_y)\ dA$$

Therefore, combining both real integrals, we get

$$\oint_C f(z)\ dz = \iint_R (-v_x - u_y)\ dA + i \iint_R (u_x - v_y)\ dA$$

Since f is holomorphic in D, the Cauchy–Riemann equations hold

$$u_x = v_y, \quad u_y = -v_x$$

Substituting these into the previous expression gives

$$-v_x - u_y = -v_x + v_x = 0$$

and $u_x - v_y = 0$. Therefore,

$$\oint_C f(z)\ dz = \iint_R 0\ dA + i \iint_R 0\ dA = 0$$

as desired. □

As noted above, the continuity of f' is not actually needed

Theorem 4.3.6 (**Cauchy–Goursat Theorem**). *If f is holomorphic in a simply connected domain D. Then for every simple closed contour C in D,*

$$\oint_C f(z)\ dz = 0$$

Since the interior of a simple closed contour is a simply connected region, the theorem is often used in the following equivalent form.

Corollary 4.3.7. *If f is holomorphic at every point within and on a simple closed contour C, then*

$$\oint_C f(z)\ dz = 0$$

This is the version used most often in computations.

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Example 4.3.8. Evaluate

$$\oint_C e^z \, dz$$

where C is any simple closed contour.

Solution. The function $f(z) = e^z$ is entire, so it is holomorphic everywhere in $\mathbb{C}$. In particular, it is holomorphic on and inside C. Therefore, by the Cauchy–Goursat Theorem (Theorem 4.3.3),

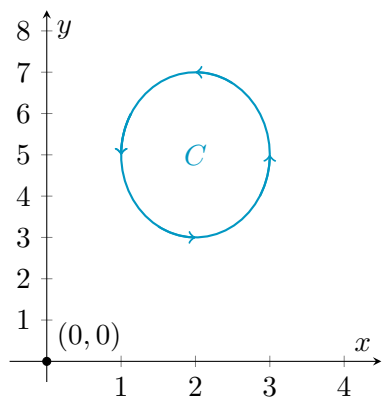
$$\oint_C e^z \, dz = 0$$

Example 4.3.9. Evaluate

$$\oint_C \frac{1}{z^2} \, dz$$

where C is the ellipse

$$(x-2)^2 + \frac{1}{4}(y-5)^2 = 1$$



Solution. The function is holomorphic for all $z \neq 0$. The ellipse is centered at $2 + 5i$, so the point $z = 0$ lies outside the contour and its interior. Hence, f is holomorphic on and inside C, and therefore

$$\oint_C \frac{1}{z^2} \, dz = 0$$

The above theorem requires holomorphicity on and inside the contour. In multiply connected domains, there may be holes where holomorphicity fails, so the integral around a closed contour need not be zero. However, one can still obtain a very useful relation between the outer boundary and the boundaries of the holes.

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Theorem 4.3.10 (**Cauchy–Goursat Theorem, Doubly Connected Domain**). *Let D be a doubly connected domain bounded externally by a positively oriented simple closed contour C and internally by a positively oriented simple closed contour C_1. Suppose f is holomorphic on and between these contours. Then*

$$\oint_C f(z)\ dz = \oint_{C_1} f(z)\ dz$$

Proof. Let R be the region between C and C_1, and choose a smooth path AB joining a point $A \in C$ to a point $B \in C_1$, lying entirely in R.

If we cut the region open along AB, the resulting region becomes simply connected. Its boundary consists of

- the outer contour C,
- one traversal of the cut AB,
- the inner contour C_1 with opposite geometric placement but induced orientation,
- and the reverse traversal $-AB$.

Since f is holomorphic throughout this simply connected cut region, the Cauchy–Goursat Theorem (Theorem 4.3.3) gives

$$\oint_C f(z)\ dz + \int_{AB} f(z)\ dz + \int_{-AB} f(z)\ dz + \oint_{C_1} f(z)\ dz = 0$$

But

$$\int_{-AB} f(z)\ dz = -\int_{AB} f(z)\ dz$$

so the two cut integrals cancel. Hence,

$$\oint_C f(z)\ dz + \oint_{C_1} f(z)\ dz$$

With the standard positive orientation convention for both boundary curves (outer boundary counterclockwise, inner boundary also taken positively with respect to its own enclosed region), this becomes

$$\oint_C f(z)\ dz = \oint_{C_1} f(z)\ dz$$

□

Theorem 4.3.10 leads to an important principle.

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Corollary 4.3.11 (Principle of Deformation of Contours). *If two simple closed contours C and C_1 lie in a region where f is holomorphic, and one can be continuously deformed into the other without crossing any point where f fails to be holomorphic, then*

$$\oint_C f(z)\ dz = \oint_{C_1} f(z)\ dz$$

This is called the principle of deformation of contours.

The value of the contour integral does not depend on the exact shape of the contour, only on the singularities enclosed, and the orientation. This principle is extremely useful because it allows us to replace a complicated contour by a simpler one.

We now apply Corollary 4.3.11 to evaluate one of the most important contour integrals in complex analysis.

Example 4.3.12. Suppose z_0 is a point interior to a simple closed contour C. Evaluate

$$\oint_C \frac{1}{z - z_0}\ dz$$

Solution. Since the integrand is holomorphic everywhere except at $z = z_0$, we may deform the contour C to a small circle C_1 centered at z_0, as long as the deformation does not cross z_0.

Take C_1 to be the contour $z = z_0 + re^{it}$ on $0 \le t \le 2\pi$. Then $dz = ire^{it}\ dt$ and $z - z_0 = re^{it}$. Therefore,

$$\begin{aligned}
\oint_C \frac{1}{z - z_0}\ dz &= \oint_{C_1} \frac{1}{z - z_0}\ dz \\
&= \int_0^{2\pi} \frac{ire^{it}}{re^{it}}\ dt \\
&= \int_0^{2\pi} i\ dt \\
&= 2\pi i
\end{aligned}$$

Theorem 4.3.13. *If z_0 is interior to a simple closed contour C, then for every $n \in \mathbb{N}$,*

$$\oint_C \frac{1}{(z - z_0)^n}\ dz = \begin{cases} 2\pi i & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$$

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Proof. For $n = 1$, this is exactly Example 4.3.12.

Now suppose $n > 1$. Then

$$\frac{1}{(z-z_0)^n} = -\frac{1}{n-1}\frac{d}{dz}\left(\frac{1}{(z-z_0)^{n-1}}\right)$$

so the integrand is the derivative of a function holomorphic away from z_0. Alternatively, deform C to a small circle $z = z_0 + re^{it}$. Then

$$dz = ire^{it}\ dt, \quad (z-z_0)^n = r^n e^{int}$$

and thus,

$$\oint_C \frac{1}{(z-z_0)^n}\ dz = \int_0^{2\pi} \frac{ire^{it}}{r^n e^{int}}\ dt = ir^{1-n}\int_0^{2\pi} e^{i(1-n)t}\ dt$$

Since $n > 1$, the exponent $1 - n \neq 0$, and therefore,

$$\int_0^{2\pi} e^{i(1-n)t}\ dt = 0.$$

Hence,

$$\oint_C \frac{1}{(z-z_0)^n}\ dz = 0$$

for $n > 1$. □

Example 4.3.14. Evaluate

$$\oint_C \frac{5z+7}{z^2+2z-3}\ dz$$

where C is the circle $|z-2| = 2$.

Solution. First factor the denominator

$$z^2 + 2z - 3 = (z-1)(z+3)$$

So

$$\frac{5z+7}{z^2+2z-3} = \frac{A}{z-1} + \frac{B}{z+3}$$

Solving for A and B, we obtain

$$\frac{5z+7}{z^2+2z-3} = \frac{3}{z-1} + \frac{2}{z+3}$$

Hence,

$$\oint_C \frac{5z+7}{z^2+2z-3}\ dz = 3\oint_C \frac{1}{z-1}\ dz + 2\oint_C \frac{1}{z+3}\ dz$$

Now the determine which singularities lie inside the contour $|z-2| = 2$

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- $z = 1$ is inside since $|1 - 2| = 1 < 2$
- $z = -3$ is outside, since $|-3 - 2| = 5 > 2$.

Therefore,

$$\oint_C \frac{1}{z-1}\, dz = 2\pi i, \quad \oint_C \frac{1}{z+3}\, dz = 0$$

Therefore

$$\oint_C \frac{5z+7}{z^2+2z-3}\, dz = 3(2\pi i) + 2(0) = 6\pi i$$

We now state the general version for a domain with several holes.

Theorem 4.3.15 (Cauchy–Goursat Theorem, Multiply Connected Domains). *Suppose C, $C_1, C_2, ..., C_n$ are simple closed contours with positive orientation such that each C_k lies interior of C, the regions interior to the C_k are pairwise disjoint, and f is holomorphic on the region inside C but outside all the C_k, as well as on each contour. Then*

$$\oint_C f(z)\, dz = \sum_{k=1}^{n} \oint_{C_k} f(z)\, dz$$

This is obtained by introducing crosscuts that connect the outer contour to each inner contour, reducing the multiply connected region to a simply connected one and then applying the Cauchy–Goursat Theorem.

Example 4.3.16. Evaluate

$$\oint_C \frac{1}{z^2+1}\, dz$$

where C is the circle $|z| = 4$.

Solution. Factor the denominator

$$z^2 + 1 = (z - i)(z + i)$$

Thus,

$$\frac{1}{z^2+1} = \frac{1}{2i}\left(\frac{1}{z-i} - \frac{1}{z-i}\right)$$

Hence

$$\oint_C \frac{1}{z^2+1}\, dz = \frac{1}{2i}\left(\oint_C \frac{1}{z-i}\, dz - \oint_C \frac{1}{z+i}\, dz\right)$$

Both singularities $z = i$ and $z = -i$ lie inside the circle $|z| = 4$. Therefore,

$$\oint_C \frac{1}{z-i}\, dz = 2\pi i, \quad \oint_C = \frac{1}{z+i}\, dz = 2\pi i$$

So

$$\oint_C \frac{1}{z^2+1}\, dz = \frac{1}{2i}(2\pi i - 2\pi i) = 0$$

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4.4 Path Independence

In real-variable calculus, one of the most powerful results is the Fundamental Theorem of Calculus, which tells us that if a function f has an antiderivative F, then

$$\int_a^b f(x)\ dx = F(b) - F(a)$$

so the value of the integral depends only on the endpoints and not on the path (interval) taken between them.

In contrast, line integrals in the plane generally depend on the specific path connecting two points. However, in complex analysis, a remarkable phenomenon occurs: for holomorphic functions, contour integrals often depend only on the endpoints and not on the path. This property is known as path independence, and it is a direct consequence of the Cauchy–Goursat Theorem.

In this section, we formalize the notion of path independence for contour integrals and establish its deep connection with holomorphicity. We show that if a function is holomorphic in a simply connected domain, then its contour integrals are independent of the path. This leads naturally to the concept of an antiderivative in the complex setting and culminates in the Fundamental Theorem for Contour Integrals, which mirrors its real-variable counterpart.

In this section, we formalize the notion of path independence for contour integrals and establish its deep connection with holomorphicity. We show that if a function is holomorphic in a simply connected domain, then its contour integrals are independent of the path. This leads naturally to the concept of an antiderivative in the complex setting and culminates in the Fundamental Theorem for Contour Integrals, which mirrors its real-variable counterpart.

Definition 4.4.1. Let z_0 and z_1 be two points in a domain D. A contour integral

$$\int_C f(z)\ dz$$

is said to be *path independent* if its value is the same for all contours C in D with initial point z_0 and terminal point z_1.

This means that the integral depends only on the endpoints z_0 and z_1, and not on the particular path connecting them.

The key result of this section follows directly from the Cauchy–Goursat Theorem (Theorem 4.3.6).

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Theorem 4.4.2. *If f is holomorphic in a simply connected domain D, and if C is any contour in D, then*

$$\int_C f(z)\ dz$$

is independent of the path of C.

Proof. Let C_1 and C_2 be two contours in D with the same initial point z_0 and terminal point z_1. Form a closed contour by traversing C_1 from z_0 to z_1, and then traversing C_2 in reverse from z_1 back to z_0. Then

$$\int_{C_1} f(z)\ dz + \int_{-C_2} f(z)\ dz = \oint_C f(z)\ dz$$

where $C = C_1 \cup (-C_2)$ is a closed contour.

Since f is holomorphic in a simply connected domain, the Cauchy–Goursat Theorem (Theorem 4.3.6) gives

$$\oint_C f(z)\ dz = 0$$

Therefore,

$$\int_{C_1} f(z)\ dz = \int_{C_2} f(z)\ dz$$

Hence, the integral is independent of the path. □

Example 4.4.3. Evaluate

$$\int_C 2z\ dz$$

where C is any contour from $z = -1$ to $z = -1 + i$.

Solution. Since $f(z) = 2z$ is entire, the integral is path independent. Choose the straight-line path $z(t) = -1 + it$ for $0 \le t \le 1$. Then $dz = i\ dt$ and

$$\begin{aligned}\int_C 2z\ dz &= \int_0^1 2(-1+it)i\ dt \\ &= \int_0^1 (-2i - 2t)\ dt \\ &= -2i\int_0^1\ dt - 2\int_0^1 t\ dt \\ &= -2i - 1\end{aligned}$$

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Definition 4.4.4. If f is continuous on a domain D, then a function F is called an *antiderivative* of f in D if

$$F'(z) = f(z)$$

for all $z \in D$.

Example 4.4.5. Since $(z^2)' = 2z$, an antiderivative of $f(z) = 2z$ is $F(z) = z^2$.

Theorem 4.4.6 (**Fundamental Theorem of Calculus, Contour Integration**). *Suppose that f is continuous on a domain D and has an antiderivative F in D. Let C be any contour in D with initial point z_0 and terminal point z_1. Then*

$$\int_C f(z)\ dz = F(z_1) - F(z_0)$$

Proof. Let C be parametrized by $z(t)$, $a \leq t \leq b$ with $z(a) = z_0$ and $z(b) = z_1$. Then

$$\int_C f(z)\ dz = \int_a^b f(z(t))z'(t)\ dt$$

Since $F'(z) = f(z)$, we have

$$f(z(t))z'(t) = \frac{d}{dt}F(z(t))$$

by the chain rule. Hence,

$$\int_C f(z)\ dz = \int_a^b \frac{d}{dt}F(z(t))\ dt = F(z(b)) - F(z(a))$$

Therefore,

$$\int_C f(z)\ dz = F(z_1) - F(z_0)$$

□

Example 4.4.7. Evaluate

$$\int_C 2z\ dz$$

from $z = -1$ to $z = -1 + i$.

Solution. Using $F(z) = z^2$,

$$\int_C 2z\ dz = \left[z^2\right]_{-1}^{-1+i} = -2i - 1$$

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Example 4.4.8. Evaluate

$$\int_C \cos(z)\ dz$$

from $z = 0$ to $z = 2 + i$.

Solution. Since $\sin'(z) = \cos(z)$, we take $F(z) = \sin(z)$. Then

$$\int_C \cos(z)\ dz = \sin(2+i) - \sin(0) = \sin(2+i)$$

Theorem 4.4.9. *If f is continuous in a domain D and $\int_C f(z)\ dz$ is independent of the path in D, then f has an antiderivative in D.*

Proof. Fix a point $z_0 \in D$. Define

$$F(z) = \int_{z_0}^{z} f(s)\ ds$$

where the integral is taken over any path in D. Since the integral is path independent, $F(z)$ is well-defined. We will show that $F'(z) = f(z)$. Consider

$$\frac{F(z+\Delta z) - F(z)}{\Delta z} = \frac{1}{\Delta z}\int_{z}^{z+\Delta z} f(s)\ ds$$

As $\Delta z \to 0$, continuity of f implies

$$\frac{F(z+\Delta z) - F(z)}{\Delta z} \to f(z)$$

Thus, $F'(z) = f(z)$, so F is an antiderivative of f. □

Theorem 4.4.10. *If f is holomorphic in a simply connected domain, then f has an antiderivative in D.*

This follows from holomorphicity, which gives us path independence, and path independence gives the existence of an antiderivative by Theorem 4.4.9.

Example 4.4.11. Evaluate

$$\int_C \frac{1}{z}\ dz$$

where C lies in the domain $x > 0$ and $y > 0$.

Solution. In this domain, we can define

$$\mathrm{Ln}(z) = \log|z| + i\arg(z)$$

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with $0 < \arg(z) < \frac{\pi}{2}$. Then

$$\frac{d}{dz}(\mathrm{Ln}(z)) = \frac{!}{z}$$

Thus,

$$\int_C \frac{1}{z}\, dz = \mathrm{Ln}(z_1) - \mathrm{Ln}(z_0)$$

Remark 4.4.12. If the domain contains a loop around 0, then $\frac{1}{z}$ has no global antiderivative, and path independence fails.

Example 4.4.13. Evaluate

$$\int_C \frac{1}{z^{\frac{1}{2}}}\, dz$$

where C lies in a domain where a branch of $z^{\frac{1}{2}}$ is defined.

Solution. Take $f(z) = z^{-\frac{1}{2}}$. Then an antiderivative is $F(z) = 2z^{\frac{1}{2}}$. Thus,

$$\int_C \frac{1}{z^{\frac{1}{2}}}\, dz = \left[2z^{\frac{1}{2}}\right]_{z_0}^{z_1}$$

Corollary 4.4.14. *If f has an antiderivative in D, then for any closed contour C in D,*

$$\oint_C f(z)\, dz = 0$$

Proof. If C is closed, then $z_0 = z_1$, so

$$\int_C f(z)\, dz = F(z_1) - F(z_0) = 0$$

□

4.5 Cauchy's Integral Formulas

In the previous sections, we developed the theory of contour integration and established the Cauchy–Goursat Theorem, which showed that the integral of an holomorphic function around a closed contour is zero. We also saw that holomorphic functions give rise to path-independent integrals and admit antiderivatives in simply connected domains.

We now arrive at one of the most powerful and far-reaching results in complex analysis: Cauchy's Integral Formulas. These formulas provide an explicit representation of an holomorphic function in terms of its values

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along a surrounding contour. Remarkably, they show that the value of an holomorphic function at a point is completely determined by its values on any simple closed contour enclosing that point.

Even more striking is the fact that these formulas extend to all derivatives of holomorphic functions. As a consequence, holomorphic functions are infinitely differentiable and possess a level of smoothness far beyond that of general real-valued functions. This leads to deep results such as Liouville's Theorem, the Fundamental Theorem of Algebra, and important inequalities governing holomorphic functions.

In this section, we first derive Cauchy's Integral Formula, then extend it to derivatives, and finally explore several powerful consequences. These results form the foundation for much of the remainder of complex analysis, including series expansions and residue theory.

Theorem 4.5.1 (**Cauchy's Integral Formula**). *Suppose that f is holomorphic in a simply connected domain D, and let C be any simple closed contour lying entirely within D. If z_0 is any point inside C, then*

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0}\, dz$$

Proof. Let C be a simple closed contour and z_0 a point in its interior. By Corollary 4.3.11, we may replace C by a small circle C_1 centered at z_0, lying entirely within C. Thus,

$$\oint_C \frac{f(z)}{z - z_0}\, dz = \oint_{C_1} \frac{f(z)}{z - z_0}\, dz$$

Now write

$$f(z) = f(z_0) + (f(z) - f(z_0))$$

Then

$$\oint_{C_1} \frac{f(z)}{z - z_0}\, dz = f(z_0) \oint_{C_1} \frac{1}{z - z_0}\, dz + \oint_{C_1} \frac{f(z) - f(z_0)}{z - z_0}\, dz$$

By Theorem 4.3.13,

$$\oint_{C_1} \frac{1}{z - z_0}\, dz = 2\pi i$$

Since f is continuous, for sufficiently small C_1, the quantity

$$\left| \frac{f(z) - f(z_0)}{z - z_0} \right|$$

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can be made arbitrarily small. Using the ML-inequality (Theorem 4.2.26), the second integral tends to zero. Hence,

$$\oint_C \frac{f(z)}{z - z_0}\, dz = 2\pi i f(z_0)$$

Therefore,

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0}\, dz$$

□

Remark 4.5.2. This theorem shows that holomorphic functions are completely determined by their boundary values.

Example 4.5.3. Evaluate

$$\oint_C \frac{z^2 - 4z + 4}{z + i}\, dz$$

where C is the circle $|z| = 2$.

Solution. Rewrite

$$\frac{z^2 - 4z + 4}{z + i} = \frac{f(z)}{z - (-i)}$$

where $f(z) = z^2 - 4z + 4$. Since $z_0 = -i$ lies inside $|z| = -2$, Cauchy's Integral Formula gives

$$\oint_C \frac{f(z)}{z - z_0} = 2\pi i f(z_0)$$

Here,

$$f(-i) = (-i)^2 - 4(-i) + 4 = 3 + 4i$$

Therefore,

$$\oint_C \frac{f(z)}{z - z_0} = 2\pi i(3 + 4i)$$

Example 4.5.4. Evaluate

$$\oint_C \frac{z}{z^2 + 9}\, dz$$

where C is the circle $|z - 2i| = 4$.

Solution. Factor the denominator $z^2 + 9 = (z - 3i)(z + 3i)$. Then

$$\frac{z}{z^2 + 9} = \frac{z}{z + 3i} \cdot \frac{1}{z - 3i}$$

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Only $z = 3i$ lies inside the contour. Thus,

$$f(z) = \frac{z}{z+3i}$$

where $z_0 = 3i$. Computing

$$f(3i) = \frac{3i}{6i} = \frac{1}{2}$$

then

$$\oint_C \frac{z}{z^2+9}\, dz = 2\pi i \cdot \frac{1}{2} = \pi i$$

Theorem 4.5.5 (Cauchy's Integral Formula, Derivatives). *Suppose f is holomorphic in a simply connected domain D, and C is a simple closed contour in D. If z_0 is inside C, then for every integer $n \geq 1$,*

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^{n+1}}\, dz$$

Proof. We prove for the case when $n = 1$. Starting with

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

Apply Cauchy's formula to both terms

$$f(z_0 + \Delta z) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - (z_0 + \Delta z)}\, dz,$$

and so

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^2}\, dz$$

Higher derivatives follow by induction. □

Example 4.5.6. Evaluate

$$\oint_C \frac{z+1}{z^4 + 2iz^3}\, dz$$

where C is the circle $|z| = 1$.

Solution. Factor the denominator $z^4 + 2iz^3 = z^3(z + 2i)$. Only $z = 0$ lies inside the contour. Rewrite

$$\frac{z+1}{z^3(z+2i)} = \frac{f(z)}{z^3}$$

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where $f(z) = \frac{z+1}{z+2i}$. Then

$$\oint_C \frac{f(z)}{z^3}\, dz = \frac{2\pi i}{2!} f''(0)$$

Compute $f''(0)$ using the quotient rule, yielding

$$f''(0) = \frac{2-4i}{(2i)^3}$$

We now present some consequences of the Cauchy's integral formula.

Theorem 4.5.7. *If f is holomorphic in a domain D, then all derivatives f', f'', $f^{(3)}$,... exist and are holomorphic in D.*

Corollary 4.5.8. *Holomorphic functions are infinitely differentiable, unlike general real functions.*

Theorem 4.5.9 (**Cauchy's Inequality**)**.** *Let f be holomorphic in a domain D, and suppose C is the circle $|z - z_0| = r$ lying in D. If $|f(z)| \leq M$ on C, then*

$$|f^{(n)}(z_0)| \leq \frac{n!M}{r^n}$$

Proof. From Cauchy's derivative formula,

$$|f^{(n)}(z_0)| \leq \frac{n!}{2\pi} \oint_C \frac{|f(z)|}{|z - z_0|^{n+1}}\, |dz| \tag{4.1}$$

Using $|z - z_0| = r$ and the ML-inequality (Theorem 4.2.26),

$$|f^{(n)}(z_0)| \leq \frac{n!M}{r^n}$$

□

Theorem 4.5.10 (**Liouville's Theorem**)**.** *If f is entire and bounded, then f is constant.*

Proof. By Cauchy's inequality,

$$|f'(z_0)| \leq \frac{M}{r}$$

Let $r \to \infty$, then $f'(z_0) = 0$. Hence, f is constant. □

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Theorem 4.5.11 (**Fundamental Theorem of Algebra**). *Every nonconstant polynomial has at least one complex root.*

Proof. Suppose $p(z) \neq 0$ for all z. Then

$$f(z) = \frac{1}{p(z)}$$

is entire and bounded. By Liouville's Theorem, f is constant, so p is constant, a contradiction. □

Theorem 4.5.12 (**Morera's Theorem**). *If f is continuous in a simply connected domain D and*

$$\oint_C f(z)\ dz = 0$$

for every closed contour C in D, then f is holomorphic in D.

This is essentially the converse of the Cauchy–Goursat Theorem.

Theorem 4.5.13 (**Maximum Modulus Theorem**). *If f is holomorphic and nonconstant in a region R, then $|f(z)|$ attains its maximum on the boundary of R.*

Example 4.5.14. Find the maximum of $|2z + 5i|$ on $|z| \leq 2$.

Solution. Maximum occurs on $|z| = 2$, giving

$$\max |2z + 5i| = 9$$

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Chapter 5

Sequences and Series

In complex analysis, sequences and series play a central role in understanding how complex functions behave locally and globally. Just as in real analysis, sequences of complex numbers allow us to formalize the idea of convergence, while infinite series provide a way to represent complicated objects through simpler building blocks. However, in the complex setting, these ideas become significantly more powerful because holomorphic functions possess remarkable holomorphic structure.

We begin in Section 5.1 by studying sequences and series of complex numbers, extending familiar notions of convergence, absolute convergence, and geometric series from the real setting to the complex plane. Since a complex number has both real and imaginary parts, convergence in $\mathbb{C}$ is closely tied to convergence in $\mathbb{R}^2$, and many familiar results remain valid with elegant geometric interpretations.

In Section 5.2, we introduce Taylor series, one of the most important tools in complex analysis. A Taylor series expresses a holomorphic function as an infinite power series centered at a point, revealing that complex differentiability is far stronger than real differentiability. Unlike in real analysis, where infinitely differentiable functions need not equal their Taylor series, holomorphic functions are automatically holomorphic: locally, they are their Taylor expansions. This gives a powerful method for computation, approximation, and theoretical analysis.

We then move in Section 5.3 to Laurent series, which generalize Taylor series by allowing negative powers. Laurent expansions are especially useful for studying functions near isolated singularities, where ordinary power series may fail. They provide a precise description of local behaviour around problematic points and form the foundation for later topics such as residues

and contour integration.

Finally, in Section 5.4, we study zeros and poles, two fundamental types of local behaviour of complex functions. Zeros describe where a function vanishes, while poles describe controlled forms of divergence. Understanding the order and structure of zeros and poles is essential for classifying singularities, analyzing meromorphic functions, and developing the deeper theory of complex integration.

Together, the ideas in this chapter show one of the most beautiful features of complex analysis: holomorphic functions are governed by their local series expansions. By studying these expansions, we gain a powerful lens through which to understand the behaviour of complex functions.

5.1 Sequences and Series of Complex Numbers

The study of sequences and series in the complex plane is in many ways analogous to what one encounters in real analysis, but it also carries an important geometric interpretation. A sequence of complex numbers may be viewed as a collection of points in the complex plane, and convergence describes whether these points approach a fixed complex number. Similarly, an infinite series of complex numbers is understood through the convergence of its sequence of partial sums.

These ideas are foundational for complex analysis because many of the most important functions in the subject — including the exponential, trigonometric, logarithmic, and rational functions — can be represented by infinite series. In particular, power series will become one of the central tools of the next sections, eventually leading to Taylor series and Laurent series expansions.

In this section, we introduce the notions of convergence and divergence for sequences and infinite series of complex numbers. We also study geometric series, absolute convergence, and several standard tests for convergence such as the ratio test and the root test. These results will serve as the holomorphical foundation for the study of holomorphic functions later in the chapter.

Definition 5.1.1. A *sequence of complex numbers* is a function $z : \mathbb{N} \to \mathbb{C}$, where to each positive integer n there corresponds a complex number z_n.

We usually denote a sequence by $(z_n)_{n=!}^{\infty}$ or (z_n). Thus, a sequence is an ordered list of complex numbers $z_1, z_2, z_3, \ldots$.

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Example 5.1.2. Consider the sequence $(1+i^n)_{n=1}^{\infty}$. Its first few terms are

$$1+i, \quad 0, \quad 1-i, \quad 2, \quad 1+i, \quad 0, ...$$

Since the powers of i repeat in the cycle

$$i, -1, -i, 1, i, -1, -i, 1, ...$$

the sequence repeats every four terms.

Definition 5.1.3. Let $(z_n)_{n=1}^{\infty}$ be a sequence in $\mathbb{C}$, and let $L \in \mathbb{C}$. We say that $(z_n)_{n=1}^{\infty}$ converges to L, written

$$\lim_{n\to\infty} z_n = L, \quad z_n \to L$$

if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|z_n - L| < \varepsilon$.

If such an L exists, the sequence is called *convergent*. Otherwise, it is *divergent*.

The condition $|z_n - L| < \varepsilon$ means that the point z_n lies inside the open disk of radius ε centered at L. Thus, the sequence $(z_n)_{n=1}^{\infty}$ converges to L if eventually all of its terms lie inside every ε-neighbourhood of L.

Example 5.1.4. Consider the sequence

$$z_n = \frac{i^{n+1}}{n}$$

Its first few terms are

$$-1, \quad -\frac{i}{2}, \quad \frac{1}{3}, \quad \frac{i}{4}, \quad -\frac{1}{5}, ...$$

Since

$$\left|\frac{i^{n+1}}{n}\right| = \frac{|i^{n+1}|}{n} = \frac{1}{n} \to 0$$

it follows that

$$\lim_{n\to\infty} \frac{i^{n+1}}{n} = 0$$

Thus, the sequence converges to 0.

Theorem 5.1.5. *Let $z_n = a_n + ib_n$ and let $L = a + ib$. Then $z_n \to L$ if and only if $a_n \to a$ and $b_n \to b$.*

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Proof. Suppose first that $z_n \to L$. Then $|z_n - L| \to 0$, but

$$z_n - L = (a_n - a) + i(b_n - b)$$

so then $|a_n - a| \leq |z_n - L|$ and $|b_n - b| \leq |z_n - L|$. Since $|z_n - L| \to 0$, it follows that $a_n \to a$ and $b_n \to a$.

Conversely, suppose that $a_n \to a$ and $b_n \to b$. Then

$$|z_n - L| = \sqrt{(a_n - a)^2 + (b_n - b)^2}$$

Since both $a_n - a \to 0$ and $b_n - b \to 0$, we conclude that $|z_n - L| \to 0$. Hence, $z_n \to L$. □

Example 5.1.6. Determine whether the sequence

$$z_n = \frac{3 + ni}{n + 2ni}$$

converges.

Solution. Multiply numerator and denominator by the complex conjugate of the denominator

$$z_n = \frac{3 + ni}{n + 2ni} \cdot \frac{n - 2ni}{n - 2ni} = \frac{2n^2 + 3n}{5n^2} + i\frac{n^2 - 6n}{5n^2}$$

As $n \to \infty$, $\mathrm{Re}(z_n) \to \frac{2}{5}$ and $\mathrm{Im}(z_n) \to \frac{1}{5}$. Therefore,

$$\lim_{n\to\infty} z_n = \frac{2}{5} + \frac{1}{5}i$$

Definition 5.1.7. An *infinite series of complex numbers* is an expression of the form

$$\sum_{n=1}^{\infty} z_n = z_1 + z_2 + z_3 + \cdots$$

To determine whether this series makes sense, we study its *partial sums*.

Definition 5.1.8. The Nth partial sum of the series $\sum_{n=1}^{\infty} z_n$ is defined by

$$S_N = \sum_{n=1}^{N} z_n = z_1 + z_2 + \cdots + z_N$$

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Definition 5.1.9. The series $\sum_{n=1}^{\infty} z_n$ is said to *converge* if the sequence of partial sums $(S_N)_{N=1}^{\infty}$ converges to some complex number L. In that case, we write

$$\sum_{n=1}^{\infty} z_n = L$$

If $(S_N)_{N=1}^{\infty}$ does not converge, then the series is said to *diverge*.

Definition 5.1.10. A *geometric series* is a series of the form

$$\sum_{n=1}^{\infty} az^{n-1} = a + az + az^2 + az^3 + \cdots$$

where $a, z \in \mathbb{C}$.

Proposition 5.1.11. *For the geometric series $a + az + az^2 + \cdots$, the nth partial sum is*

$$S_N = a + az + az^2 + \cdots + az^{N-1}$$

and satisfies

$$S_N = \frac{a(1 - z^N)}{1 - z}$$

with $z \neq 1$.

Proof. Multiply S_N by z

$$zS_N = az + az^2 + \cdots + az^N$$

Subtract

$$\begin{aligned} S_N - zS_N &= (a + az + az^2 + \cdots + az^{N-1}) - (az + az^2 + az^3 + \cdots + az^N) \\ (1 - z)S_N &= a - az^N = a(1 - z^N) \\ S_N &= \frac{a(1 - z^N)}{1 - z} \end{aligned}$$

□

Theorem 5.1.12. *Let $\sum_{n=1}^{\infty} az^{n-1}$ be a geometric series. Then*

(1) If $|z| < 1$, then the series converges and

$$\sum_{n=1}^{\infty} az^{n-1} = \frac{a}{1 - z}$$

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(2) If $n \geq 1$, the series diverges.

Proof. From Proposition 5.1.11,

$$S_N = \frac{a(1 - z^N)}{1 - z}$$

If $|z| < 1$, then $z^n \to 0$, so

$$S_N \to \frac{a}{1 - z}$$

Thus, the series converges. If $|z| \geq 1$, then in general z^n does not end to zero, and hence, the partial sums do not converge. Therefore, the series diverges. □

Example 5.1.13. Determine whether $\sum_{n=1}^{\infty} \frac{(1+2i)^n}{5^n}$ converges, and if so, find its sum.

Solution. Rewrite the series as

$$\sum_{n=1}^{\infty} \left(\frac{1 + 2i}{5}\right)^n$$

This is geometric with common ratio $z = \frac{1+2i}{5}$. Its modulus is

$$|z| = \frac{|1 + 2i|}{5} = \frac{\sqrt{5}}{5} < 1$$

Hence, the series converges. Using Theorem 5.1.12,

$$\sum_{n=1}^{\infty} \left(\frac{1 + 2i}{5}\right)^n = \frac{\frac{1+2i}{5}}{1 - \frac{1+2i}{5}} = \frac{1 + 2i}{4 - 2i} = \frac{i}{2}$$

The geometric series formula yields several important expansions.

Proposition 5.1.14. *If $|z| < 1$, then*

$$\frac{1}{1 - z} = \sum_{n=1}^{\infty} z^{n-1}$$

Proof. Take $a = 1$ in the geometric series formula. □

Corollary 5.1.15. *If $|z| < 1$, then*

$$\frac{1}{1 + z} = \sum_{n=1}^{\infty} (-z)^{n-1}$$

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Proof. Replace z in $-z$ in the formula in Proposition 5.1.14. □

Proposition 5.1.16. *For $z \neq 1$,*

$$\sum_{k=1}^{n} z^{k-1} = \frac{1-z^n}{1-z}$$

Equivalently,

$$\frac{1}{1-z} = 1 + z + z^2 + \cdots + z^{n-1} + \frac{z^n}{1-z}$$

This identity is often useful in estimates and proofs.

Theorem 5.1.17. *If the series $\sum_{n=1}^{\infty} z_n$ converges, then*

$$\lim_{n\to\infty} z_n = 0$$

Proof. Let $S_N = \sum_{n=1}^{N} z_n$. If the series converges, then $S_N \to L$ for some $L \in \mathbb{C}$. But $z_n = S_N - S_{N-1}$. Taking limits

$$\lim_{n\to\infty} z_n = \lim_{N\to\infty} (S_N - S_{N-1}) = L - L = 0$$

□

Remark 5.1.18. The converse is not true. That is, even if $z_n \to 0$, the series $\sum_{n=1}^{\infty} z_n$ need not converge. This only gives a necessary condition, but not a sufficient one.

Theorem 5.1.19. *If $\lim_{n\to\infty} z_n \neq 0$ or does not exist, then $\sum_{n=1}^{\infty} z_n$ diverges.*

Proof. This is the contrapositive of Theorem 5.1.17. □

Example 5.1.20. Consider the series

$$\sum_{n=1}^{\infty} \frac{in+5}{n}$$

Then

$$z_n = \frac{in+5}{n} = i + \frac{5}{n}$$

Since $\lim_{n\to\infty} z_n = i \neq 0$, then by Theorem 5.1.19, the series diverges.

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Definition 5.1.21. A series $\sum_{n=1}^{\infty} z_n$ is said to be *absolutely convergent* if the real series $\sum_{n=1}^{\infty} |z_n|$ converges. The series is said to be *conditionally convergent* if $\sum_{n=1}^{\infty} z_n$ converges but $\sum_{n=1}^{\infty} |z_n|$ diverges.

Example 5.1.22. Consider

$$\sum_{n=1}^{\infty} \frac{i^n}{n^2}$$

Then $\left|\frac{i^n}{n^2}\right| = \frac{1}{n^2}$, and so

$$\sum_{n=1}^{\infty} \left|\frac{i^n}{n^2}\right| = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

which is a convergent p-series with $p = 2 > 1$. Hence,

$$\sum_{n=1}^{\infty} \frac{i^n}{n^2}$$

is absolutely convergent.

Theorem 5.1.23. *If a complex series converges absolutely, then it converges.*

Proof. Let $z_n = a_n + ib_n$. Then $|a_n| \leq |z_n|$ and $|b_n| \leq |z_n|$. If $\sum_{n=1}^{\infty} |z_n|$ converges, then by comparison, both $\sum_{n=1}^{\infty} |a_n|$ and $\sum_{n=1}^{\infty} |b_n|$ also converge. Therefore, $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge as real series. Hence,

$$\sum_{n=1}^{\infty} z_n = \sum_{n=1}^{\infty} (a_n + ib_n) = \sum_{n=1}^{\infty} a_n + i \sum_{n=1}^{\infty} b_n$$

converges in $\mathbb{C}$. □

We now record two important convergence tests for complex series.

Theorem 5.1.24 (The Ratio Test). *Let $\sum_{n=1}^{\infty} z_n$ be a series of nonzero complex numbers. Suppose*

$$\lim_{n\to\infty} \left|\frac{z_{n+1}}{z_n}\right| = L$$

Then

(1) If $L < 1$, the series converges absolutely.
(2) If $L > 1$ or $L = \infty$, the series diverges.
(3) If $L = 1$, the test is inconclusive.

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Proof. This follows directly from the corresponding ratio test for the real nonnegative series

$$\sum_{n=1}^{\infty} |z_n|$$

Indeed,

$$\left| \frac{|z_{n+1}|}{|z_n|} \right| = \left| \frac{z_{n+1}}{z_n} \right|$$

Thus, the convergence behaviour is determined by applying the real ratio test to $\sum_{n=1}^{\infty} |z_n|$. □

Theorem 5.1.25 (The Root Test). *Let $\sum_{n=1}^{\infty} z_n$ be a series of complex numbers such that*

$$\lim_{n\to\infty} \sqrt[n]{|z_n|} = L$$

Then

(1) If $L < 1$, the series converges absolutely.
(2) If $L > 1$ or $L = \infty$, the series diverges.
(3) If $L = 1$, the test is inconclusive.

Proof. Again, apply the real root test to the nonnegative real series $\sum_{n=1}^{\infty} |z_n|$. □

Power series will be studied in great detail in the next section, but we introduce them here since they naturally arise from series of complex numbers.

Definition 5.1.26. A *power series centered at z_0* is a series of the form

$$\sum_{n=0}^{\infty} a_n(z - z_0^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$

where each $a_k \in \mathbb{C}$.

Definition 5.1.27. Every power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has a number $R \in [0, \infty]$ called its *radius of convergence* such that

- the series converges absolutely for all $z \in \mathbb{C}$ satisfying $|z - z_0| < R$
- the series diverges for all $z \in \mathbb{C}$ satisfying $|z - z_0| > R$.

The set $D(z_0, R) = \{z \in \mathbb{C} : |z - z_0| < R\}$ is called the *circle of convergence* (or more precisely, the open disk of convergence).

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Remark 5.1.28. On the boundary $|z - z_0| = R$, no general conclusion can be made. Some boundary points may yield convergence, while others may yield divergence.

Within their common region of convergence, power series behave much like polynomials.

Proposition 5.1.29. *Suppose* $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ *and* $\sum_{n=0}^{\infty} b_n(z - z_0)^n$ *are two power series with positive radii of convergence. Then, inside the disk where both converge,*

(1) $\displaystyle\sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=0}^{\infty} b_n(z - z_0)^n = \sum_{n=0}^{\infty}(a_n + b_n)(z - z_0)^n$

(2) For any complex constant $\alpha \in \mathbb{C}$, $\displaystyle\alpha \sum_{n=0}^{\infty} a_n(z - z_0)^n = \sum_{n=0}^{\infty} \alpha a_n(z - z_0)^n$

(3) They may be rearranged term-by-term because they converge absolutely inside the circle of convergence.

These facts become extremely important when building holomorphic functions from known series expansions.

5.2 Taylor Series

In the previous section, we studied power series and saw that every power series has a radius of convergence within which it defines a complex-valued function. A natural question now arises: *If a function is holomorphic on a domain, can it always be represented by a power series?*

One of the central achievements of complex analysis is that the answer is yes. In fact, holomorphic functions are not merely differentiable: they are locally equal to their Taylor series. This is one of the main ways in which complex analysis differs from real analysis. In the real-variable setting, even infinitely differentiable functions need not equal their Taylor series. In the complex setting, holomorphicity and power series representation go hand in hand.

In this section, we first study the basic properties of power series, showing that within their circle of convergence they may be differentiated and integrated term-by-term. These results imply that any function represented by a power series is automatically holomorphic. We then prove Taylor's Theorem, which shows that every holomorphic function admits a Taylor series expansion about any point in its domain. We also discuss the radius of convergence of Taylor series, its connection with singularities, and several

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important Maclaurin series expansions that will be used throughout the remainder of the course.

We begin with a power series centered at a point $z_0 \in \mathbb{C}$ given by

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

Recall from the previous section that such a series has a radius of convergence $R \in [0, \infty]$ and it converges absolutely for all z satisfying $|z - z_0| < R$ and diverges for all z satisfying $|z - z_0| > R$.

The first goal of this section is to show that the power series behave like polynomials inside their circle of convergence.

Theorem 5.2.1. *Let $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ be a power series with radius of convergence $R > 0$. Then f is continuous at every point z such that $|z - z_0| < R$.*

Proof. Fix a point z with $|z - z_0| < R$. Since the power series converges at z, it also converges absolutely at every point sufficiently close to z. Let $(S_N(z))_{N=1}^{\infty}$ denote the sequence of partial sums at z

$$S_N(z) = \sum_{n=0}^{N} a_n (z - z_0)^n$$

Each S_N is a polynomial in z, so $S_N(z)$ is continuous. Moreover, the convergence of the series on compact subsets of the open disk $|z - z_0| < R$ is uniform. Therefore, f is the uniform limit of continuous functions on sufficiently small closed disks contained in $|z - z_0| < R$, and so f is continuous there. □

Theorem 5.2.2. *Let $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ be a power series with radius of convergence $R > 0$. Then f may be differentiated term-by-term inside its circle of convergence*

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$$

for $|z - z_0| < R$. Moreover, the differentiated series has the same radius of convergence R.

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Proof. Formally differentiating term-by-term gives

$$\begin{aligned}\frac{d}{dz}\sum_{n=0}^{\infty} a_n(z-z_0)^n &= \sum_{n=0}^{\infty} a_n \frac{d}{dz}(z-z_0)^n \\ &= \sum_{n=0}^{\infty} n a_n (z-z_0)^{n-1} \\ &= \sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1}\end{aligned}$$

The term with $n = 0$ disappears because the derivative of the constant term is zero.

It remains to justify that this differentiation is valid and that the radius of convergence remains unchanged. To see this, apply the ratio test to the original series and the differentiated series. For the original series, the ratio of successive terms is controlled by

$$\left|\frac{a_{n+1}(z-z_0)^{n+1}}{a_n(z-z_0)^n}\right| = \left|\frac{a_{n+1}}{a_n}\right| |z-z_0|$$

For the differentiated series

$$\sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1}$$

the extra factor $\frac{n+1}{n}$ tends to 1 so the ratio test yields the same radius of convergence. Hence, the differentiated series converges in the same disk $|z-z_0| < R$, and term-by-term differentiation is valid there. □

Remark 5.2.3. Since the derivative of a power series is again a power series with the same radius of convergence, we may differentiate repeatedly. Therefore, a power series defines a function that is infinitely differentiable inside its circle of convergence.

In particular, every power series represents a holomorphic function in its circle of convergence.

Theorem 5.2.4. *Let $f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n$ be a power series with radius of convergence $R > 0$. Then for every contour C lying entirely inside the disk $|z-z_0| < R$,*

$$\int_C \sum_{n=0}^{\infty} a_n(z-z_0)^n \, dz = \sum_{n=0}^{\infty} a_n \int_C (z-z_0)^n \, dz$$

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Equivalently, indefinite integration may be carried out term-by-term

$$\int \sum_{n=0}^{\infty} a_n (z-z_0)^n \, dz = \sum_{n=0}^{\infty} \frac{a_n}{n+1}(z-z_0)^{n+1} + C$$

Moreover, the integrand series has the same radius of convergence R.

Proof. Since the series converges uniformly on compact subsets of the disk $|z-z_0| < R$, contour integration may be interchanged with summation. Thus

$$\int_C \sum_{n=0}^{\infty} a_n (z-z_0)^n \, dz = \sum_{n=0}^{\infty} a_n \int_C (z-z_0)^n \, dz$$

For the indefinite integration, integrating term by term yields

$$\int a_n (z-z_0)^n \, dz = \frac{a_n}{n+1}(z-z_0)^{n+1}$$

so

$$\int \sum_{n=0}^{\infty} a_n (z-z_0)^n \, dz = \sum_{n=0}^{\infty} \frac{a_n}{n+1}(z-z_0)^{n+1} + C$$

As with differentiation, the ratio test shows that the radius of convergence is unchanged. □

Suppose now that a function f is represented inside $|z-z_0| < R$ by a power series

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$$

By differentiating term by term, we obtain

$$\begin{aligned} f'(z) &= \sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1} \\ f''(z) &= \sum_{n=2}^{\infty} n(n-1) a_n (z-z_0)^{n-2} \\ f^{(3)}(z) &= \sum_{n=3}^{\infty} n(n-1)(n-2) a_n (z-z_0)^{n-3} \end{aligned}$$

and more generally,

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1) a_n (z-z_0)^{n-k}$$

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Evaluating at $z = z_0$, all terms vanish except the one with $n = k$, so

$$f(z_0) = a_0, \quad f'(z_0) = 1!a_1, \quad f''(z_0) = 2!a_2, \quad ..., \quad f^{(k)}(z_0) = k!a_k$$

Thus, the coefficients are determined by

$$a_n = \frac{f^{(n)}(z_0)}{n!}$$

for $n \geq 0$. This gives the familiar Taylor expansion.

Definition 5.2.5. Let f be a holomorphic function in a neighbourhood of z_0. The *Taylor series of f centered at z_0* is the power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n$$

That is,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots$$

whenever the series converges to $f(z)$.

Definition 5.2.6. A Taylor series centered at $z_0 = 0$ is called a *Maclaurin series*. The Maclaurin series of f is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n$$

We now prove the fundamental theorem of this section.

Theorem 5.2.7 (Taylor's Theorem). *Let f be holomorphic in a domain D, and let $z_0 \in D$. Then f has the series representation*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n$$

for all z in the largest open disk centered at z_0 that lies entirely inside D.

Proof. Let C be a positively oriented circle centered at z_0 with radius R lying entirely inside D. Fix a point z inside C. By Cauchy's integral formula (Theorem 4.5.1),

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(s)}{s - z}\, ds$$

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Write

$$s - z = (s - z_0) - (z - z_0)$$

so that

$$\frac{1}{s-z} = \frac{1}{(s-z_0)-(z-z_0)} = \frac{1}{s-z_0} \cdot \frac{1}{1-\frac{z-z_0}{s-z_0}}$$

Since z lies inside C, we have

$$\left|\frac{z-z_0}{s-z_0}\right| < 1$$

Therefore, we may use the geometric series expansion

$$\frac{1}{1-w} = 1 + w + w^2 + \cdots + w^{n-1} + \frac{w^n}{1-w}$$

$|w| < 1$ with $w = \frac{z-z_0}{s-z_0}$. Thus,

$$\frac{1}{s-z} = \frac{1}{s-z_0}\left[1 + \frac{z-z_0}{s-z_0} + \frac{(z-z_0)^2}{(s-z_0)^2} + \cdots + \frac{(z-z_0)^{n-1}}{(s-z_0)^{n-1}}\right] + \frac{(z-z_0)^n}{(s-z)(s-z_0)^n}$$

Substituting into Cauchy's formula gives

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(s)}{s-z_0}\,ds + \frac{z-z_0}{2\pi i}\int_C \frac{f(s)}{(s-z_0)^2}\,ds + \cdots + \frac{(z-z_0)^{n-1}}{2\pi i}\int_C \frac{f(s)}{(s-z_0)^n}\,ds + R_n(z)$$

where

$$R_n(z) = \frac{(z-z_0)^n}{2\pi i}\int_C \frac{f(s)}{(s-z)(s-z_0)}\,ds$$

By Cauchy's integral formula for derivatives (Theorem 4.5.5),

$$\frac{1}{2\pi i}\int_C \frac{f(s)}{(s-z_0)^{n+1}} = \frac{f^{(n)}(z_0)}{n!}$$

Therefore,

$$f(z) = f(z_0) + \frac{f'(z_0)}{1!}(z-z_0) + \frac{f''(z_0)}{2!}(z-z_0)^2 + \cdots + \frac{f^{(k-1)}(z_0)}{(k-1)!}(z-z_0)^{k-1} + R_k(z)$$

It remains to show that $R_k(z) \to 0$ as $k \to \infty$. Since f is continuous on the circle C, there exists $M > 0$ such that $|f(s)| \leq M$ for all $s \in C$. Let $d = |z - z_0|$. Since z lies strictly inside the circle C, we have $d < R$. Also, for $s \in C$, $|s - z_0| = R$. Moreover,

$$|s - z| = |s - z_0 - (z - z_0)| \geq |s - z_0| + |z - z_0| = R - d$$

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Therefore,

$$|R_k(z)| \leq \frac{|z-z_0|^k}{2\pi} \int_C \frac{|f(s)|}{|s-z||s-z_0|^k} \, |ds| \leq \frac{d^k}{2\pi} \cdot \frac{M}{(R-d)R^k} \cdot 2\pi R$$

Hence,

$$|R_k(z)| \leq \frac{MR}{R-d} \left(\frac{d}{R}\right)^k$$

Since $\frac{d}{R} < 1$, it follows that $\left(\frac{d}{R}\right)^k \to 0$, and therefore, $R_k(z) \to 0$. Therefore,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$$

for every $z \in C$. Since C may be any circle centered at z_0 lying entirely inside D, the result holds in the largest such disk. □

Taylor's Theorem guarantees that a holomorphic function has a Taylor series expansion around each point of holomorphicity. The next natural question is: *how far does that series converge?* The answer is governed by the nearest singularity.

Remark 5.2.8. The radius of convergence of a Taylor series centered at z_0 is the distance from z_0 to the nearest isolated singularity of the function. If the function is entire, then the radius of convergence is infinite.

Example 5.2.9. Suppose

$$f(z) = \frac{3-i}{1-i+z}$$

Find the radius of convergence of the Taylor series centered at $z_0 = 4-2i$.

Solution. The function is holomorphic everywhere except where the denominator vanishes

$$1-i+z=0 \implies z=-1+i$$

Thus, the only singularity is at $z=-1+i$. The radius of convergence is the distance from the center $z_0 = 4-2i$ to this singularity

$$R = |(-1+i)-(4-2i)| = |-5+3i| = \sqrt{34}$$

Hence, the Taylor series centered at $4-2i$ has radius of convergence $R = \sqrt{34}$.

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The following standard expansions are fundamental.

Example 5.2.10. Show that

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

Solution. Since every derivative of e^z is again e^z, we have $f^{(n)}(0) = 1$ for all $n \geq 0$. Therefore, the Maclaurin series is

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots$$

Since e^z is entire, the radius of convergence is ∞.

Example 5.2.11. Show that

$$\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

Solution. The derivatives of $\sin(z)$ cycle as

$$\sin(z), \quad \cos(z), \quad -\sin(z), \quad -\cos(z), \quad \sin(z), ...$$

Evaluating at $z = 0$ gives $0, 1, 0, -1, 0, 1, ...$ Hence, the nonzero coefficients occur only for odd powers, and the Maclaurin series becomes

$$\sin(z) = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

Since $\sin(z)$ is entire, the radius of convergence is infinite.

Example 5.2.12. Show that

$$\cos(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}$$

Solution. The derivatives of $\cos(z)$ cycle as

$$\cos(z), \quad -\sin(z), \quad -\cos(z), \quad \sin(z), \quad \cos(z), ...$$

Evaluating at $z = 0$ gives 1, 0, -1, 0, 1,.... Thus, only even powers occur and

$$\cos(z) = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}$$

Again, $R = \infty$.

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A Taylor expansion is uniquely determined by the function and the center.

Theorem 5.2.13. *Suppose two power series centered at z_0*

$$\sum_{n=0}^{\infty} a_n(z-z_0)^n, \quad \sum_{n=0}^{\infty} b_n(z-z_0)^n$$

represent the same function f in a neighbourhood of z_0 and both have the same nonzero radius of convergence. Then

$$a_n = b_n = \frac{f^{(n)}(z_0)}{n!}$$

for all $n \geq 0$.

Proof. If $f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n$, then differentiating term-by-term and evaluating at z_0, we obtain

$$a_n = \frac{f^{(n)}(z_0)}{n!}, \quad b_n = \frac{f^{(n)}(z_0)}{n!}$$

Hence, $a_n = b_n$ for all n. □

Remark 5.2.14. This means that once a function is holomorphic at z_0, its Taylor series about z_0 is unique.

In practice, Taylor series are often found not by repeated differentiation, but by using known series and algebraic manipulation.

Example 5.2.15. Find the Maclaurin series for

$$f(z) = \frac{1}{(1-z)^2}$$

Solution. From the geometric series formula, for $|z| < 1$,

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

Differentiate both sides term-by-term

$$\frac{d}{dz}\left(\frac{1}{1-z}\right) = \frac{d}{dz}\sum_{n=0}^{\infty} z^n$$

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$$\frac{1}{(1-z)^2} = \sum_{n=1}^{\infty} nz^{n-1}$$

Equivalently,

$$\frac{1}{(1-z)^2} = \sum_{n=0}^{\infty} (n+1)z^n$$

Since differentiation does not change the radius of convergence, the radius is $R = 1$.

Example 5.2.16. Find the Maclaurin series for

$$f(z) = \frac{z^3}{(1-z)^2}$$

Solution. Using the previous example, multiplying by z^3 gives

$$\frac{z^3}{(1-z)^2} = z^3 \sum_{n=0}^{\infty} (n+1)z^n = \sum_{n=0}^{\infty} (n+1)z^{n+3}$$

The radius of convergence remains $R = 1$.

Example 5.2.17. Expand

$$f(z) = \frac{1}{1-z}$$

in a Taylor series centered at $z_0 = 2i$.

Solution. Rewrite the denominator in terms of $z - 2i$

$$1 - z = 1 - 2i - (z - 2i)$$

Hence

$$\frac{1}{1-z} = \frac{1}{1-2i-(z-2i)} = \frac{1}{1-2i} \cdot \frac{1}{1 - \frac{z-2i}{1-2i}}$$

Now use the geometric series formula with $\frac{z-2i}{1-2i}$ to get

$$\frac{1}{1-z} = \frac{1}{1-2i} \sum_{n=0}^{\infty} \left(\frac{z-2i}{1-2i}\right)^n = \sum_{n=0}^{\infty} \frac{(z-2i)^n}{(1-2i)^{n+1}}$$

This expansion converges when $\left|\frac{z-2i}{1-2i}\right| < 1$, so $R = \sqrt{5}$.

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Remark 5.2.18 (Hadamard's Formula). If a power series is written in the form

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

then its radius of convergence R may also be found from the coefficients by Hadamard's formula

$$\frac{1}{R} = \limsup_{n\to\infty} \sqrt[n]{|a_n|}$$

Equivalently

$$R = \frac{1}{\limsup_{n\to\infty} \sqrt[n]{|a_n|}}$$

This result is often called the Cauchy–Hadamard formula. It gives a direct way to compute the radius of convergence from the coefficients of the Taylor series, and it is especially useful when the location of singularities is not immediately apparent.

Example 5.2.19. Consider the power series

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} (z-1)^n$$

Find its radius of convergencce using Hadamard's formula.

Solution. We identify the coefficients $a_n = \frac{n^2}{3^n}$. Note that

$$\sqrt[n]{|a_n|} = \sqrt[n]{\frac{n^2}{3^n}} = \frac{\sqrt[n]{n^2}}{3}$$

Now observe that $\sqrt[n]{n^2} = n^{\frac{2}{n}} \to 1$ as $n \to \infty$. Therefore,

$$\frac{1}{R} = \limsup_{n\to\infty} \sqrt[n]{\frac{n^2}{3^n}} = \frac{1}{3}$$

Therefore, $R = 3$.

5.3 Laurent Series

In the previous section, we established that every holomorphic function can be represented locally by a Taylor series. However, Taylor series expansions require the function to be holomorphic at the center point. This raises an

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important question: *What happens if a function is not holomorphic at a point, but it is holomorphic in a neighbourhood around that point?*

Such points are called *singularities*, and they arise naturally in complex analysis. For example, functions like

$$f(z) = \frac{1}{z}, \quad f(z) = \frac{1}{z-1}, \quad f(z) = \ln(z)$$

fail to be holomorphic at certain points, but are holomorphic elsewhere nearby.

To study functions near such points, we introduce a new type of series expansion: the Laurent series, which generalizes Taylor series by allowing negative powers of $z - z_0$. Unlike Taylor series, Laurent series describe functions on annular regions rather than disks.

Laurent series provide a powerful tool for analyzing the behavior of functions near singularities and form the foundation for the study of residues, poles, and complex integration techniques that follow.

Definition 5.3.1. Let f be a complex function and $z_0 \in \mathbb{C}$. The point z_0 is called a *singularity* of f if f is not holomorphic at z_0.

Definition 5.3.2. A point z_0 is called an *isolated singularity* of f, if there exists $R > 0$ such that

- f is holomorphic in the punctured disk $0 < |z - z_0| < R$,
- but f is not holomorphic at z_0.

Example 5.3.3. Let $f(z) = \frac{z}{z^2+4}$. Then f is not holomorphic at $z = 2i$ and $z = -2i$ since the denominator vanishes there. Both are isolated singularities, because f is holomorphic around each point excluding the point itself.

Definition 5.3.4. Let $z_0 \in \mathbb{C}$. A *Laurent series* centered at z_0 is a series of the form

$$\sum_{k=-\infty}^{\infty} a_k(z-z_0)^k = \cdots + \frac{a_{-2}}{(z-z_0)^2} + \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \cdots$$

A Laurent series can be written as

$$\sum_{k=1}^{\infty} \frac{a_{-k}}{(z-z_0)^k} + \sum_{k=0}^{\infty} a_k(z-z_0)^k$$

The first sum (negative powers) is called the *principal part* and the second sum (nonnegative powers) is called the *holomorphic part*.

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Definition 5.3.5. A Laurent series converges on an *annular domain* of the form

$$r < |z - z_0| < R$$

where $0 \leq r < R \leq \infty$.

A Laurent series is essentially a Taylor series with negative power terms, allowing us to represent functions near singularities.

Example 5.3.6. Find the Laurent series of

$$f(z) = \frac{\sin(z)}{z^4}$$

about $z = 0$.

Solution. Recall the Maclaurin series

$$\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

Divide by z^4 to get

$$f(z) = \frac{\sin(z)}{z^4} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n-3}}{(2n+1)!} = \underbrace{\frac{1}{z^3} - \frac{1}{6z}}_{\text{principal part}} + \underbrace{\frac{z}{120} - \cdots}_{\text{holomorphic part}}$$

This series is valid for $0 < |z| < \infty$.

We now state the central result of this section.

Theorem 5.3.7 (Laurent's Theorem). *Let f be holomorphic in an annular domain*

$$D = \{z \in \mathbb{C} : r < |z - z_0| < R\}$$

Then f can be represented as a Laurent series

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k$$

valid for $r < |z - z_0| < R$. The coefficients are given by

$$a_k = \frac{1}{2\pi i} \int_C \frac{f(s)}{(s - z_0)^{k+1}} \, ds$$

for $k \in \mathbb{Z}$, where C is a positively oriented simple closed curve in the annulus.

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Proof (Sketch). The proof follows the same idea as Taylor's Theorem (Theorem 5.2.7), but uses two contours.

(1) Choose two circles C_1 and C_2 centered at z_0 with radii satisfying

$$r < r_1 < |z - z_0| < r_2 < R$$

(2) Use Cauchy's integral formula in the annulus

$$f(z) = \frac{1}{2\pi i}\int_{C_2} \frac{f(s)}{s-z}\, ds - \frac{1}{2\pi i}\int_{C_1} \frac{f(s)}{s-z}\, ds$$

(3) Expand the integrand using geometric series
   - One expansion gives positive powers
   - The other gives negative powers

(4) Combine both expansions to obtain the Laurent series.

(5) Estimate the remainder term using the ML-inequality (Theorem 4.2.26) to show convergence.

□

In practice, Laurent series are rarely computed using the integral formula. Instead, we use algebraic manipulation, geometric series, or partial fractions.

Example 5.3.8. Find Laurent expansions of

$$f(z) = \frac{1}{z(z-1)}$$

in different regions.

Solution. Write

$$f(z) = \frac{1}{z} \cdot \frac{1}{1-z}$$

(1) If $0 < |z| < 1$, then

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

Then

$$f(z) = \frac{1}{z} \cdot \frac{1}{1-z} = \frac{1}{z}\sum_{n=0}^{\infty} z^n = \sum_{n=0}^{\infty} z^{n-1} = \frac{1}{z} + 1 + z + z^2 + \cdots$$

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(2) If $|z| > 1$, rewrite

$$\frac{1}{1-z} = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}}$$

Then

$$\frac{1}{1-\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{z^n}$$

So

$$f(z) = -\sum_{n=0}^{\infty} \frac{1}{z^{n+2}}$$

The same function has different Laurent expansions, depending on the region.

Example 5.3.9. Find a Laurent expansion of

$$f(z) = \frac{8z+1}{z(1-z)}$$

for $0 < |z| < 1$.

Solution. Using partial fractions,

$$\frac{8z+1}{z(1-z)} = \frac{1}{z} + \frac{9}{1-z}$$

Expand

$$\frac{9}{1-z} = 9\sum_{n=0}^{\infty} z^n$$

Therefore,

$$f(z) = \frac{1}{z} + 9\sum_{n=0}^{\infty} z^n$$

Remark 5.3.10. If all negative coefficients vanish, the Laurent series reduces to a Taylor series. The presence of negative powers indicates a singularity at z_0. Laurent series are unique in their domain of convergence.

Remark 5.3.11. Laurent series allow us to define integrals of the form

$$\oint_C f(z)\, dz = 2\pi i a_{-1}$$

where a_{-1} is the coefficient of $\frac{1}{z-z_0}$. This will be central in residue theory.

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5.4 Zeros and Poles

In the previous section, we introduced Laurent series as a way of representing holomorphic functions near isolated singularities. One of the most important insights from Laurent series is that the behaviour of a function near a singular point is encoded in the principal part (the portion of involving negative powers of $z - z_0$).

This naturally leads to a classification of isolated singularities into three important types

- removable singularities,
- poles, and
- essential singularities.

At the same time, when a function is holomorphic, another fundamental notion is the idea of a zero of a function, a point where the function vanishes. Just as poles measure how a function "blow up", zeros measure how a function "vanishes", and these two ideas are deeply connected.

Suppose $z = z_0$ is an isolated singularity of a complex function f, and suppose that f has the Laurent expansion

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k = \underbrace{\sum_{k=1}^{\infty} a_{-k} (z - z_0)^{-k}}_{\text{principal part}} + \underbrace{\sum_{k=0}^{\infty} a_k (z - z_0)^k}_{\text{holomorphic part}}$$

valid in the punctured disk

$$0 < |z - z_0| < R$$

The classification of the singularity z_0 depends entire on this principal part.

Definition 5.4.1. Let z_0 be an isolated singularity of f.

(1) If the principal part is identically zero, that is, $a_{-k} = 0$ for all $k \in \mathbb{N}$, then z_0 is called a *removable singularity*.

(2) If the principal part contains only finitely many nonzero terms, then z_0 is called a *pole*. If the last nonzero negative power term is

$$\frac{a_{-n}}{(z - z_0)^n}$$

with $a_{-n} \neq 0$, then z_0 is called a *pole of order n*.

(3) If the principal part contains infinitely many nonzero terms, then z_0 is called an *essential singularity*.

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Example 5.4.2. Consider

$$f(z) = \frac{\sin(z)}{z}$$

We know from the Maclaurin series of $\sin(z)$ that

$$\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

Dividing by z, we obtain

$$f(z) = \frac{\sin(z)}{z} = \frac{1}{z}\sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n+1)!} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots$$

There are no negative powers of z, so the principal part is zero. Hence, $z = 0$ is a removable singularity of $\frac{\sin(z)}{z}$.

Remark 5.4.3. If f has a removable singularity at $z = z_0$, then we can define $f(z_0)$ appropriately so that f becomes holomorphic at z_0. For example, since

$$\frac{\sin(z)}{z} \to 1$$

as $z \to 0$, we may define $f(0) = 1$, and thereby obtain a function holomorphic at $z = 0$.

Example 5.4.4. Consider

$$f(z) = \frac{\sin(z)}{z^2}$$

Using the Maclaurin series for $\sin(z)$,

$$f(z) = \frac{\sin(z)}{z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n-1}}{(2n+1)!} = \frac{1}{z} - \frac{z}{3!} + \frac{z^3}{5!} - \cdots$$

The principal part is $\frac{1}{z}$. Thus, $z = 0$ is a *simple pole* (pole of order 1).

Example 5.4.5. Consider

$$f(z) = \frac{\sin(z)}{z^4}$$

Then

$$f(z) = \frac{\sin(z)}{z^4} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n-3}}{(2n+1)!} = \frac{1}{z^3} - \frac{1}{3!z} + \frac{z}{5!} - \cdots$$

The highest negative power is $\frac{1}{z^3}$, so $z = 0$ is a pole of order 3.

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Example 5.4.6. Consider

$$f(z) = e^{\frac{3}{z}}$$

Using the Maclaurin series for e^w,

$$e^{\frac{3}{z}} = \sum_{n=0}^{\infty} \frac{\left(\frac{3}{z}\right)^n}{n!} = \sum_{n=0}^{\infty} \frac{3^n}{n! z^n} = 1 + \frac{3}{z} + \frac{3^2}{2! z^2} + \frac{3^3}{3! z^3} + \cdots$$

This Laurent series contains infinitely many nonzero negative powers, so $z = 0$ is an essential singularity.

Definition 5.4.7. A complex number z_0 is called a *zero* of a function f, if $f(z_0) = 0$.

Definition 5.4.8. Let f be holomorphic at z_0. We say that f has a *zero of order* n at $z = z_0$ if for all $0 \leq k \leq n-1$, $f^{(k)}(z_0) = 0$, but $f^{(n)}(z_0) \neq 0$. A zero of order 1 is called a *simple zero*.

Example 5.4.9. Consider $f(z) = (z-5)^3$. Then

$$f(5) = 0, \quad f'(5) = 0, \quad f''(5) = 0, \quad f'''(5) = 6 \neq 0$$

Hence, $z = 5$ is a zero of order 3.

Theorem 5.4.10. *Let f be holomorphic in a disk $|z - z_0| < R$. Then f has a zero of order n if and only if $f(z) = (z - z_0)^n \varphi(z)$ where φ is holomorphic and $\varphi(z_0) \neq 0$.*

Proof. Suppose first that f has a zero of order n at z_0. Since f is holomorphic, it has a Taylor series expansion

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k, \quad a_k = \frac{f^{(k)}(z_0)}{k!}$$

Since f has a zero of order n, we have $a_k = 0$ for all $0 \leq k \leq n-1$ and $a_n \neq 0$. Hence

$$f(z) = \sum_{k=n}^{\infty} a_k (z - z_0)^k = (z - z_0)^n \sum_{k=n}^{\infty} a_k (z - z_0)^{n-k}$$

Define $\varphi(z) = \sum_{k=n}^{\infty} a_n (z - z_0)^{n-k}$. Then φ is holomorphic at z_0, and $\varphi(z_0) = a_n \neq 0$. Therefore,

$$f(z) = (z - z_0)^n \varphi(z)$$

Conversely, if $f(z) = (z - z_0)^n \varphi(z)$ with φ holomorphic and $\varphi(z_0) \neq 0$, then the Taylor expansion of f begins with the term $(z - z_0)^n$, so f has a zero of order n at z_0. □

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Example 5.4.11. Determine the order of the zero of $f(z) = z\sin(z^2)$ at $z = 0$.

Solution. Using the Maclaurin series

$$\sin(z^2) = \sum_{n=0}^{\infty} \frac{(-1)^n (z^2)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{4n+2}}{(2n+1)!}$$

Then

$$f(z) = z\sin(z^2) = z\sum_{n=0}^{\infty} \frac{(-1)^n z^{4n+2}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{4n+3}}{(2n+1)!}$$

Factoring out z^3,

$$f(z) = z^3 \sum_{n=0}^{\infty} \frac{(-1)^n z^{4n}}{(2n+1)!}$$

Since the factor in the parentheses is holomorphic and nonzero at $z = 0$, it follows from Theorem 5.4.10 that z_0 is a zero of order 3.

Remark 5.4.12. Zeros and poles are reciprocal notions. Indeed, if f has a zero at z_0, then $\frac{1}{f}$ has a pole at z_0.

Theorem 5.4.13. *A function f holomorphic in a punctured disk $0 < |z - z_0| < R$ has a pole of order n at $z = z_0$ if and only if $f(z) = \frac{\varphi(z)}{(z-z_0)^n}$ where φ is holomorphic at z_0 and $\varphi(z_0) \neq 0$.*

Proof. Suppose f has a pole of order n at z_0. Then its Laurent series has the form

$$f(z) = \sum_{k=1}^{n} \frac{a_{-k}}{(z-z_0)^k} + \sum_{k=0}^{\infty} a_k (z-z_0)^k$$

with $a_{-n} \neq 0$. Factor out $(z - z_0)^{-n}$, we get

$$f(z) = \frac{1}{(z-z_0)^n}\left[a_{-n} + a_{-(n-1)}(z-z_0) + \cdots + a_0(z-z_0)^n + \cdots\right]$$

Define $\varphi(z) = a_{-n} + a_{-(n-1)}(z-z_0) + a_{-(n-2)}(z-z_0)^2 + \cdots$. Then φ is holomorphic at z_0 with $\varphi(z_0) = a_{-n} \neq 0$. Hence,

$$f(z) = \frac{\varphi(z)}{(z-z_0)^n}$$

Conversely, if $f(z) = \frac{\varphi(z)}{(z-z_0)^n}$ with φ holomorphic and $\varphi(z_0) \neq 0$, then the Laurent expansion has finitely many negative power terms, ending with $(z - z_0)^{-n}$. Thus, z_0 is a pole of order n. □

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Theorem 5.4.14. *Let g and h be holomorphic at z_0 and suppose h has a zero of order n at z_0 and $g(z_0) \neq 0$. Then*

$$f(z) = \frac{g(z)}{h(z)}$$

has a pole of order n at $z = z_0$.

Proof. Since h has a zero of order n, by Theorem 5.4.10,

$$h(z) = (z - z_0)^n \varphi(z)$$

where φ is holomorphic and $\varphi(z_0) \neq 0$. Thus,

$$f(z) = \frac{g(z)}{(z - z_0)^n \varphi(z)} = \frac{\frac{g(z)}{\varphi(z)}}{(z - z_0)^n}$$

Since both g and φ are holomorphic and $\varphi(z_0) \neq 0$, the quotient

$$\psi(z) = \frac{g(z)}{\varphi(z)}$$

is holomorphic at z_0, and moreover $\psi(z_0) \neq 0$. Therefore,

$$f(z) = \frac{\psi(z)}{(z - z_0)^n}$$

so by Theorem 5.4.13, f hs a pole of order n at z_0. □

Example 5.4.15. Determine the poles of

$$f(z) = \frac{2z + 5}{(z - 1)(z + 5)(z - 2)^4}$$

Solution. The denominator vanishes at $z = 1$, $z = -5$, and $z = 2$. The order of these zeros of the denominator are order 1, order 1, and order 4, respectively.

Since the numerator $2z + 5$ is nonzero at each of these points, Theorem 5.4.14 implies that f has a simple pole at $z = 1$, a simple pole at $z = -5$, and a pole of order 4 at $z = 2$.

Example 5.4.16. In Example 5.4.11, we showed that $z = 0$ is a zero of order 3 of $f(z) = z \sin(z^2)$. Therefore,

$$\frac{1}{z \sin(z^2)}$$

has a pole of order 3 at $z = 0$.

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Remark 5.4.17. If f has a pole at $z = z_0$, then intuitively, $|f(z)| \to \infty$ as $z \to z_0$. Thus, poles correspond to points where the function "blows up."

Remark 5.4.18. A function f is called *meromorphic* on a domain D if it is holomorphic throughout D except possibly for poles. Thus, for rational functions such as $f(z) = \frac{1}{z^2+1}$, are meromorphic on $\mathbb{C}$. Meromorphic functions are among the most important objects in complex analysis.

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Chapter 6

Residue Theory

In previous chapters, we developed the theory of complex integration and established powerful results such as the Cauchy–Goursat Theorem and Cauchy's Integral Formula. These results reveal a remarkable fact: the behavior of a holomorphic function inside a region is completely determined by its behavior on the boundary. However, many functions of interest are not holomorphic everywhere—they may have isolated singularities. The theory of residues provides a systematic and elegant way to analyze such functions and compute complex integrals in their presence.

The central idea of this chapter is that the contribution of a singularity to a contour integral is encoded in a single complex number called its residue. This leads to one of the most powerful tools in complex analysis, the Cauchy Residue Theorem, which allows contour integrals to be evaluated by summing residues at isolated singularities. This result not only simplifies many complex integrals but also provides a bridge between local behavior near singularities and global properties of integrals.

Beyond its theoretical importance, residue theory has striking applications. It enables the evaluation of real integrals that are otherwise difficult or impossible to compute using elementary methods. In particular, we will see how contour integration can be used to compute trigonometric integrals, handle improper integrals via Cauchy principal values, and evaluate Fourier transforms of rational functions. The method extends further to integrals involving multi-valued functions, where special care must be taken along branch cuts.

6.1 Residues and Cauchy's Residue Theorem

In the previous chapter, we studied isolated singularities and saw that a complex function can often be expanded in a Laurent series near such a point. Among all the terms in that expansion, one coefficient turns out to be especially important: the coefficient of $\frac{1}{z-z_0}$. This coefficient is called the residue of the function at the singularity z_0.

At first glance, the residue may seem like just one coefficient among many, but it carries remarkable power. It provides a direct connection between the local behavior of a function near its singularities and the global value of contour integrals. This connection is made precise in one of the most important results in complex analysis: Cauchy's Residue Theorem.

The residue theorem gives a systematic and efficient method for evaluating contour integrals by reducing the problem to the computation of residues at the singularities inside the contour. In practice, this often turns a difficult integral into a manageable algebraic calculation. For this reason, residues are not only theoretically elegant, but also one of the most useful computational tools in the subject.

In this section, we define residues carefully, learn how to compute them in different situations, and prove Cauchy's Residue Theorem. We then illustrate how the theorem can be used to evaluate contour integrals involving poles of various orders and even essential singularities.

Recall that if f has an isolated singularity at $z = z_0$, then f admits a Laurent series expansion of the form

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z-z_0)^k = \sum_{k=1}^{\infty} \frac{a_{-k}}{(z-z_0)^k} + \sum_{k=0}^{\infty} a_k(z-z_0)^k$$

valid in some punctured disk $0 < |z - z_0| < R$.

Definition 6.1.1. Let f have an isolated singularity at $z = z_0$, and suppose its Laurent series about z_0 is

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z-z_0)^k$$

The *residue* of f at z_0 is the coefficient a_{-1} of $(z-z_0)^{-1}$. We write

$$\mathrm{Res}(f, z_0) = a_{-1}$$

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Remark 6.1.2. Thus, if

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k$$

then $\operatorname{Res}(f, z_0) = a_{-1}$. So the residue is simply the coefficient of the $(z-z_0)^{-1}$ term in the Laurent expansion.

Recall that

- If the principal part contains only finitely many nonzero negative power terms, then z_0 is a pole.
- If the principal part contains infinitely many nonzero negative power terms, then z_0 is an essential singularity.
- If there are no negative power terms at all, then z_0 is a removable singularity.

In all of these cases, the residue is just the coefficient a_{-1} whenever the Laurent series exists.

Example 6.1.3. Suppose

$$f(z) = \frac{1}{(z-1)^2(z-3)}$$

and from a Laurent expansion about $z = 1$ (valid for $0 < |z - 1| < 2$), we have

$$f(z) = \frac{-\frac{1}{2}}{(z-1)^2} + \frac{-\frac{1}{4}}{z-1} - \frac{1}{8} - \frac{z-1}{16} - \cdots$$

Find $\operatorname{Res}(f, 1)$.

Solution. The residue is the coefficient of $(z-1)^{-1}$. From the above expansion, that coefficient is $-\frac{1}{4}$. Therefore,

$$\operatorname{Res}(f, 1) = -\frac{1}{4}$$

Example 6.1.4. Suppose $f(z) = e^{\frac{3}{z}}$. Recall from Example 5.4.6, its Laurent series about $z = 0$ is

$$e^{\frac{3}{z}} = \sum_{n=0}^{\infty} \frac{3^n}{n! z^n} = 1 + \frac{3}{z} + \frac{3^2}{2! z^2} + \frac{3^3}{3! z^3} + \cdots$$

for $0 < |z| < \infty$. Find $\operatorname{Res}(f, 0)$.

Solution. The residue is the coefficient of $\frac{1}{z}$. From the series above, the coefficient is 3. Hence, $\operatorname{Res}(f, 0) = 3$.

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In many problems, computing a full Laurent expansion is tedious. Fortunately, there are formulas for residues at poles.

Theorem 6.1.5. *If f has a simple pole at $z = z_0$, then*

$$\mathrm{Res}(f, z_0) = \lim_{z \to z_0} (z - z_0) f(z)$$

Proof. Since f has a simple pole at $z = z_0$, its Laurent expansion near z_0 has the form

$$f(z) = \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$

with $a_{-1} \neq 0$. Multiply both sides by $z - z_0$ we get

$$(z - z_0) f(z) = a_{-1} + a_0(z - z_0) + a_1(z - z_0)^2 + \cdots$$

Now let $z \to z_0$. Every term except for the first term vanishes, so

$$\lim_{z \to z_0} (z - z_0) f(z) = a_{-1}$$

But then $a_{-1} = \mathrm{Res}(f, z_0)$, and thus,

$$\mathrm{Res}(f, z_0) = \lim_{z \to z_0} (z - z_0) f(z)$$

□

Theorem 6.1.6. *If f has a pole of order n at $z = z_0$, then*

$$\mathrm{Res}(f, z_0) = \frac{1}{(n-1)!} \lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} [(z - z_0)^n f(z)]$$

Proof. Since f has a pole of order n at $z = z_0$, its Laurent expansion near z_0 is

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots$$

with $a_{-n} \neq 0$. Multiply by $(z - z_0)^n$ to get

$$(z - z_0)^n f(z) = a_{-n} + a_{-(n-1)}(z - z_0) + + \cdots + a_{-1}(z - z_0)^{n-1} + a_0(z - z_0)^n + \cdots$$

Now differentiate both sides $n - 1$ times. The term involving $a_{-1}(z - z_0)^{n-1}$ becomes $(n - 1)! a_{-1}$ while every term after that still contains a factor of $z - z_0$, which vanishes as $z \to z_0$. Thus,

$$\lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} [(z - z_0)^n f(z)] = (n - 1)! a_{-1}$$

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Since $a_{-1} = \text{Res}(f, z_0)$, we obtain

$$\text{Res}(f, z_0) = \frac{1}{(n-1)!} \lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} [(z - z_0)^n f(z)]$$

□

Remark 6.1.7. In the case when $n = 1$, this reduces to the simple pole formula (Theorem 6.1.5).

Example 6.1.8. Consider

$$f(z) = \frac{1}{(z-1)^2(z-3)}$$

Compute the residues at each pole.

Solution. The denominator vanishes at $z = 1$ where there is a pole of order 2, and $z = 3$ where there is a simple pole.

Since $z = 3$ is a simple pole, using Theorem 6.1.5,

$$\text{Res}(f, 3) = \lim_{z \to 3} (z-3) \frac{1}{(z-1)^2(z-3)} = \lim_{z \to 3} \frac{1}{(z-1)^2} = \frac{1}{4}$$

Since $z = 1$ is a pole of order 2, using Theorem 6.1.6 with $n = 2$,

$$\begin{aligned}
\text{Res}(f, 1) &= \lim_{z \to 1} \frac{d}{dz} \left((z-1)^2 \frac{1}{(z-1)^2(z-3)} \right) \\
&= \lim_{z \to 1} \frac{d}{dz} \left(\frac{1}{z-3} \right) \\
&= \lim_{z \to 1} \left(-\frac{1}{(z-3)^2} \right) \\
&= -\frac{1}{4}
\end{aligned}$$

There is an especially convenient formula when the denominator has a simple zero.

Proposition 6.1.9. *Suppose $f(z) = \frac{g(z)}{h(z)}$ where g and h are holomorphic at $z = z_0$, and suppose $h(z_0) = 0$ and $h'(z_0) \neq 0$. Then z_0 is a simple pole of f, and*

$$\text{Res}(f, z_0) = \frac{g(z_0)}{h'(z_0)}$$

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Proof. Since $h(z_0)$ and $h'(z_0) \neq 0$, the function h has a zero of order 1 at z_0. Therefore, $f(z)$ has a simple pole at z_0.

Using the simple pole formula

$$\operatorname{Res}(f, z_0) = \lim_{z \to z_0} (z - z_0) \frac{g(z)}{h(z)} = \lim_{z \to z_0} g(z) \frac{z - z_0}{h(z)}$$

Since g is holomorphic, $g(z) \to g(z_0)$. Also, by the definition of the derivative

$$h'(z_0) = \lim_{z \to z_0} \frac{h(z) - h(z_0)}{z - z_0} = \lim_{z \to z_0} \frac{h(z)}{z - z_0}$$

Therefore,

$$\lim_{z \to z_0} \frac{z - z_0}{h(z)} = \frac{1}{h'(z_0)}$$

Therefore,

$$\operatorname{Res}(f, z_0) = g(z_0) \cdot \frac{1}{h'(z_0)} = \frac{g(z_0)}{h'(z_0)}$$

□

Remark 6.1.10. This formula is often the fastest way to compute residues at simple poles of rational functions and quotients of holomorphic functions.

Example 6.1.11. Consider

$$f(z) = \frac{1}{z^4 + 1}$$

Find the residues at its poles.

Solution. The poles occur where $z^4 + 1 = 0$. Thus, $z^4 = -1 = e^{i\pi}$, so the four roots are

$$z_1 = e^{\frac{i\pi}{4}}, \quad z_2 = e^{\frac{3i\pi}{4}}, \quad z_3 = e^{\frac{5i\pi}{4}}, \quad z_4 = e^{\frac{7i\pi}{4}}$$

These are all simple roots, so each pole is simple. Let $g(z) = 1$, $h(z) = z^4 + 1$. Then $h'(z) = 4z^3$, so by Proposition 6.1.9,

$$\operatorname{Res}(f, z_k) = \frac{1}{4z_k^3}$$

Computing each residue

- At $z_1 = e^{\frac{i\pi}{4}}$,

$$\operatorname{Res}(f, e^{\frac{i\pi}{4}}) = \frac{1}{4e^{\frac{3i\pi}{4}}} = \frac{1}{4} e^{-\frac{3i\pi}{4}} = -\frac{1}{4\sqrt{2}} - \frac{i}{4\sqrt{2}}$$

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- At $z_2 = e^{\frac{3i\pi}{4}}$,

$$\text{Res}(f, e^{\frac{3i\pi}{4}}) = \frac{1}{4e^{\frac{9i\pi}{4}}} = \frac{1}{4}e^{-\frac{9i\pi}{4}} = \frac{1}{4\sqrt{2}} - \frac{i}{4\sqrt{2}}$$

- At $z_3 = e^{\frac{5i\pi}{4}}$,

$$\text{Res}(f, e^{\frac{5i\pi}{4}}) = \frac{1}{4e^{\frac{15i\pi}{4}}} = \frac{1}{4}e^{-\frac{15i\pi}{4}} = \frac{1}{4\sqrt{2}} + \frac{i}{4\sqrt{2}}$$

- At $z_4 = e^{\frac{7i\pi}{4}}$,

$$\text{Res}(f, e^{\frac{7i\pi}{4}}) = \frac{1}{4e^{\frac{21i\pi}{4}}} = \frac{1}{4}e^{-e^{\frac{21i\pi}{4}}} = -\frac{1}{4\sqrt{2}} + \frac{i}{4\sqrt{2}}$$

Remark 6.1.12. This example shows how much easier residue calculations become simple when the simple pole quotient formula is available.

So far, residues may seem like just a coefficient to compute. Their true importance come from the fact that contour integrals around singularities are determined entirely by these coefficients. This is the content of the residue theorem.

Theorem 6.1.13 (Cauchy's Residue Theorem). *Let D be a simply connected domain, and let C be a simple closed contour lying entirely within D. Suppose f is holomorphic on and inside C, except at finitely many isolated singularities $z_1, z_2, ..., z_n$. Then*

$$\oint_C f(z)\ dz = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

This theorem says, to evaluate a contour integral, it is enough to

(1) Locate the singularities inside the contour,
(2) Compute the residues there,
(3) Add them
(4) Multiply by $2\pi i$

This is one of the most powerful computational tools in all of complex analysis.

Proof. Let $z_1, z_2, ..., z_n$ be the isolated singularities of f inside C. Around each singularity z_k, draw a circle C_k centered at z_k, chosen small enough so that each C_k lies entirely inside C, and the circles are pairwise disjoint.

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Now remove the interior of each small circle from the region enclosed by C. On the remaining region, the function f is holomorphic.

By the generalized form of Cauchy's theorem (or contour deformation (Corollary 4.3.11), or multiply connected version of Cauchy–Goursat (Theorem 4.3.15)), we have

$$\oint_C f(z)\ dz = \sum_{k=1}^{n} \oint_{C_k} f(z)\ dz$$

Now, near each z_k, write the Laurent expansion

$$f(z) = \sum_{m=-\infty}^{\infty} a_m^{(k)} (z - z_k)^m$$

When integrating around C_k, all terms integrate to zero except the $(z - z_k)^{-1}$ term. Thus,

$$\oint_{C_k} f(z)\ dz = 2\pi i a_{-1}^{(k)} = 2\pi i \operatorname{Res}(f, z_k)$$

Summing over all singularities gives

$$\oint_C f(z)\ dz = \sum_{k=1}^{n} 2\pi i \operatorname{Res}(f, z_k) = 2\pi i \sum_{k=1}^{n} \operatorname{Res}(f, z_k)$$

□

Example 6.1.14. Evaluate

$$\oint_C \frac{1}{(z-1)^2(z-3)}\ dz$$

where C is the rectangle bounded by $x = 0$, $x = 4$, $y = -1$, and $y = 1$.

Solution. The integrand has poles at $z = 1$ of order 2, and a simple pole at $z = 3$. We already computed

$$\operatorname{Res}(f, 1) = -\frac{1}{4}, \quad \operatorname{Res}(f, 3) = \frac{1}{4}$$

in Example 6.1.8.

In C, both poles lie inside the rectangle. Therefore, by the residue theorem (Theorem 6.1.13),

$$\oint_C \frac{1}{(z-1)^2(z-3)}\ dz = 2\pi i(\operatorname{Res}(f, 1) + \operatorname{Res}(f, 3)) = 2\pi i \left(-\frac{1}{4} + \frac{1}{4}\right) = 0$$

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Example 6.1.15. Evaluate

$$\oint_C \frac{1}{(z-1)^2(z-3)}\,dz$$

where C is the circle $|z| = 2$.

Solution. Note that now, only $z = 1$ lies inside the contour. Therefore,

$$\oint_C \frac{1}{(z-1)^2(z-3)}\,dz = 2\pi i\,\mathrm{Res}(f,1) = 2\pi i\left(-\frac{1}{4}\right) = -\frac{\pi i}{2}$$

Example 6.1.16. Evaluate

$$\oint_C \frac{2z+6}{z^2+4}\,dz$$

where C is the circle $|z - i| = 2$.

Solution. Factor the denominator to get $z^2 + 4 = (z - 2i)(z + 2i)$. So the poles are at $z = 2i$ and $z = -2i$. The contour is centered at i with radius 2, so

- the distance from i to $2i$ is 1, so $2i$ lies inside, and
- the distance from i to $-2i$ is 3, so $-2i$ lies outside.

Thus, only $z = 2i$ contributes.

Since this is a simple pole,

$$\mathrm{Res}(f,2i) = \lim_{z\to 2i}(z-2i)\frac{2z+6}{(z-2i)(z+2i)} = \lim_{z\to 2i}\frac{2z+6}{z+2i} = \frac{3+2i}{2i}$$

By the Cauchy Residue Theorem (Theorem 6.1.13),

$$\oint_C \frac{2z+6}{z^2+4}\,dz = 2\pi i\,\mathrm{Res}(f,2i) = 2\pi i\left(\frac{3+2i}{2i}\right) = \pi(3+2i)$$

Example 6.1.17. Evaluate

$$\oint_C \frac{e^z}{z^4+5z^3}\,dz$$

where C is the circle $|z| = 2$.

Solution. Factor the denominator $z^4 + 5z^3 = z^3(z + 5)$, so the poles are $z = 0$ of order 3, and $z = -5$ which is a simple pole. Only $z = 0$ lies inside $|z| = 0$. Computing the residue at $z = 0$,

$$\mathrm{Res}(f,0) = \frac{1}{2!}\lim_{z\to 0}\frac{d^2}{dz^2}\left(z^3\frac{e^z}{z^3(z+5)}\right)$$

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$$= \frac{1}{2} \lim_{z \to 0} \frac{d^2}{dz^2} \left(\frac{e^z}{z+5} \right)$$
$$= \frac{17}{125}$$

Thus,

$$\oint_C \frac{e^z}{z^4 + 5z^3} \, dz = 2\pi i \cdot \frac{17}{125} = \frac{34\pi i}{125}$$

Example 6.1.18. Evaluate

$$\oint_C \tan(z) \, dz$$

where C is the circle $|z| = 2$.

Solution. Since $\tan(z) = \frac{\sin(z)}{\cos(z)}$, its poles occur where $\cos(z) = 0$, that is,

$$z = \frac{(2n+1)\pi}{2}$$

for $n \in \mathbb{Z}$. Inside $|z| = 2$, the only such points are $z = -\frac{\pi}{2}$ and $z = \frac{\pi}{2}$. Both of these are simple poles. Taking $g(z) = \sin(z)$ and $h(z) = \cos(z)$ with $h'(z) = -\sin(z)$, then

$$\operatorname{Res}(\tan(z), z_0) = \frac{\sin(z_0)}{-\sin(z_0)} = -1$$

at each simple zero z_0 of $\cos(z)$. Thus,

$$\operatorname{Res}\left(\tan(z), \pm\frac{\pi}{2}\right) = -1$$

By the Cauchy Residue Theorem (Theorem 6.1.13),

$$\oint_C \tan(z) \, dz = 2\pi i(-1 + (-1)) = -4\pi i$$

Example 6.1.19. Evaluate

$$\oint_C e^{\frac{3}{z}} \, dz$$

where C is the circle $|z| = 1$.

Solution. The function $e^{\frac{3}{z}}$ has an essential singularity at $z = 0$. Its Laurent expansion is

$$e^{\frac{3}{z}} = 1 + \frac{3}{z} + \frac{3^2}{2!z^2} + \frac{3^3}{3!z^3} + \cdots$$

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Then $\text{Res}(e^{\frac{3}{z}}, 0) = 3$.

Since $z = 0$ lies inside $|z| = 1$,

$$\oint_C e^{\frac{3}{z}}\ dz = 2\pi i \cdot 3 = 6\pi i$$

6.2 Trigonometric Integrals

One of the most important applications of the residue theorem is the evaluation of definite real integrals. Among the classical of these integrals involving trigonometric expressions, especially integrals of the form

$$\int_0^{2\pi} F(\cos(\theta), \sin(\theta))\ d\theta$$

where F is a rational function of $\cos(\theta)$ and $\sin(\theta)$. At first sight, such integrals belong entirely to real-variable calculus. However, by using the parametrization of the unit circle $z = e^{i\theta}$ with $0 \leq \theta \leq 2\pi$, we can rewrite $\cos(\theta)$, $\sin(\theta)$, and $d\theta$ in terms of the complex variable z. This transforms the original real integral into a contour integral around the unit circle, where the methods of complex analysis and residues become available.

This section develops that method carefully. We begin by showing how to convert trigonometric integrals into contour integrals on the unit circle. We then evaluate representative examples using residues, including integrals already in the standard 0 to 2π form and integrals that must first be rewritten to fit that framework. The central idea throughout is that a trigonometric integral can often be reduced to the computation of residues of a rational function inside the unit disk.

The techniques in this section are fundamental not only because they provide elegant computations, but also because they illustrate a central theme of complex analysis: a real problem can often be solved most efficiently by embedding it into the complex plane.

We begin with the standard type of trigonometric integral considered in this section.

Definition 6.2.1. A *trigonometric integral of standard form* is an integral of the form

$$\int_0^{2\pi} F(\cos(\theta), \sin(\theta))\ d\theta$$

where $F(\cos(\theta), \sin(\theta))$ is a rational function of $\cos(\theta)$ and $\sin(\theta)$

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The key observation is that the unit circle $C : |z| = 1$ can be parametrized by $z = e^{i\theta}$ for $0 \leq \theta \leq 2\pi$. Under this substitution, $\cos(\theta)$, $\sin(\theta)$, and $d\theta$ can all be written in terms of z. This allows us to convert the real integral into a contour integral around the unit circle.

Theorem 6.2.2. *Let $z = e^{i\theta}$ for $0 \leq \theta \leq 2\pi$. Then the following identities hold*

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{1}{2}\left(z + \frac{1}{z}\right),$$
$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{1}{2i}\left(z - \frac{1}{z}\right)$$

and $dz = ie^{i\theta}\ d\theta = iz\ d\theta$. Hence, $d\theta = \frac{dz}{i\theta}$.

Proof. Since $z = e^{i\theta}$, we have $z^{-1} = e^{-i\theta}$. From Euler's formula, the two identities hold. Differentiating $z = e^{i\theta}$ with respect to θ, we get

$$\frac{dz}{d\theta} = ie^{i\theta}$$

so $dz = ie^{i\theta}\ d\theta = iz\ d\theta$. Thus, $d\theta = \frac{dz}{iz}$. □

Corollary 6.2.3. *If*

$$I = \int_0^{2\pi} F(\cos(\theta), \sin(\theta))\ d\theta$$

then

$$I = \oint_C F\left(\frac{1}{2}\left(z + \frac{1}{z}\right), \frac{1}{2i}\left(z - \frac{1}{z}\right)\right)\frac{dz}{iz}$$

where C is the unit circle $|z| = 1$, oriented counterclockwise.

Proof. By Theorem 6.2.2, as θ runs from 0 to 2π, the point $z = e^{i\theta}$ traces the unit circle once counterclockwise. Replacing $\cos(\theta)$, $\sin(\theta)$, and $d\theta$ by their expressions in z gives exactly the formula above. □

Remark 6.2.4. After substitution, the integrand becomes a rational function of z. The original real integral is then evaluated by computing residues of the poles inside the circle.

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Example 6.2.5. Evaluate

$$\int_0^{2\pi} \frac{1}{(2+\cos(\theta))^2}\, d\theta$$

Solution. Let

$$I = \int_0^{2\pi} \frac{1}{(2+\cos(\theta))^2}\, d\theta$$

Using the substitutions from Theorem 6.2.2,

$$I = \oint_C \frac{1}{\left(2+\frac{1}{2}\left(z+\frac{1}{z}\right)\right)^2} \frac{dz}{iz}$$

We now simplify the integrand algebraically. Indeed,

$$2+\frac{1}{2}\left(z+\frac{1}{z}\right) = \frac{z^2+4z+1}{2z}$$
$$\left(2+\frac{1}{2}\left(z+\frac{1}{z}\right)\right)^2 = \frac{(z^2+4z+1)^2}{4z^2}$$

so the integrand becomes

$$\frac{1}{\left(2+\frac{1}{2}\left(z+\frac{1}{z}\right)\right)^2} \frac{dz}{iz} = \frac{4z^2}{(z^2+4z+1)^2} \cdot \frac{1}{iz}\, dz = \frac{4}{i} \cdot \frac{z}{(z^2+4z+1)^2}\, dz$$

Hence,

$$I = \frac{4}{i} \oint_C \frac{z}{(z^2+4z+1)^2}\, dz$$

Now factor the quadratic $z^2+4z+1 = (z-z_1)(z-z_2)$, where $z_1 = -2-\sqrt{3}$ and $z_2 = -2+\sqrt{3}$. Thus,

$$\frac{z}{(z^2+4z+1)^2} = \frac{z}{(z-z_1)^2(z-z_2)^2}$$

Among these roots, only $z_2 = -2+\sqrt{3}$ lies inside the unit circle. Therefore, by Cauchy Residue Theorem (Theorem 6.1.13),

$$I = \frac{4}{i} \cdot 2\pi i \operatorname{Res}\left(\frac{z}{(z^2+4z+1)^2}, z_2\right)$$

Since z_2 is a pole of order 2, we use the residue formula for a double pole

$$\operatorname{Res}(f, z_2) = \lim_{z\to z_2} \frac{d}{dz}[(z-z_2)^2 f(z)]$$

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Here,

$$(z - z_2)^2 f(z) = \frac{z}{(z - z_1)^2}$$

Differentiating

$$\frac{d}{dz}\left(\frac{z}{(z - z_1)^2}\right) = -\frac{z + z_1}{(z - z_1)^3}$$

Evaluating at $z = z_2$ and using $z_2 - z_1 = 2\sqrt{3}$ gives

$$\operatorname{Res}(f, z_2) = \frac{1}{6\sqrt{3}}$$

Thus,

$$I = \frac{4}{i} \cdot 2\pi i \operatorname{Res}\left(\frac{z}{(z^2 + 4z + 1)^2}, z_2\right) = \frac{4\pi}{3\sqrt{3}}$$

Therefore

$$\int_0^{2\pi} \frac{1}{(2 + \cos(\theta))^2}\, d\theta = \frac{4\pi}{3\sqrt{3}}$$

Remark 6.2.6. This example illustrates the general procedure.

(1) Write everything in terms of $z = e^{i\theta}$.
(2) Simplify to a rational function of z.
(3) Locate the poles inside $|z| < 1$.
(4) Compute residues.
(5) Apply the Cauchy Residue Theorem.

Example 6.2.7. Evaluate

$$\int_0^{2\pi} \frac{\sin^2(\theta)}{5 + 4\cos(\theta)}\, d\theta$$

Solution. Let

$$I = \int_0^{2\pi} \frac{\sin^2(\theta)}{5 + 4\cos(\theta)}\, d\theta$$

Using the identities in Theorem 6.2.2,

$$\sin^2(\theta) = -\frac{1}{4}\frac{(z^2 - 1)^2}{z^2},$$
$$5 + 4\cos(\theta) = \frac{2z^2 + 5z + 2}{z}$$

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Therefore,

$$I = \oint_C \frac{-\frac{1}{4}\frac{(z^2-1)^2}{z^2}}{\frac{2z^2+5z+2}{z}} \frac{dz}{iz} = -\frac{1}{4i}\oint_C \frac{(z^2-1)^2}{z^2(2z^2+5z+2)}\, dz$$

Set

$$f(z) = \frac{(z^2-1)^2}{z^2(2z^2+5z+2)}$$

Factoring the quadratic $2z^2+5z+2 = (2z+1)(z+2)$. Thus, f has a simple pole at $z = -\frac{1}{2}$, a simple pole at $z = -2$, and a pole of order 2 at $z = 0$. Inside the unit circle, the poles are $z = -\frac{1}{2}$ and $z = 0$. So by Cauchy's Residue Theorem (Theorem 6.1.13),

$$I = -\frac{1}{4}2\pi i\left(\operatorname{Res}\left(f, -\frac{1}{2}\right) + \operatorname{Res}(f, 0)\right)$$

We compute these residues.

- Since $z = -\frac{1}{2}$ is a simple pole,

$$\operatorname{Res}\left(f, -\frac{1}{2}\right) = \lim_{z\to -\frac{1}{2}}\left(z+\frac{1}{2}\right)\frac{(z^2-1)^2}{z^2(2z+1)(z+2)} = \lim_{z\to -\frac{1}{2}}\frac{(z^2-1)^2}{2z^2(z+2)} = \frac{3}{4}$$

- Since $z = 0$ is a pole of order 2,

$$\operatorname{Res}(f, 0) = \lim_{z\to 0}\frac{d}{dz}[z^2 f(z)] = \lim_{z\to 0}\frac{d}{dz}\left(\frac{(z^2-1)^2}{2z^2+5z+2}\right) = -\frac{5}{4}$$

  Therefore,

$$\operatorname{Res}\left(f, -\frac{1}{2}\right) + \operatorname{Res}(f, 0) = -\frac{1}{2}$$

  Hence,

$$I = \int_0^{2\pi}\frac{\sin^2(\theta)}{5+4\cos(\theta)}\, d\theta = -\frac{1}{4i}\cdot 2\pi i\left(-\frac{1}{2}\right) = \frac{\pi}{4}$$

Sometimes the given trigonometric integral is not over the interval $[0, 2\pi]$, but it can be transformed into one that is.

Theorem 6.2.8. *A trigonometric integral over $[0, \pi]$ or another interval can sometimes rewritten as an integral over $[0, 2\pi]$ by a suitable symmetry substitution, such as $\phi = 2\pi - \theta$.*

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Remark 6.2.9. This is not a single formula to memorize, but rather a useful strategy: if the integral is not already in standard form, try to use trigonometric identities or interval symmetries to convert it to one that is.

Example 6.2.10. Evaluate

$$I = \int_0^{\pi} \frac{1}{2 - \cos(\theta)}\, d\theta$$

Solution. This is not the standard $[0, 2\pi]$ form, so first we transform it. Using the substitution $\phi = 2\pi - \theta$, we have $d\phi = -d\theta$, and $\cos(2\pi - \theta) = \cos(\phi)$. Therefore,

$$I = \int_{\pi}^{2\pi} \frac{1}{2 - \cos(\theta)}\, d\theta$$

Adding this to the original integral yields

$$2I = \int_0^{2\pi} \frac{1}{2 - \cos(\theta)}\, d\theta$$

Now apply the substitution $z = e^{i\theta}$. Then

$$\begin{aligned} 2I &= \oint_C \frac{1}{2 - \frac{1}{2}\left(z + \frac{1}{z}\right)} \frac{dz}{iz} \\ &= \frac{2z}{4z - z^2 - 1} \cdot \frac{dz}{iz} \\ &= -\frac{2}{i} \oint_C \frac{1}{z^2 - 4z + 1}\, dz \end{aligned}$$

Let

$$f(z) = \frac{1}{z^2 - 4z + 1}$$

The poles are at $z = 2 \pm \sqrt{3}$. Among these, $z = 2 - \sqrt{3}$ lies inside the unit circle. Therefore,

$$2I = -\frac{2}{i} \cdot 2\pi i \operatorname{Res}(f, 2 - \sqrt{3})$$

Since $z = 2 - \sqrt{3}$ is a simple pole

$$\operatorname{Res}(f, 2 - \sqrt{3}) = \lim_{z \to 2 - \sqrt{3}} \frac{z - (2 - \sqrt{3})}{(z - (2 - \sqrt{3}))(z - (2 + \sqrt{3}))} = -\frac{1}{2\sqrt{3}}$$

Therefore, as

$$2I = -\frac{2}{i} \cdot 2\pi i \left(-\frac{1}{2\sqrt{3}}\right) = \frac{2\pi}{\sqrt{3}}$$

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Therefore,

$$\int_0^{\pi} \frac{1}{2-\cos(\theta)}\, d\theta = \frac{\pi}{\sqrt{3}}$$

6.3 Cauchy Principal Values of Improper Integrals

In the previous section, we used residue theory to evaluate trigonometric integrals by converting them into contour integrals on the unit circle. We now turn to another important application of residues: the evaluation of improper integrals over the real line, especially those taken over the interval $(-\infty, \infty)$. These integrals arise naturally in analysis, differential equations, and Fourier theory, but they do not always coverge in the ordinary sense.

A new idea is therefore needed. In many cases, even when the improper integral

$$\int_{-\infty}^{\infty} f(x)\, dx$$

fails to exist as an ordinary convergent integral, the symmetric limit

$$\lim_{R\to\infty} \int_{-R}^{R} f(x)\, dx$$

may still exist. This symmetric limit is called the Cauchy principal value. It captures a kind of balanced cancellation between the positive and negative halves of the real axis, and it is especially well suited to contour integration methods.

The main strategy of this section is to integrate a complex function $f(z)$ over a contour consisting of a real interval $[-R, R]$ together with a large semicircle in the upper-half plane. If the contribution from the semicircle tends to zero as $R \to \infty$, then the residue theorem relates the real integral, or its principal value, to the sum of residues of the poles in the upper half-plane. This idea leads to elegant evaluations of many improper integrals that would otherwise be difficult to compute directly.

We begin by reviewing the definitions of improper integrals and principal values, then establish the contour method for principal values, study the behaviour of integrals over large semicircles, and finally consider the important case where the integrand has poles on the real axis, which requires the use of indented contours.

We first review the ordinary notion of an improper integral over an infinite interval.

Definition 6.3.1. Let f be a function.

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(1) If f is a continuous function on $[0, \infty)$, the *improper integral of f* over $[0, \infty)$ is defined by

$$\int_0^\infty f(x)\ dx = \lim_{R\to\infty} \int_0^R f(x)\ dx$$

provided the limit exists. If the limit exists, the integral is said to *converge*. Otherwise it diverges.

(2) If f is a continuous function on $(-\infty, 0]$, the *improper integral of f* over $(-\infty, 0]$ is defined by

$$\int_{-\infty}^0 f(x)\ dx = \lim_{R\to\infty} \int_{-R}^0 f(x)\ dx$$

(3) If f is a continuous function on $(-\infty, \infty)$, the *improper integral of f* over $(-\infty, \infty)$ is defined by

$$\int_{-\infty}^\infty f(x)\ dx = \int_{-\infty}^0 f(x)\ dx + \int_0^\infty f(x)\ dx$$

provided both integrals on the right converge.

Remark 6.3.2. It is important to note that $\int_{-\infty}^\infty f(x)\ dx$ is not, by definition, the same thing as the symmetric limit

$$\lim_{R\to\infty} \int_{-R}^R f(x)\ dx$$

The ordinary improper integral requires the two one-sided improper integrals to converge separately.

Example 6.3.3. Consider $f(x) = x$ on $(-\infty, \infty)$. Then

$$\int_0^\infty x\ dx = \lim_{R\to\infty} \int_0^R x\ dxx = \lim_{R\to\infty} \frac{R^2}{2} = \infty$$

so $\int_{-\infty}^\infty x\ dx$ does not exist as an improper integral. However,

$$\int_{-R}^R x\ dx = \frac{R^2}{2} - \frac{R^2}{2} = 0$$

Thus, the symmetric limit exists and is zero, even though the improper integral diverges. This motivates the next definition.

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Definition 6.3.4. Let f be continuous on $(-\infty, \infty)$. The *Cauchy principal value* of the integral of f over $(-\infty, \infty)$ is defined by

$$\text{pv} \int_{-\infty}^{\infty} f(x)\ dx\ dx = \lim_{R\to\infty} \int_{-R}^{R} f(x)\ dx$$

provided the limit exists.

Remark 6.3.5. The Cauchy principal value uses a symmetric limiting process. It may exist even when the usual improper integral does not.

Theorem 6.3.6. *If the improper integral*

$$\int_{-\infty}^{\infty} f(x)\ dx$$

exists, then the Cauchy principal value also exists and

$$\text{pv} \int_{-\infty}^{\infty} f(x)\ dx = \int_{-\infty}^{\infty} f(x)\ dx$$

Proof. Assume the improper integral exists. Then both

$$\int_{-\infty}^{0} f(x)\ dx, \quad \int_{0}^{\infty} f(x)\ dx$$

exist. Hence,

$$\lim_{R\to\infty} \int_{-R}^{R} f(x)\ dx = \lim_{R\to\infty} \left(\int_{-R}^{0} f(x)\ dx + \int_{0}^{R} f(x)\ dx \right)$$

Since the two limits exist separately, we may pass to the limit term by term

$$\lim_{R\to\infty} \int_{-R}^{R} f(x)\ dx = \int_{-\infty}^{0} f(x)\ dx + \int_{0}^{\infty} f(x)\ dx = \int_{-\infty}^{\infty} f(x)\ dx$$

Therefore, the principal value exists and is equal to the improper integral. □

Example 6.3.7. Let $f(x) = x$. Then, as seen above,

$$\int_{-\infty}^{\infty} x\ dx$$

does not exist, but

$$\text{pv} \int_{-\infty}^{\infty} x\ dx = \lim_{R\to\infty} \int_{-R}^{R} x\ dx = 0$$

So the converse of Theorem 6.3.6 is false.

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Remark 6.3.8. If f is even, so that $f(-x) = f(x)$, then

$$\int_{-R}^{R} f(x)\ dx = 2\int_{0}^{R} f(x)\ dx$$

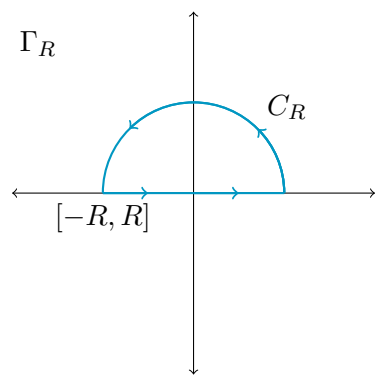
Therefore, whenever the principal value exists

$$\text{pv}\int_{-\infty}^{\infty} f(x)\ dx = 2\int_{0}^{\infty} f(x)\ dx$$

provided the integral on $[0, \infty)$ converges. In particular, for even functions, the existence of the principal value often leads directly to the value of the original improper integral on $[0, \infty)$.

We now explain how residues are used to evaluate principal values.

Suppose $f(z)$ is a rational function with no poles on the real axis and whose poles in the upper half-plane are $z_1, z_2, ..., z_n$. Let C_R be the upper semicircle of radius R, and let Γ_R be the closed contour formed by $[-R, R]$ together with C_R, oriented counterclockwise.



Then by the residue theorem (Theorem 6.1.13),

$$\int_{-R}^{R} f(x)\ dx + \int_{C_R} f(z)\ dz = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

If we can show that

$$\int_{C_R} f(z)\ dz \to 0$$

as $R \to \infty$, then letting $R \to \infty$ gives

$$\text{pv}\int_{-\infty}^{\infty} f(x)\ dx = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

This is the basic contour formula for principal values.

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Theorem 6.3.9. *Let $f(z)$ be rational, and suppose*

(1) f has no poles on the real axis,
(2) all poles in the upper-half plane are $z_1, z_2, ..., z_n$, and
(3) the integral over the upper semicircle C_R satisfies

$$\int_{C_R} f(z)\ dz \to 0$$

as $R \to \infty$.

Then

$$\text{pv} \int_{-\infty}^{\infty} f(x)\ dx = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

Proof. For each sufficiently large R, the contour Γ_R encloses exactly the poles $z_1, z_2, ..., z_n$ in the upper half plane. Hence, by the residue theorem (Theorem 6.1.13),

$$\int_{-R}^{R} f(x)\ dx + \int_{C_R} f(z)\ dz = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

Taking the limit as $R \to \infty$, and using condition (3), we obtain

$$\lim_{R\to\infty} \int_{-R}^{R} f(x)\ dx = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

Therefore,

$$\text{pv} \int_{-\infty}^{\infty} f(x)\ dx = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

□

To apply Theorem 6.3.9, we need conditions ensuring that the integral over the large semicircle tends to zero.

Theorem 6.3.10. *Suppose $f(z) = \frac{p(z)}{q(z)}$ is a rational function, where $\deg(p(z)) = n$ and $\deg(q(z)) = m$. If $m \geq n + 2$, then for the upper semicircle C_R given by $z = Re^{i\theta}$ for $0 \leq \theta \leq \pi$,*

$$\int_{C_R} f(z)\ dz \to 0$$

as $R \to \infty$.

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Proof. Because $\deg(q(z)) - \deg(m(z)) \geq 2$, the rational function behaves like $\frac{1}{z^2}$ or faster than infinity. More precisely, there exists $M > 0$ and $R_0 > 0$ such that for all $|z| \geq R_0$,

$$|f(z)| \leq \frac{M}{|z|^2}$$

On the semicircle C_R, we have $|z| = R$, so

$$|f(z)| \leq \frac{M}{R^2}$$

The length of C_R is πR, so by the ML-inequality (Theorem 4.2.26)

$$\left| \int_{C_R} f(z)\ dz \right| \leq \frac{M}{R^2} \cdot \pi R = \frac{M\pi}{R}$$

As $R \to \infty$, the right side tends to zero. Hence,

$$\int_{C_R} f(z)\ dz \to 0.$$

□

Remark 6.3.11. This theorem is the standard tool for rational integrands. It says that if the denominator has degree at least two more than the numerator, then the semicircular contribution vanishes.

Example 6.3.12. Evaluate

$$\text{pv} \int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\ dx$$

Solution. Let

$$f(z) = \frac{1}{(z^2+1)(z^2+9)}$$

Then factoring the denominator gives

$$(z^2+1)(z^2+9) = (z-i)(z+i)(z-3i)(z+3i)$$

The poles in the upper half plane are $z = i$ and $z = 3i$, both are simple.

Since the degree of the denominator is 4 and the degree of the numerator is zero, Theorem 6.3.10 applies, so the semicircle integral tends to zero. Thus,

$$\text{pv} \int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\ dx = 2\pi i(\text{Res}(f,i) + \text{Res}(f,3i))$$

Now compute the residues:

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- At $z = i$,

$$\operatorname{Res}(f, i) = \lim_{z \to i} \frac{1}{(z+i)(z-3i)(z+3i)} = \frac{1}{16i}$$

- At $z = 3i$,

$$\operatorname{Res}(f, 3i) = \lim_{z \to 3i} \frac{1}{(z-i)(z+i)(z+3i)} = -\frac{1}{48i}$$

Therefore,

$$\operatorname{pv} \int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\, dx = 2\pi i \left(\frac{1}{24i}\right) = \frac{\pi}{12}$$

Because the integrand is even and in fact, convergent on $(-\infty, \infty)$, we also obtain

$$\int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\, dx = \frac{\pi}{12}$$

Example 6.3.13. Evaluate

$$\operatorname{pv} \int_{-\infty}^{\infty} \frac{1}{x^4+1}\, dx$$

Solution. Let

$$f(z) = \frac{1}{z^4+1}$$

From the factorization of $z^4 + 1$, the poles in the upper half plane are

$$z_1 = e^{\frac{i\pi}{4}}, \quad z_2 = e^{\frac{3i\pi}{r}}$$

both simple. By Example 6.1.11, their residues are

$$\operatorname{Res}(f, z_1) = -\frac{1}{4\sqrt{2}} - \frac{i}{4\sqrt{2}}, \quad \operatorname{Res}(f, z_2) = \frac{1}{4\sqrt{2}} - \frac{i}{4\sqrt{2}}$$

Adding and multiplying by $2\pi i$ gives

$$2\pi i(\operatorname{Res}(f, z_1) + \operatorname{Res}(f, z_2)) = \frac{\pi}{\sqrt{2}}$$

Therefore, since $\deg(q(z)) - \deg(p(z)) = 4 \geq 2$, the semicircle integral tends to 0. Thus,

$$\operatorname{pv} \int_{-\infty}^{\infty} \frac{1}{x^4+1}\, dx = \frac{\pi}{\sqrt{2}}$$

Since the integrand is even and convergent, it follows

$$\int_{-\infty}^{\infty} \frac{1}{x^4+1}\, dx = \frac{\pi}{\sqrt{2}}$$

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Many important improper integrals involve factors such as $\cos(\alpha x)$ and $\sin(\alpha x)$. These are often called Fourier integrals. The starting formula

$$e^{i\alpha x} = \cos(\alpha x) + i\sin(\alpha x)$$

Then

$$\int_{-\infty}^{\infty} f(x)e^{i\alpha x}\ dx = \int_{-\infty}^{\infty} f(x)\cos(\alpha x)\ dx + i\int_{-\infty}^{\infty} f(x)\sin(\alpha x)\ dx$$

whenever these integrals make sense.

To evaluate such expressions, one considers the contour integral of $f(z)e^{i\alpha z}$ over the upper semicircle when $\alpha > 0$. Under suitable conditions, the semiciricular contribution again tends to zero.

Theorem 6.3.14. *Let $f(z) = \frac{p(z)}{q(z)}$ be a rational function with $\deg(q(z)) \geq \deg(p(z)) + 2$ and let $\alpha > 0$. Then for the upper semicircle C_R*

$$\int_{C_R} f(z)e^{i\alpha z}\ dz \to 0$$

as $R \to \infty$.

Remark 6.3.15. This is the key fact behind contour evaluations of integrals involving sine and cosine. The exponential factor $e^{i\alpha z}$ decays in the upper half-plane when $\alpha > 0$, which helps force the semicircle contribution to vanish.

Example 6.3.16. Evaluate

$$\int_0^{\infty} \frac{x\sin(x)}{x^2+9}\ dx$$

Solution. Since the integrand is even, we have

$$\int_0^{\infty} \frac{x\sin(x)}{x^2+9}\ dx = \frac{1}{2}\text{pv}\int_{-\infty}^{\infty} \frac{x\sin(x)}{x^2+9}\ dx$$

Now consider

$$\text{pv}\int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2+9}\ dx$$

Let

$$f(z) = \frac{ze^{iz}}{z^2+9}$$

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The only pole in the upper half-plane is at $z = 3i$, which is simple. Since the exponential factor ensures decay on the upper semicircle, the residue theorem gives

$$\text{pv} \int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2+9}\, dx = 2\pi i \operatorname{Res}(f, 3i)$$

Compute the residue

$$\operatorname{Res}(f, 3i) = \lim_{z \to 3i} \frac{ze^{iz}}{z+3i} = \frac{e^{-3}}{2}$$

Thus,

$$\text{pv} \int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2+9}\, dx = 2\pi i \frac{e^{-3}}{2} = \pi i e^{-3}$$

Now write

$$\frac{xe^{ix}}{x^2+9} = \frac{x\cos(x)}{x^2+9} + i\frac{x\sin(x)}{x^2+9}$$

So

$$\text{pv} \int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2+9}\, dx = \text{pv} \int_{-\infty}^{\infty} \frac{x\cos(x)}{x^2+9}\, dx + \text{pv} \int_{-\infty}^{\infty} \frac{x\sin(x)}{x^2+9}\, dx$$

Comparing real and imaginary parts with $\pi i e^{-3}$, we get

$$\text{pv} \int_{-\infty}^{\infty} \frac{x\cos(x)}{x^2+9}\, dx = 0, \quad \text{pv} \int_{-\infty}^{\infty} \frac{x\sin(x)}{x^2+9}\, dx = \pi e^{-3}$$

Therefore,

$$\int_{0}^{\infty} \frac{x\sin(x)}{x^2+9}\, dx = \frac{\pi}{2e^3}$$

So far we assumed there were no poles on the real axis. If the integrand has a pole on the real axis, we must deform the contour to avoid it. This leads to the use of an indented contour.

Suppose f has a simple pole at $z = c \in \mathbb{R}$. We consider a contour consisting of

- The interval from $-R$ to R,
- a large upper semicircle,
- a small semicicular indentation of radius r above the point c.

The small indentation is traversed clockwise, so that the overall contour remains positively oriented around the upper half-plane with the pole excluded.

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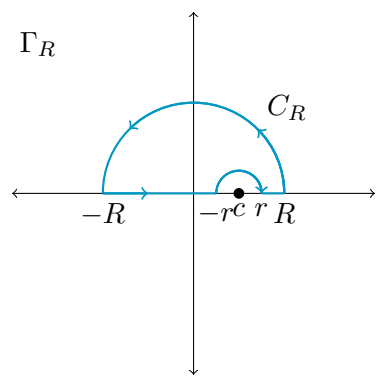

Theorem 6.3.17. *Suppose f has a simple pole at $z = c \in \mathbb{R}$. Let C_r be the small upper semicircle $z = c + re^{i\theta}$, $0 \leq \theta \leq \pi$ traversed clockwiseas part of the indented contour. Then*

$$\lim_{r \to 0} \int_{C_r} f(z)\ dz = -\pi i \operatorname{Res}(f, c)$$

Equivalently, if one denotes the small semicircle with positive orientation centered at c, then its limit is $\pi i \operatorname{Res}(f, c)$. In the indented contour used for principal values, the actual contribution appears with a minus sign.

Proof. Since f has a simple pole at $z = c$, its Laurent expansion near c is

$$f(z) = \frac{a_{-1}}{z - c} + g(z)$$

where $a_{-1} = \operatorname{Res}(f, c)$ and g is holomorphic near c. Thus,

$$\int_{C_r} f(z)\ dz = a_{-1} \int_{C_r} \frac{1}{z - c}\ dz + \int_{C_r} g(z)\ dz$$

Parametrize the upper semicircle by $z = c + re^{i\theta}$, $0 \leq \theta \leq \pi$. Then $dz = ire^{i\theta}\ d\theta$ and $z - c = re^{i\theta}$. Hence,

$$\int_{C_r} \frac{1}{z - c}\ dz = \int_0^\pi \frac{ire^{i\theta}}{re^{i\theta}}\ d\theta = i \int_0^\pi d\theta = \pi i$$

Also, since g is bounded near c, the ML-inequality (Theorem 4.2.26) shows that

$$\lim_{r \to 0} \left| \int_{C_r} g(z)\ dz \right| = 0$$

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Therefore,

$$\lim_{r\to 0}\int_{C_r} f(z)\ dz = \pi i a_{-1} = \operatorname{Res}(f,c)$$

If the arc is traversed in the opposite direction, the sign changes. In the standard indented contour used for principal values, the small arc is clockwise, giving $-\pi i \operatorname{Res}(f,c)$. □

Example 6.3.18. Evaluate

$$\operatorname{pv}\int_{-\infty}^{\infty} \frac{\sin(x)}{x(x^2-2x+2)}\ dx$$

Solution. Consider the complex integral

$$\operatorname{pv}\int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2-2x+2)}\ dx$$

Let

$$f(z) = \frac{e^{iz}}{z(z^2-2z+2)}$$

The poles are $z=0$ (a simple pole on the real axis), $z=1+i$ (a simple pole in the upper-half plane), and $z=1-i$ (a pole in the lower half plane). Using an indented contour above the real axis, the residue theorem gives

$$\operatorname{pv}\int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2-2x+2)}\ dx = 2\pi i \operatorname{Res}(f,1+i) + \pi i \operatorname{Res}(f,0)$$

Computing the residues

$$\begin{aligned}\operatorname{Res}(f,0) &= \lim_{z\to 0}\frac{e^{iz}}{z^2-2z+2} = \frac{1}{2},\\ \operatorname{Res}(f,1+i) &= \lim_{z\to 1+i}\frac{e^{iz}}{z(z-(1-i))} = \frac{e^{i(1+i)}}{(1+i)(2i)} = -\frac{e^{-1+i}}{4}(1+i)\end{aligned}$$

Therefore,

$$\operatorname{pv}\int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2-2x+2)}\ dx = \pi i\left(\frac{1}{2}\right) + 2\pi i\left(-\frac{e^{-1+i}}{4}(1+i)\right)$$

Separating real and imaginary parts, and using

$$e^{-1+i} = e^{-1}(\cos(1) + i\sin(1))$$

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we obtain

$$\text{pv}\int_{-\infty}^{\infty} \frac{\cos(x)}{x(x^2-2x+2)}\,dx = \frac{\pi}{2}e^{-1}(\sin(1)+\cos(1))$$

and

$$\text{pv}\int_{-\infty}^{\infty} \frac{\sin(x)}{x(x^2-2x+2)}\,dx = \frac{\pi}{2}[1+e^{-1}(\sin(1)-\cos(1)]$$

6.4 Fourier Transforms of Rational Functions

In the previous section, we used contour integration and residues to evaluate improper integrals and Cauchy principal values on the real line. A particularly important class of such integrals arises when the integrand contains the oscillatory factor $e^{i\alpha x}$ or equivalently, $\cos(\alpha x)$ and $\sin(\alpha x)$. These integrals are central in analysis because they lead directly to the Fourier transform, one of the most fundamental tools in pure and applied mathematics.

The Fourier transform converts a function of the real variable x into a function of the frequency variable α. In this way, it allows us to analyze a function in terms of its oscillatory components. For rational functions, or functions closely related to them, residue theory provides a powerful method for computing the transform explicitly. The exponential factor $e^{i\alpha z}$ behaves differently in the upper and lower half-planes depending on the sign of α, so the choice of contour becomes an essential part of computation.

In this section, we develop the residue method for evaluating Fourier transforms of rational functions. We begin by recalling the definitions of the Fourier transform and inverse Fourier transform, then establish a key estimate known as Jordan's Lemma, which guarantees that the contribution from a large semicircle tends to zero. With this result in hand, we compute several Fourier-type integrals by residues and then use these ideas to recover inverse transforms explicitly. These examples illustrate how complex analysis gives not only elegant formulas, but also a deeper understanding of why those formulas take the form they do.

Definition 6.4.1. Let $f(x)$ be a real-valued or complex-valued function defined on $(-\infty,\infty)$. Its *Fourier transform* is defined by

$$\hat{f}(\alpha) = \int_{-\infty}^{\infty} f(x)e^{i\alpha x}\,dx$$

whenever the integral exists.

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Definition 6.4.2. If $\hat{f}(\alpha)$ is the Fourier transform of f, then the *inverse Fourier transform* is given by

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\alpha) e^{-i\alpha x} \, d\alpha$$

Remark 6.4.3. In these notes, the Fourier transform is written without normalization in the forward transform, and with the factor $\frac{1}{2\pi}$ in the inverse transform. Other conventions exist, but we will follow this one throughout.

Using Euler's formula

$$e^{i\alpha x} = \cos(\alpha x) + i \sin(\alpha x)$$

we obtain

$$\int_{-\infty}^{\infty} f(x) e^{i\alpha x} \, dx = \int_{-\infty}^{\infty} f(x) \cos(\alpha x) \, dx + i \int_{-\infty}^{\infty} f(x) \sin(\alpha x) \, dx$$

Thus, Fourier transforms are closely related to the cosine and sine transforms studied in the previous section.

To compute Fourier transforms by contour integration, we need a result that guarantees the integral over a large semicircle tends to zero. This is the key estimate that makes the residue method work for oscillatory integrals.

Theorem 6.4.4 (Jordan's Lemma). *Let $f(z) = \frac{p(z)}{q(z)}$ be a rational function such that $\deg(q(z)) \geq \deg(p(z)) + 1$. Then*

(1) If $\alpha > 0$, for the upper semicircle C_R^+ of radius R oriented counterclockwise,

$$\lim_{R \to \infty} \int_{C_R^+} e^{i\alpha z} \frac{p(z)}{q(z)} \, dz = 0$$

(2) If $\alpha < 0$, for the lower semicircle C_R^- of radius R, oriented counterclockwise,

$$\lim_{R \to \infty} \int_{C_R^-} e^{i\alpha z} \frac{p(z)}{q(z)} \, dz = 0$$

Proof. We prove the case $\alpha > 0$ and the upper semicircle C_R^+. The other case is analogous. Parametrize the upper semicircle by $z = Re^{it}$ for $0 \leq t \leq \pi$. Then $dz = iRe^{it} \, dt$, and so

$$\int_{C_R^+} e^{i\alpha z} \frac{p(z)}{q(z)} \, dz = \int_0^{\pi} e^{i\alpha Re^{it}} \frac{p(Re^{it})}{q(Re^{it})} iRe^{it} \, dt$$

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Now,

$$e^{i\alpha Re^{it}} = e^{i\alpha R(\cos(t)+i\sin(t))} = e^{i\alpha R\cos(t)-\xi R\sin(t)}$$

so its modulus is

$$|e^{i\alpha Re^{it}}| = e^{-\alpha R\sin(t)}$$

Since $\deg(q(z)) - \deg(p(z) \geq 1$, there exists $M > 0$ such that for sufficiently large R,

$$\left|\frac{p(Re^{it})}{q(Re^{it})}\right| \leq \frac{M}{R}$$

Therefore,

$$\left|e^{i\alpha Re^{it}}\frac{p(Re^{it})}{q(Re^{it})}iRe^{it}\right| \leq Me^{-\alpha R\sin(t)}$$

Hence,

$$\left|\int_{C_R^+} e^{i\alpha z}\frac{p(z)}{q(z)}\ dz\right| \leq M\int_0^\pi e^{-i\alpha R\sin(t)}\ dt$$

Now using the symmetry of $\sin(t)$ on $[0,\pi]$

$$\int_0^1 e^{-\alpha R\sin(t)}\ dt = 2\int_0^{\frac{\pi}{2}} e^{-\alpha R\sin(t)}\ dt$$

On $[0,\frac{\pi}{2}]$, we have the inequality $\sin(t) \geq \frac{2}{\pi}t$. Thus,

$$e^{-\alpha R\sin(t)} \leq e^{-\frac{2\alpha Rt}{\pi}}$$

So

$$\int_0^\pi e^{-\alpha R\sin(t)}\ dt \leq 2\int_0^{\frac{\pi}{2}} e^{-\frac{2\alpha Rt}{\pi}}\ dt = 2\left[-\frac{\pi}{2\alpha R}e^{-\frac{2\alpha Rt}{\pi}}\right]_0^{\frac{\pi}{2}} < \frac{\pi}{\alpha R}$$

Therefore,

$$\left|\int_{C_R^+} e^{i\alpha z}\frac{p(z)}{q(z)}\ dz\right| \leq M\frac{\pi}{\alpha R} \to 0$$

as $R \to \infty$. This proves the result. □

Remark 6.4.5. Jordan's Lemma is stronger than the degree condition used earlier for plain rational functions. The exponential factor $e^{i\alpha z}$ supplies decay in teh correct half-plane, so one extra power in the denominator is needed.

Sometimes a Fourier transform can be computed directly from the definition before turning to residue theory.

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Example 6.4.6. Find the Fourier transform of

$$f(x) = e^{-|x|}$$

Solution. First write f in piecewise form

$$f(x) = \begin{cases} e^{x} & \text{if } x < 0 \\ e^{-x} & \text{if } x \geq 0 \end{cases}$$

Therefore,

$$\hat{f}(\alpha) = \int_{-\infty}^{0} e^{x} e^{i\alpha x} \, ddx + \int_{0}^{\infty} e^{-x} e^{i\alpha x} \, dx$$

Call these integrals I_1 and I_2, respectively. For I_1,

$$I_1 = \int_{-\infty}^{0} e^{-(1+i\alpha)x} \, dx = \left[\frac{e^{(1+i\alpha)x}}{1+i\alpha} \right]_{-\infty}^{0} = \frac{1}{1+i\alpha}$$

since $e^{(1+i\alpha)x} \to 0$ as $x \to -\infty$. For I_2,

$$I_2 = \int_{0}^{\infty} e^{(-1+i\alpha)x} \, dx = \left[\frac{e^{(-1+i\alpha)x}}{-1+i\alpha} \right]_{0}^{\infty} = \frac{1}{1-i\alpha}$$

Adding

$$\hat{f}(\alpha) = \frac{1}{1+i\alpha} + \frac{1}{1-i\alpha} = \frac{2}{1+\alpha^2}$$

Remark 6.4.7. This example is important because it gives a classical transform pair

$$e^{-|x|} \longleftrightarrow \frac{2}{1+\alpha^2}$$

We will recover the inverse transform of this rational function by contour integration.

We now use residues to invert the transform found in Example 6.4.6.

Example 6.4.8. Find the inverse Fourier transform of

$$\hat{f}(\alpha) = \frac{2}{1+\alpha^2}$$

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Solution. We want to evaluate

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{2}{1+\alpha^2} e^{-i\alpha x}\, d\alpha$$

Rename the integration variable as z, and write

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{2}{1+z^2} e^{-ixz}\, dz$$

The integrand has simple poles at $z = i$ and $z = -i$. The contour we choose depends on the sign of x.

(1) If $x < 0$, then $-ixz$ has exponential decay in the upper half plane, so we close the contour with a large semicircle. By Jordan's Lemma (Theorem 6.4.4), the integral over the semicircle tends to 0. Only the pole $z = i$ lies inside the contour. Thus,

$$\int_{-\infty}^{\infty} \frac{2}{1+z^2} e^{-ixz}\, dz = 2\pi i \operatorname{Res}\left(\hat{f}e^{-ixz}, i\right)$$

Computing the residue,

$$\operatorname{Res}\left(\hat{f}e^{-ixz}, i\right) = \lim_{z\to i}(z-i)\frac{2e^{-ixz}}{(z-i)(z+i)} = \frac{e^x}{i}$$

Therefore,

$$f(x) = \frac{1}{2\pi}\cdot 2\pi i \frac{e^x}{i} = e^x$$

for $x < 0$.

(2) If $x > 0$, then $-ixz$ decays in the lower half plane, so we close the contour with a large lower semicircle. By Jordan's Lemma (Theorem 6.4.4), the integral over the semicircle tends to 0. Only the pole $z = -i$ lies inside the contour. Thus,

$$\int_{-\infty}^{\infty} \frac{2}{1+z^2} e^{-ixz}\, dz = -2\pi i \operatorname{Res}\left(\hat{f}e^{-ixz}, -i\right)$$

Now,

$$\operatorname{Res}\left(\hat{f}e^{-ixz}, -i\right) = \lim_{z\to -i}(z+i)\frac{2e^{-ixz}}{(z-i)(z+i)} = -\frac{e^{-x}}{i}$$

Therefore,

$$f(x) = \frac{1}{2\pi}\left(-2\pi i\left(-\frac{e^{-x}}{i}\right)\right) = e^{-x}$$

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Combining the two cases, we have $f(x) = e^{-|x|}$. At $x = 0$, the integral is 1.

We now emphasize a principal-value version of the Fourier transform and the use of Jordan's Lemma to compute cosine and sine transforms of rational functions.

Definition 6.4.9. For a suitable function f, one may define the Fourier transform by

$$\hat{f}(\alpha) = \frac{1}{\sqrt{2\pi}} \text{pv} \int_{-\infty}^{\infty} e^{-ix\alpha} f(x)\ dx$$

for $\alpha \in \mathbb{R}$, provided the Cauchy principal value exists.

Remark 6.4.10. This normalization differs from the one used earlier in the chapter, but the residue method is the same. The central idea is to choose the upper or lower semicircle according to the sign of the frequency variable α, so that the exponential factor decays.

Example 6.4.11. Compute

$$\text{pv} \int_{-\infty}^{\infty} \frac{x \sin(x)}{1 + x^2}\ dx$$

Solution. Since

$$\text{pv} \int_{-\infty}^{\infty} \frac{x \sin(x)}{1 + x^2}\ dx = \text{Im} \left(\text{pv} \int_{-\infty}^{\infty} \frac{xe^{ix}}{1 + x^2}\ dx \right)$$

we study the complex integral

$$\text{pv} \int_{-\infty}^{\infty} \frac{xe^{ix}}{1 + x^2}\ dx$$

Let

$$f(z) = \frac{ze^{iz}}{1 + z^2}$$

Take the upper semicircular contour, since the exponential factor e^{iz} decays in the upper half-plane. By Jordan's Lemma, the semicircular contribution tends to zero. The only pole in the upper half plane is $z = i$. Thus,

$$\text{pv} \int_{-\infty}^{\infty} \frac{xe^{ix}}{1 + x^2}\ dx = 2\pi i \operatorname{Res}(f, i)$$

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Computing the residue,

$$\operatorname{Res}(f, i) = \lim_{z \to i} (z - i) \frac{ze^{iz}}{(z-i)(z+i)} = \frac{1}{2e}$$

Therefore,

$$\operatorname{pv} \int_{-\infty}^{\infty} \frac{xe^{ix}}{1+x^2}\, dx = 2\pi i \cdot \frac{1}{2e} = \pi i e^{-1}$$

Taking imaginary parts,

$$\operatorname{pv} \int_{-\infty}^{\infty} \frac{x \sin(x)}{1+x^2}\, dx = \frac{\pi}{e}$$

Remark 6.4.12. Because the integrand $\frac{x\sin(x)}{1+x^2}$ is even, this also implies

$$\int_{0}^{\infty} \frac{x \sin(x)}{1+x^2}\, dx = \frac{\pi}{2e}$$

Example 6.4.13. Compute

$$\operatorname{pv} \int_{-\infty}^{\infty} \frac{\cos(x)}{x+i}\, dx$$

Solution. We cannot simply write this as the real and imaginary part of a single upper half plane contour integral without paying attention to which exponential decays in which half plane. Instead we use

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

Hence

$$\operatorname{pv} \int_{-\infty}^{\infty} \frac{\cos(x)}{x+i}\, dx = \frac{1}{2}\operatorname{pv} \int_{-\infty}^{\infty} \frac{e^{ix}}{x+i}\, dx + \frac{1}{2}\operatorname{pv} \int_{-\infty}^{\infty} \frac{e^{-ix}}{x+i}\, dx$$

Call these integrals I_1 and I_2.

Consider I_1. The pole is at $z = -i$, which lies in the lower half-plane. Since e^{iz} decays in the upper half-plane, we close in the upper half-plane. There are no poles inside, so by Cauchy's Theorem and Jordan's Lemma,

$$I_1 = \frac{1}{2}\operatorname{pv} \int_{-\infty}^{\infty} \frac{e^{ix}}{x+i}\, dx = 0$$

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Now consider I_2. Since e^{-iz} decays in the lower half plane, we close in the lower half-plane. The pole $z = -i$ lies inside. Because the contour is clockwise, we get

$$\text{pv} \int_{-\infty}^{\infty} \frac{e^{-ix}}{x+i}\, dx = -2\pi i \operatorname{Res}(f, -i)$$

Computing the residue

$$\operatorname{Res}(f, -i) = e^{-i(-i)} = e^{-1}$$

Thus,

$$I_2 = \frac{1}{2}\text{pv} \int_{-\infty}^{\infty} \frac{e^{-ix}}{x+i}\, dx = -\pi i e^{-1}$$

Therefore,

$$\text{pv} \int_{-\infty}^{\infty} \frac{\cos(x)}{x+i}\, dx = -\pi i e^{-1}$$

Remark 6.4.14. This example shows that one must be careful when separating oscillatory integrals into exponential pieces. The correct contour depends on the sign in the exponential.

6.5 Integrals on Branch Cuts

In the previous sections, we used residues to evaluate contour integrals around isolated singularities such as poles and essential singularities. However, not every important complex integral comes from a function that is single-valued on a punctured domain. Many algebraic and logarithmic functions, such as z^α, $\sqrt{z}$, and $\ln(z)$ are naturally multivalued, and their singular behaviour is no longer described only by isolated poles. Instead, they possess branch points, and to work with them analytically we must introduce a branch cut to make the function single-valued on a suitable region.

This leads to a new type of contour method. Rather than integrating around poles alone, we now integrate around contours that run along both sides of a branch cut. The values of the function on the upper and lower sides of the cut are generally different, and this difference is precisely what allows us to extract real integrals from the contour integral. In this way, branch cuts provide a powerful extension of residue methods to integrals involving fractional powers and other multivalued expressions.

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In this section, we study contour integration for functions with branch points, focusing especially on integrals of the form

$$\int_0^\infty f(x)\ dx$$

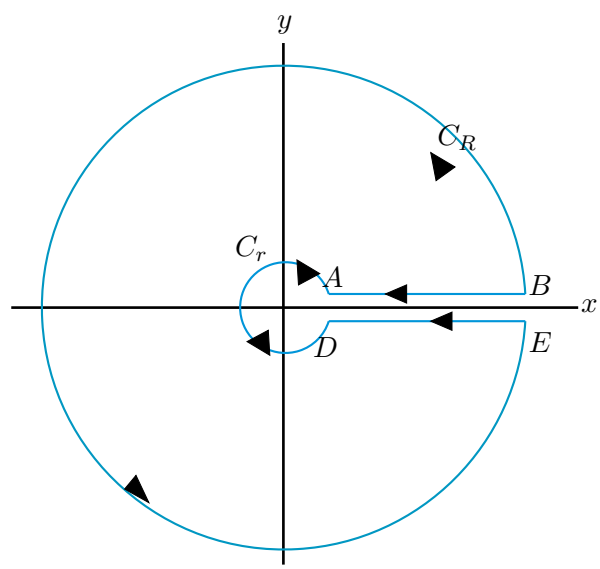
where the complexified integrand contains factors such as z^α or $z^{-\alpha}$.

To evaluate integrals along branch cuts, we use a contour that circles the origin while running along both sides of the positive real axis.

Definition 6.5.1. A *keyhole contour* adapted to a branch cut along the positive real axis consists of

(1) A large circle C_R of radius R,
(2) A small circle C_r of radius r around the branch point at 0,
(3) An upper line segment AB along the upper side of the positive real axis.
(4) A lower line segment ED along the lower side of the positive real axis.

This contour is traversed positively around the cut region.



Example 6.5.2. Evaluate

$$\int_0^\infty \frac{1}{\sqrt{x}(1+x)}\ dx$$

Solution. The integral is improper for two reasons:

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- the integrand is singular at $x = 0$
- the interval is unbounded.

Indeed, near $x = 0$,

$$\frac{1}{\sqrt{x}(x+1)} \sim x^{-\frac{1}{2}}$$

and near ∞,

$$\frac{1}{\sqrt{x}(x+1)} \sim x^{-\frac{3}{2}}$$

Both of these are integrable in the improper sense, so the integral converges.

Define

$$f(z) = \frac{1}{z^{\frac{1}{2}}(z+1)} \tag{6.1}$$

where $z^{\frac{1}{2}}$ is taken with the branch $z^{\frac{1}{2}} = e^{\frac{1}{2}(\ln|z|+i\theta)}$ for $0 < \theta < 2\pi$. Then f is single-valued and holomorphic on and inside the keyhole contour, except for a simple pole at $z = -1$.

Let C be the keyhole contour. Since the only singularity inside C is the simple pole at $z = -1$,

$$\oint_C \frac{1}{z^{\frac{1}{2}}(z+1)}\, dz = 2\pi i \operatorname{Res}(f, -1)$$

Writing the contour as a sum of its four parts give

$$\int_{C_R} f(z)\, dz + \int_{ED} f(z)\, dz + \int_{C_r} f(z)\, dz + \int_{AB} f(z)\, dz = 2\pi i \operatorname{Res}(f, -1)$$

On the upper segment AB, we have $z = x$ with $r \le x \le R$ and $z^{\frac{1}{2}} = x^{\frac{1}{2}}$ and $dz = dx$. Therefore,

$$\int_{AB} \frac{1}{z^{\frac{1}{2}}(z+1)}\, dz = \int_r^R \frac{x^{-\frac{1}{2}}}{x+1}\, dx$$

On the lower segment ED, $z = xe^{2\pi i}$ with $R \ge x \ge r$, so

$$z^{\frac{1}{2}} = (xe^{2\pi i})^{\frac{1}{2}} = x^{\frac{1}{2}} e^{i\pi} = -x^{\frac{1}{2}}$$

Also, $dz = dx$, but the path runs from R back to r. Hence,

$$\int_{ED} \frac{1}{z^{\frac{1}{2}}(z+1)}\, dz = \int_R^r \frac{1}{-x^{\frac{1}{2}}(x+1)}\, dx = \int_r^R \frac{x^{-\frac{1}{2}}}{x+1}\, dx$$

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Adding the two line integrals together give

$$\int_{AB} f(z)\ dz + \int_{ED} f(z)\ dz = 2\int_r^R \frac{x^{-\frac{1}{2}}}{x+1}\ dx$$

Now we show that the circular arcs vanish. On $|z| = R$, we have

$$|f(z)| = \left|\frac{1}{z^{\frac{1}{2}}(z+1)}\right| \leq \frac{1}{\sqrt{R}(R+1)}$$

Since the length of C_R is $2\pi R$, the ML-inequality (Theorem 4.2.26) gives

$$\left|\int_{C_R} f(z)\ dz\right| \leq \frac{2\pi R}{\sqrt{R}(R+1)} = \frac{2\pi\sqrt{R}}{R-1} \to 0$$

as $R \to \infty$.

On the smaller circle $|z| = r$,

$$|f(z)| = \left|\frac{1}{z^{\frac{1}{2}}(z+1)}\right| \leq \frac{1}{\sqrt{r}(1-r)}$$

for sufficiently small r. Since the length of C_r is $2\pi r$,

$$\left|\int_{C_r} f(z)\ dz\right| \leq \frac{2\pi r}{\sqrt{r}(1-r)} = \frac{2\pi\sqrt{r}}{1-r} \to 0$$

as $r \to 0$.

Since $z = -1 = e^{\pi i}$, we have

$$\text{Res}(f, -1) = \lim_{z\to -1}(z+1)\frac{1}{z^{\frac{1}{2}}(z+1)} = (-1)^{-\frac{1}{2}}$$

Using the chosen branch,

$$\text{Res}(f, -1) = (-1)^{-\frac{1}{2}} = (e^{i\pi})^{-\frac{1}{2}} = e^{-\frac{\pi i}{2}} = -i$$

Let $R \to \infty$ and $r \to 0$. The circular integrals vanish, so the residue equation becomes

$$2\int_0^\infty \frac{1}{\sqrt{x}(x+1)}\ dx = 2\pi i(-1) = 2\pi$$

Therefore,

$$\int_0^\infty \frac{1}{\sqrt{x}(x+1)}\ dx = \pi$$

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The previous example illustrates the standard mechanism.

Theorem 6.5.3. *Let $f(z)$ be a function involving a branch z^α with branch cut along the positive real axis, and suppose f is holomorphic inside a keyhole contour except for finitely many poles away from the cut. If the integrals over the large and small circles tend to zero, then the contour integral reduces to the difference between the values of the function on the two sides of the branch cut. In particular, if the values of the two sides differ by a constant factor, then the contour integral yields a multiple of a real improper integral over $(0, \infty)$.*

We now consider a second example, where the integrand has both a branch point at 0 and a pole on the positive real axis.

Example 6.5.4. Let $\alpha \in (0, 1)$. Evaluate

$$\text{pv} \int_0^\infty \frac{1}{x^\alpha (x-4)}\, dx$$

Solution. Let

$$f(z) = \frac{1}{z^\alpha (z-4)}$$

with $z^\alpha = e^{\alpha(\ln|z| + i\theta)}$ for $0 < \theta < 2\pi$. Then the branch cut is along the positive real axis. Because of the pole at $z = 4$, we must also indent the contour around 4.

The contour consists of a large circle C_R, a small circle C_r around the origin, a small upper indentation C_ρ^+ around 4, a small lower indentation C_ρ^- around 4, and upper and lower line segments along the cut. Since the contour avoids both the branch point and the pole on the cut, the function is holomorphic inside the contour, and Cauchy's theorem gives total contour integral zero.

On the upper side of the cut, $z^\alpha = x^\alpha$, so

$$\frac{1}{z^\alpha (z-4)} = \frac{1}{x^\alpha (x-4)}$$

On the lower side of the cut,

$$z^\alpha = x^\alpha e^{2\pi i \alpha}$$

so

$$\frac{1}{z^\alpha (z-4)} = \frac{e^{-2\pi i \alpha}}{x^\alpha (x-4)}$$

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Hence, the sum of the line integrals along the cut becomes

$$(1 - e^{-2\pi i\alpha})\text{pv}\int_0^\infty \frac{1}{x^\alpha(x-4)}\,dx$$

As in Example 6.5.2, one shows by the ML-inequality (Theorem 4.2.26) that the large circle contribution tends to zero as $R \to \infty$, and the small circle around the origin tends to zero as $r \to 0$.

The integrand has a simple pole at $z = 4$. The small upper and lower semicircular indentations contribute terms involving πi times the residue, with opposite branch values. Now,

$$\text{Res}(f, 4) = \lim_{z\to 4} \frac{1}{z^\alpha} = 4^{-\alpha}$$

Using the upper and lower branch values, the two indentation contributions combine to give $-\pi i 4^{-\alpha}$ and $-\pi i 4^{-\alpha} e^{-2\pi i\alpha}$. Thus, the total contribution from the indentation is $-\pi i 4^{-\alpha}(1 + e^{-2\pi i\alpha}$.

Since the total contour integral is zero, we obtain

$$\text{pv}\int_0^\infty \frac{1}{x^\alpha(x-4)}\,dx(1 - e^{-2\pi i\alpha}) = \pi i 4^{-\alpha}(1 + e^{-2\pi i\alpha})$$

Therefore,

$$\text{pv}\int_0^\infty \frac{1}{x^\alpha(x-4)}\,dx = \pi i 4^{-\alpha} \cdot \frac{1 + e^{-2\pi i\alpha}}{1 - e^{-2\pi i\alpha}}$$

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